MUHS final-year BPTh (IV BPTh), 2012 syllabus

Cardio-Vascular and Respiratory Physiotherapy

Short notes for revision: one sheet per topic, with the page of the full notes for each.



Sheets	58 sheets: the 23 Chapter 6 conditions, the technique cards, the other syllabus chapters and exam skills
Built from	The full notes and 20 MUHS question papers, Summer 2016 to Summer 2026 (code 51413)
Content checked	19 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/

Educational notes for exam preparation. Not medical advice. Sources for every fact are listed in the full notes.

How to use the short notes

These sheets are for revision, after you have read a topic once in the full notes. Each topic has its own sheet of one to three pages, in the same order as the full notes, and the last sheet collects the numbers to know.

1. **Read the yellow line first.** It says how the topic is asked in the MUHS paper.
2. **Cover a box and recall it,** then check. Bold words and numbers are the ones to write in the answer.
3. **Tick a square at the top right of a sheet** each time you revise it. Aim for day 1, day 7 and day 21.
4. **Need more?** Each sheet gives the page of the same topic in the full notes.

What the colours mean

Grey
Definitions, types, medical and surgical management

Orange
Disease process, clinical features and the problems they cause

Blue
Investigations, scales and normal values

Green
Physiotherapy: assessment, ICF, goals, treatment and doses

Amber
Precautions, contraindications, stop criteria and complications

Yellow
Exam plan: answer structure, what to draw, keywords

The paper (20 papers, 2016–2026): 80 marks in 3 hours, no MCQs. Short answers: any 5 of 6 at 3 marks and any 5 of 6 at 7 marks. Long answers: two questions, each any 1 of 2 at 15 marks, always a patient case, almost always asking ICF, a functional diagnosis or impairments. Since 2023 a ventilated ICU patient appears as a long case and an ICU short question is in every paper.

Contents

Part A: Chapter 6: Physiotherapy management

Tier 1: full notes, asked almost every year

IHD, MI, angioplasty and CABG, with cardiac rehab 3
Burns: acute care, positioning, splinting and inhalation injury 6
Mastectomy, lymphoedema and cancer surgery 8
Peripheral arterial disease, Buerger's disease and other arterial disorders 10
COPD: chronic bronchitis and emphysema 12
Lung surgery, thoracotomy, chest drain and chest trauma 14
Bronchiectasis, including after TB 16
Amputation and stump care 18
Varicose veins, venous ulcer, DVT and thrombophlebitis 20
Asthma in adults and children 22

Tier 2: asked sometimes

Abdominal surgery: pre- and post-operative care 24
Diabetes, diabetic foot, wound healing, pressure sores and diabetes insipidus 26
Obesity and metabolic syndrome 28
Tuberculosis, thoracoplasty and post-TB lung disease 29
Interstitial and occupational lung disease 30
Pleural effusion, pneumothorax, empyema and pleurisy 31
Hypertension 32
Valve disease, heart failure, cardiomyopathy, pacemaker and ICD 33
Congenital heart disease: tetralogy of Fallot, ASD, VSD, PDA 34

Tier 3: short notes, rarely or never asked on their own

Pneumonia, lung abscess and atelectasis 35
Neonatal and paediatric respiratory conditions 36
Cystic fibrosis 37
Flail chest, scoliosis and kyphosis 38

Part B: Technique cards

ICU and ventilation

Mechanical ventilation: indications, modes, CPAP and PEEP, complications, weaning 39
Physiotherapy in the ICU: assessment, safety, equipment and ventilated cases 41
Suctioning and manual hyperinflation 43
Therapeutic positioning, V/Q and the zones of the lung 44
Effects of bed rest and early mobilisation 45
ICU conditions: head injury, tetanus, spinal cord injury and neuromuscular disease 46
ICU conditions: pulmonary oedema, shock, organ failure, poisoning, aspiration and drowning 47

Airway clearance

Airway clearance techniques: overview and choice 48
ACBT, forced expiration, huffing and autogenic drainage 49
Postural drainage, percussion and vibration 50
PEP, flutter, Acapella and other airway-clearance devices 51

Breathing and breathlessness

Breathing exercises, lung expansion and breathing retraining 52
Neurophysiological facilitation and PNF of respiration 53
Respiratory muscle training 54
Dyspnoea, relieving positions, energy conservation and pacing 55

Oxygen, humidification and aerosols

Oxygen therapy: devices, targets and hazards 56
Humidification, nebulisers and inhalers 57

Electrotherapy, exercise, yoga and handling skills

Electrotherapy and therapeutic exercise in heart, lung and vascular conditions 58
Applied yoga in cardio-respiratory conditions 59
Handling skills and exercise equipment: stockings, bandaging, taping, splints, monitors 60

Exercise testing and exercise physiology

Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests 61
Exercise physiology and prescription: METs, energy systems, intensity, training 62

Investigations and scales

Investigations: chest X-ray, PFT, ECG and bedside tests 63
Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires 64
ABG interpretation and why patients desaturate 65
Abnormal breathing patterns and chest signs 66

Part C: Other syllabus chapters

Cardiac and pulmonary rehabilitation

Cardiac rehabilitation: definition, indications, contraindications, phases, outcome measures 67
Pulmonary rehabilitation: definition, indications, contraindications, components, outcome measures 68

Basics: anatomy, physiology and drugs

Respiratory anatomy, physiology and mechanics of the thorax 69
Cardiac anatomy and physiology for physiotherapists 70
Cardiac and respiratory drugs: what physiotherapists need to know 71

Basic life support

Basic life support (CPR) for physiotherapists 72

Part D: Exam skills

Exam skills

Answer writing: the paper, ICF, goals and documentation 73
Practical exam pack: long case, short case, spots and journal 74

End sheets

Numbers to know 76
Abbreviations 78

IHD, MI, angioplasty and CABG, with cardiac rehab

6(a). In 18 of 20 papers; the most common long answer (16 cases). Full notes p. 7

Asked: a 15-mark case in almost every paper (ICF 5 + cardiac rehab 10, or 3 + 5 + 7). **7 marks:** Phase I after MI, risk stratification, cardiac markers, Phase II after PCI for a sportsman, lifestyle education. **3 marks:** risk factors, CABG grafts and indications, Karvonen, ECG of acute and old MI.

Definitions

- **IHD (CAD):** heart muscle gets less blood and oxygen than it needs, usually from coronary **atherosclerosis**.
- **Angina:** temporary ischaemia, no permanent damage. **MI:** death of heart muscle from prolonged ischaemia.
- **MI (Fourth Universal Definition, 2018):** troponin rise and/or fall with a value above the 99th percentile, plus evidence of ischaemia (symptoms, new ECG changes or Q waves, new wall-motion defect, clot on angiography).
- **ACS** = unstable angina, NSTEMI, STEMI. **PCI** (angioplasty, PTCA): balloon and stent from inside. **CABG:** a graft carries blood around the block. NYHA and CCS grades: Scales (p. 64).

Type	Pain	ECG and troponin
Stable angina	On exertion; settles in minutes with rest or GTN	Normal at rest; troponin normal
Unstable angina	New, worse, or at rest	May show ST depression; troponin normal
NSTEMI	Prolonged, at rest	ST depression or T inversion; troponin raised
STEMI	Severe, over 20 min, no relief	ST elevation, later Q waves; troponin raised

Risk factors (3m S17, W17)

Non-modifiable: age (men over 45, women over 55), male sex, family history of early heart disease (male relative under 55, female under 65), South Asian origin.

Modifiable: smoking and all tobacco, hypertension, diabetes, abnormal lipids (high LDL, low HDL), obesity (waist), inactivity, salty and fried diet, alcohol, stress and depression.

For "3 + 3" write: age, male sex, family history; smoking, hypertension, diabetes.

Plaque to impairments (clinical reasoning)

Plaque narrows the artery → ruptures, clot forms → O_2 supply falls below demand = **ischaemia** (brief: angina) → over **20–30 min**: MI, from the inner wall outwards → ↓ pumping (EF, cardiac output), **arrhythmias**, pain and fear → bed rest (and after CABG the sternotomy and drains) add deconditioning, shallow breathing and DVT risk.

Impairments: ↓ exercise tolerance, fatigue, breathlessness; atelectasis and DVT risk; ↓ ADL, work and confidence.

Scar is firm in **6–8 weeks**: early activity is safe, heavy straining is not. **Rate–pressure product** = HR × systolic BP = the heart's O_2 demand; angina starts at the same value in the same patient.

Extra after CABG: wound pain → shallow breaths and weak cough; basal atelectasis; guarding → stiff shoulders and thoracic spine, rounded posture; painful or swollen graft site.

Clinical features

- **Angina:** central tightness or heaviness spreading to the left arm, jaw, neck or back; brought on by exertion, cold, meals, emotion; eased by rest or GTN in minutes.
- **MI:** pain over 20 min not relieved by rest or GTN; sweating, nausea, breathlessness, sense of doom; fast or slow pulse, low BP, basal crackles if the LV fails.
- **Inferior MI → bradycardia or heart block**: the right coronary supplies the SA and AV nodes.
- **Silent MI** in diabetes, the elderly and women: breathlessness, tiredness, epigastric pain.
- **After CABG:** wound and graft-site pain, drains, ↓ basal breath sounds, weak cough, swollen leg or forearm, stiff shoulders (S19: "cannot raise his hands above his head").

Investigations: what changes physio

- **STEMI:** new ST elevation in 2 neighbouring leads, ≥ 1 mm (V2–V3: 2 mm men ≥ 40, 2.5 mm men under 40, 1.5 mm women). Sequence: tall T → ST elevation → Q waves → T inversion.
- **NSTEMI, unstable angina:** ST depression or T inversion.
- **Old (evolved) MI (3m S24):** pathological Q waves (≥ 0.03 s and 1 mm deep; older rule over 0.04 s or over ¼ of R), T inversion, ST back to baseline.

Leads changed	Wall	Artery
II, III, aVF	Inferior	Right coronary
V1–V2	Septal	LAD
V3–V4	Anterior	LAD
I, aVL, V5–V6	Lateral	Circumflex
ST depression V1–V3, tall R	Posterior	RCA or circumflex

- **Troponin:** most specific; rises in hours, peaks ~1 day, stays up ~2 weeks. **Mobilise once stable or falling.** CK-MB is normal in 2–3 days, so a new rise means re-infarction.
- **Echo EF below 40% = high risk**; BNP high = heart failure (lower intensity, slower progress). **Angiography:** triple vessel or left main → CABG. **Exercise test or 6MWT:** safe HR range for Phase II.
- **Markers in rehab (7m W25):** confirm MI; infarct size and risk group; when to start; a fresh rise during rehab means stop and reset; some rise after CABG is expected; BNP guides intensity.

Medical and surgical management

- **STEMI:** primary PCI within **90 min** (120 if transferred), otherwise thrombolysis within 30 min; aspirin plus a second antiplatelet and an anticoagulant; **oxygen only if SpO_2 is below 90%**; nitrates; opioid only for severe pain. MONA for every MI is the old answer.

- **Drugs:** beta-blocker → blunted HR, so use **RPE and the talk test**; nitrates → dizzy on standing, keep GTN at hand; antiplatelets → bleeding; ACE inhibitors, diuretics → low BP.
- **After PCI:** femoral → leg straight a few hours, check the groin before walking; radial → walk early, no heavy wrist loading for ~1 week. Driving 1 week after PCI for ACS if EF ≥ 40%, otherwise 4 weeks (DVLA).
- **CABG indications (3m W25):** left main disease; triple vessel disease (especially with diabetes or low EF); 2 vessels including the proximal LAD; angina uncontrolled when PCI is unsuitable; failed PCI; with valve surgery or mechanical complications of MI.
- **Grafts (3m W24): LIMA to LAD** (stays open longest), RIMA, radial artery, great saphenous vein. Median sternotomy, wired; on- or off-pump.
- **Course:** ICU and extubation on day 0, sitting out day 1, ward walking days 2–4, home ~day 5–7.

Complications and what physio does

- **Arrhythmia** (first 48 h; AF after CABG, days 2–4): pulse before every session; stop and report a new one.
- **Heart failure, shock:** sit upright, keep activity low, report. **Re-infarction:** stop, rest, GTN, report.
- **Rupture, embolism, stroke:** emergency. **Pericarditis** (sharp, worse lying flat) or Dressler's: report.
- **DVT, PE:** ankle pumps, early walking, stockings. **Lung collapse after CABG:** deep breaths, supported huff, sit and walk early.
- **Sternal infection or instability** (clicking, discharge; more likely with diabetes, obesity, both IMAs): report at once.
- **Graft leg swelling:** elevate, walk. **Stiff shoulders:** early pain-free movement.

PT assessment

- **Subjective:** chest pain (site, spread, triggers, relief), breathlessness, palpitations; event and treatment (date, PCI or CABG, post-op day, grafts, ICU course); risk factors; drugs; **social history for ICF**: job, driving, earning role, stairs, toilet type, family, tobacco, diet; goals and fears.
- **Objective:** HR and rhythm, BP lying and standing, RR, SpO_2 at rest and with activity; lines and drains; chest expansion, auscultation at the bases, cough; wounds; shoulder range, posture, calf; bed mobility, walking distance with vitals, stairs.
- **Outcome measures:** Borg RPE, 6MWT, ISWT, NYHA and CCS, DASI, HADS, MacNew or HeartQoL.

Risk stratification before Phase II (AACVPR; 7m S21)

Low	EF ≥ 50%; ≥ 7 METs; no symptoms or complex arrhythmia; uncomplicated event	Standard programme
Moderate	EF 40–49%; symptoms only at ≥ 7 METs; ST depression under 2 mm; capacity under 5 METs	ECG for the first sessions
High	EF under 40%; symptoms under 5 METs; ST depression ≥ 2 mm; BP falls with effort; complex arrhythmia; survived arrest; heart failure; depression	Continuous ECG; start low, go slow

One high-risk feature makes the patient high risk.

ICF: draw the 6 boxes first (W17 case)

- **Health condition:** triple vessel CAD; CABG day 2 (LIMA, RIMA, radial).
- **Body structure:** wired sternum; both IMAs taken; forearm wound.
- **Body function:** incisional pain; ↓ lung expansion, atelectasis risk; weak cough; ↓ exercise tolerance; AF risk; guarded shoulders; anxiety.
- **Activity:** bed mobility and sit-to-stand without the arms, walking, stairs, upper-body dressing, lifting.
- **Participation:** cannot drive to work (~4 weeks) or run his business; family and social roles.
- **Environment:** facilitators: family, rehab services, medicines; barriers: daily drive, stress (ask about stairs and toilet).
- **Personal:** 50, businessman, motivated; smoking, diet, stress.

Functional diagnosis in one line: age, job, surgery and day + main impairments + the activities and roles they limit + one facilitator and one barrier.

Goals

Short-term (by discharge, ~1 week): pain low enough to breathe deeply and cough with support; clear chest; independent sternal-safe bed mobility; corridor walking and one flight within Phase I limits; pain-free shoulder movement; knows the home programme and warning signs.

Long-term (6–12 weeks): walks 30 min continuously at RPE 12–13; full shoulder range, upright posture; back to driving (~4 weeks) and work; risk factors controlled; exercise for life.

Phases of cardiac rehab (S26)

I	In hospital: stable day to discharge	Prevent bed-rest effects; self-care and walking to ~3 METs; education, GTN, warning signs; refer on
II	From discharge, 8–12 weeks	Supervised, monitored exercise; risk factors; psychological support; return to work, driving, sex
III	Months after II	Less supervision; self-monitoring
IV	Lifelong	Independent exercise and healthy lifestyle

Before CABG: explain ICU and drains; deep breaths with a hold; supported huff and cough with a pillow; sternal-safe bed mobility; ankle pumps; baseline walk.

Phase I

Start when stable: no new chest pain, no new ECG changes or dangerous arrhythmia, troponin stable or falling, no uncontrolled heart failure (usually day 1). After CABG: day 1.

13 / 20 / 30 / 120

RPE **below 13**; HR **≤ rest + 20** and **≤ 120** after MI; HR **≤ rest + 30** after surgery (ACSM).

Dose: bouts of 3–5 min with 1–2 min rest, ~20 min in all; 3–4 times a day on days 1–3, then twice; build to 10–15 min continuous first, then intensity; activities up to ~3 METs.

- **CCU (1–2 METs):** breathing exercises, ankle pumps, active limbs, sitting up, self-feeding, commode. After CABG add deep breaths with a 3-s hold, 5–10 an hour, supported huff, sitting out on day 1, log-roll, both arms moving together.

- **Ward (2–3 METs):** standing, walking in the room then the corridor (3–4 times a day), bathroom, posture, elevate the graft leg.
- **Before discharge (~3 METs):** 10–15 min walk, one flight of stairs, own pulse and RPE, sternal and wound care, home walking plan.

Monitor HR, rhythm, BP, SpO_2, RPE and symptoms before, during and after. Teach warning signs and GTN use. Refer to Phase II.

Progress only if (S24): HR rise within limits; systolic rises a little (a fall over 10 mmHg or diastolic ≥ 110 stops it); no new arrhythmia, angina, dizziness or marked breathlessness; RPE under 13; SpO_2 steady; pain controlled, sternum stable, Hb and chest X-ray checked; recovers within minutes and not unduly tired next morning; consider age, fitness, EF, diabetes.

Current evidence: mobilise within 12–24 h once stable (long bed rest is the old answer). After CABG the incentive spirometer is an adjunct only, never alone (AARC 2011).

Phase II (FITT-VP)

Risk-stratify and test first (exercise test or 6MWT). Warm-up 5–10 min, conditioning 20–60 min, cool-down 5–10 min.

	Aerobic	Resistance
F	3–5 days a week	2–3 non-consecutive days
I	40–80% HRR (start low) or RPE 12–16 (start 11–13); ≥ 10 beats below the ischaemic HR	10–15 reps at RPE 11–13 (about 40–60% 1RM)
T	20–60 min	8–10 exercises, 1–3 sets
T	Walk, treadmill, cycle; arm ergometer when the sternum allows	Bands, light weights; legs first after CABG
V, P	Build to 150 min a week; duration first, then intensity	"2 for 2" rule (AHA 2023)

Karvonen: age 50, resting HR 70 → max 170, HRR 100 → 40–60% = **110–130/min**. On a beta-blocker use RPE 11–13 and the talk test. Breathe out on effort; no Valsalva.

After PCI, young sportsman (7m W24): no sternal limits; protect the puncture site ~1 week; symptom-limited test first; 40–60% rising to 60–80% HRR, intervals later; sport skills at low intensity first; competitive sport only after cardiology review (ESC 2020).

Sternal precautions: for 6–8 weeks no heavy lifting, pushing, pulling or one-sided arm loading; keep arms close to the body and pain-free ("keep your move in the tube"). No heavy lifting for ~3 months.

Phases III–IV: ≥ 150 min a week of moderate activity + resistance on 2 days; self-monitoring.

Physiotherapy for each impairment (S19)

Incisional pain	Pillow splinting for cough and movement; treat after analgesia; TENS beside the wound if allowed (not with pacing)
↓ Lung expansion	Deep breaths with 3-s hold and thoracic expansion, 5–10 every hour; sit upright; walk early
Weak cough	ACBT; supported huff and cough
↓ Exercise tolerance	Graded walking within Phase I limits, then Phase II training
Stiff shoulders	Both-arm flexion and abduction, thoracic rotation and extension, scapular setting; 10 reps, 3–4 times a day; posture
DVT risk, graft leg	Ankle pumps 10 an hour; walking; elevation; stockings if prescribed
Anxiety	Explain each step; relaxation; graded success; involve family
Risk factors	Lifestyle counselling; aerobic training for weight and sugar

Do not start exercise (ACSM)

Unstable angina; resting BP over 180/110; symptomatic fall over 20 mmHg on standing; significant aortic stenosis; uncontrolled arrhythmia or sinus rate over 120; uncompensated heart failure; third-degree block without a pacemaker; active peri- or myocarditis; recent embolism or DVT; fever; uncontrolled diabetes.

Stop the session

Diastolic ≥ 110; systolic falls over 10 mmHg as work rises; significant arrhythmia; second- or third-degree block; angina, marked breathlessness, ischaemic ECG; dizziness, confusion, pallor, cold sweat; sternal movement or wound bleeding.

Angina in a session: stop, sit, GTN; not better 5 min after the first dose → emergency help. No exercise within 1–2 h of a heavy meal or in extreme heat or cold.

Lifestyle and education (S16, W20, S23, S25)

- Stop all tobacco (bidi, gutkha, second-hand smoke). Salt under 5 g a day; less fried food, vanaspati, ghee, sugar.
- ≥ 150 min of activity a week. BMI under 23; waist under 90 cm (men), 80 cm (women). BP under 130/80. Control sugar and lipids; limit alcohol; manage stress.
- **Obesity grades (S21):** WHO normal 18.5–24.9, overweight 25–29.9, obese I 30–34.9, II 35–39.9, III ≥ 40. Asian Indian: normal 18.5–22.9, overweight 23–24.9, obese ≥ 25.
- **Return:** driving ~4 weeks after CABG; sex when 3–5 METs are symptom-free (~6–8 weeks after CABG); no heavy lifting ~3 months; work ~6 weeks (desk work sooner).
- **Do's and don'ts (S25):** walk daily at RPE 11–13, take medicines, carry GTN, join Phase II. Don't hold the breath, exercise to exhaustion, stop antiplatelets, smoke, lift heavy or drive until cleared.

Prognosis: rehab cuts MI by ~28% and admissions by ~35% (Cochrane 2026). **Phase I discharge:** stable vitals, walking at ~3 METs, one flight of stairs, knows the home programme.

Exam plan: the 15-mark case

1. Restate the case in 2 lines.
2. **ICF (5):** draw 6 boxes, fill them from the case, one-line functional diagnosis.
3. Problem list → short- and long-term goals tied to his job and home.
4. **Management (7–10):** Phase I with 13/20/30/120 and progression; Phase II FITT-VP with a Karvonen example; III–IV in 2 lines.
5. Stop criteria; sternal precautions if CABG.
6. **Add-on (3):** lifestyle with reasons, obesity grades or outcome measures.

Draw: ICF, phases timeline, plaque-to-impairment chain. **Underline:** early mobilisation, METs, RPE, FITT-VP, Karvonen, risk stratification, sternal precautions.

Burns: acute care, positioning, splinting and inhalation injury

6(g). In 19 of 20 papers; 12 long answers. Full notes p. 31

Asked: a 15-mark case in most papers (ICF 5 + short-term 5 + long-term 5, or 5 + 7 + 3 with an add-on). **7 marks:** anti-deformity positioning, splinting, care after grafting, inhalation burns, wound-healing factors and modalities, lower-limb burns. **3 marks:** rule of nines, depth types, three splints, stages of wound healing, inhalation burn complications.

Definition and types

Burn: injury to skin and deeper tissue from heat, electricity, chemicals, radiation or friction. Described by **cause, depth and %TBSA**. **Inhalation injury:** airway and lung damage from hot gases, smoke or toxic products.

Causes: flame (stove, dupatta catching fire, firecrackers), scald (children), contact, electrical, chemical (alkali goes deeper).

Depth	Look	Heals
Superficial (1st°)	Red, dry, painful, no blisters, blanches	3–7 days, no scar
Superficial partial (2nd°)	Blisters, pink, moist, very painful, quick refill	2–3 weeks, little scar
Deep partial (2nd°)	Blotchy red-white, drier, less pain, slow blanching	Over 3 weeks; hypertrophic scar; often grafted
Full thickness (3rd°)	White, brown or black leathery eschar; **painless** (nerves destroyed)	Only from edges: **graft**
4th°	Into muscle, tendon, bone	Flaps, amputation

Rule of nines (3m W18, S21)

Head and neck **9**; each arm **9**; front of trunk **18**; back of trunk **18**; each leg **18**; perineum **1**. Patient's palm with fingers ≈ **1%** (small or patchy burns). Children: **Lund and Browder** chart (bigger head, smaller legs). Count only partial and full thickness. Show the working: both arms 18 + front of head 4.5 + upper front trunk 9 = 31.5%.

Burn unit if: partial thickness over 10%, any full thickness, face, hands, feet, perineum or major joints, electrical, chemical or inhalation injury, other serious injury.

Burn to impairments (clinical reasoning)

- **Jackson's zones:** coagulation (dead), **stasis** (can be saved or lost: resuscitation, infection), hyperaemia (recovers).
- **Capillary leak** → oedema; over ~20% TBSA → **burn shock** in 24–48 h.
- **Hypermetabolism** → muscle breakdown for months; ↓ immunity → infection, sepsis.
- **Inhalation:** upper airway swelling, lower airway and alveolar damage, CO and cyanide poisoning.
- **Scar contracts for months** (myofibroblasts) → flexion and adduction. **Position of comfort = position of contracture.**
- **Impairments:** pain, ↓ ROM, contracture, hypertrophic scar; weakness, ↓ endurance, ↓ ADL; ↓ ventilation, secretions.

Wound healing (3m S19; 7m S25)

Haemostasis (minutes–hours) → **inflammation** (days 1–5) → **proliferation** (day 4–week 3: granulation, collagen, epithelium, contraction) → **remodelling** (week 3 to 1–2 years; scar never as strong as skin).

Slow healing: local: infection, poor blood supply, oedema, dead tissue, pressure and shear, too dry or wet, radiotherapy. General: age, poor nutrition (protein, vitamin C, zinc), diabetes, smoking, anaemia, obesity, steroids, organ failure. Modalities: see Diabetes and wounds (p. 26).

Clinical features

- Pain worst in superficial partial; swelling peaks **24–48 h** (hands, face, neck). Circumferential deep chest burns restrict breathing; on a limb they cut off circulation.
- **Suspect inhalation injury:** closed space; face and neck burns; singed nasal hair; soot in mouth or nose; black sputum; hoarse voice, stridor, wheeze; confusion (CO). Swelling worsens over day 1, so early intubation.
- **Electrical:** small entry and exit wounds, deep muscle damage; arrhythmia, compartment syndrome, dark urine (myoglobin, kidney injury), nerve damage, fractures.

Investigations: burn chart; **ABG with carboxyhaemoglobin** (SpO_2 reads falsely normal in CO poisoning); chest X-ray (later collapse, ARDS); bronchoscopy; ECG, CK, urine myoglobin after electrical injury; Hb, albumin, swabs.

Medical and surgical management

- **First aid:** stop the burning; cool running water **20 min** (no ice, toothpaste, butter); cover with cling film; keep warm.
- Airway first; high-flow oxygen for CO; fluids if ~20% TBSA or more; analgesia; tetanus; high-protein feeds; silver dressings.
- **Parkland:** 4 mL × kg × %TBSA in 24 h, half in the first 8 h. Current: start 2–4 mL and titrate to urine output **0.5 mL/kg/h**.
- **Escharotomy** (tight eschar on chest or limb), fasciotomy; **early excision and split-thickness grafting** (meshed; donor site often thigh); full-thickness grafts and flaps for face, hands, joints; later contracture release, Z-plasty.

Complications and physio role

- **Contracture** (neck, axilla, elbow, hand, knee, ankle): positioning, splints, ROM from day 1.
- **Hypertrophic scar** (within wound) and **keloid** (beyond it): pressure, silicone, massage, stretch.
- Chest: pneumonia, collapse, ARDS. Infection and sepsis (hand hygiene, gowns).
- **Heterotopic ossification** (elbow): warm, painful, stiffening joint: gentle active only, no forceful passive.
- Compartment syndrome: elevate, report urgently. **Exposed extensor hood**: splint the finger straight (boutonnière risk).
- Weakness, pressure sores, DVT; neuropathy (electrical); anxiety, body image; **Marjolin's ulcer** (cancer in an old scar).

PT assessment

Subjective: how and where (closed space, electrical, first aid), pain, itch, cough and sputum, grafts and donor sites, job and hand dominance, earning role, house and toilet, goals and fears about appearance.

Objective: vitals (CO caution), breathing, chest expansion (eschar), auscultation, cough; burn chart (sites, depth, joints crossed); oedema; active then passive ROM with goniometer; strength, sensation; bed mobility, walking, self-care, hand function; scar in rehab.

Outcome measures: Vancouver Scar Scale, POSAS, goniometry, BSHS-B, VAS, grip, 6MWT, Barthel or FIM.

ICF (S21: housemaid, stove burns, face, upper chest, both arms, day 3)

- **Body function:** pain; hand and face oedema; ↓ neck, shoulder, elbow, wrist, hand range, contracture risk; ↓ chest expansion, inhalation risk; microstomia risk; ↓ strength; anxiety about appearance.
- **Activity:** feeding, washing, dressing, toileting (both hands), reaching, lifting.
- **Participation:** housemaid work (sweeping, washing, cooking), family and social roles.
- **Environment:** burn-unit care, family (facilitators); lost wages, floor-level stove, Indian toilet (barriers). **Personal:** 35, manual worker, distress about the face.

Goals: short-term (2–3 weeks): clear chest, ↓ oedema, full range kept with no contracture, grafts taken, independent feeding and toileting. Long-term (3–6 months): full range, flat pliable scar, full self-care, back to work, kitchen safety.

PT management by phase

- **Emergent (first 72 h):** head up ≥ 30°; deep breathing, thoracic expansion, ACBT, assisted cough, suction; **elevate burnt limbs** with active finger and ankle pumps; anti-deformity positions and splints from **day 1**; active ROM of every burnt joint **hourly** while awake (after analgesia); sit out and walk once stable.
- **Acute (until wounds close):** continue; exercise at dressing changes. **After grafting: immobilise the graft ~3–5 days** (surgeon decides), move all other joints, then gentle active ROM; no shear. Donor site: move early. Legs: **compression bandage before standing**.
- **Rehabilitation (scar matures 1–2 years):** slow sustained stretch until the scar blanches, 20–30 s or longer, several times a day; **pressure garments ~23 h a day for 12–18 months** (20–25 mmHg), silicone gel, massage, moisturiser; night splints, serial casts; strength, aerobic, hand therapy; work tasks; sun protection; counselling.

Anti-deformity positioning (7m S17, W18)

Area	Contracts into	Position or splint
Front of neck	Flexion	Extension: no pillow, roll under shoulders; neck conformer
Axilla	Adduction, IR	Abduction ~90° + 15° flexion; aeroplane splint
Front of elbow	Flexion, pronation	Extension, forearm supinated
Back of hand	Claw	**Safe position:** wrist 20–30° ext, MCP 70–90° flex, IP straight, thumb abducted
Palm	Finger flexion	Fingers and thumb extended
Hip, groin	Flexion, adduction	Extension, 10–20° abduction; prone when allowed
Back of knee	Flexion	Extension; no pillow under knee
Ankle	Plantarflexion	90°; ankle–foot splint
Face, mouth	Microstomia	Mouth stretches, mouth splint

Splints (3m S19; role 7m S18): resting hand (safe position), neck conformer, axillary, elbow and knee extension, ankle–foot, mouth; dynamic splints and serial casts later. They hold the position the patient cannot, protect tendons and grafts, keep range gained; worn at rest and night; check fit daily.

Special cases

- **Inhalation burns (7m W22):** sit upright; humidified oxygen (high-flow for CO); deep breathing, ACBT, huff, assisted cough, suction; early mobilisation; if ventilated: positioning, clearance; report stridor or falling SpO_2 early.
- **Lower limb (7m W25):** superficial: elevate, hourly ankle pumps and active movement, walk early in compression. Deep: hip extension with slight abduction, knee straight, ankle 90°, splints, heels free; after grafting stand and walk in compression bandages; later stretches (hip flexors, hamstrings, calf) and strengthening. Check toes in circumferential burns.
- **Pregnancy (S24):** two patients; left side-lying or wedge (no flat supine after 20 weeks); DVT prevention; breathing exercises; no heat or electrotherapy over the trunk; report bleeding, pain, contractions, reduced fetal movement.

Precautions and stop

No movement of fresh grafts until allowed; no shear. No aggressive stretch over exposed tendons. No forceful passive range with heterotopic ossification. Check cardiac stability after electrical burns. Infection control. **Stop and report:** stridor, harder breathing, falling SpO_2, falling BP or fast irregular pulse, a bleeding or lifting graft, a tight numb limb.

Education and prognosis

Stretch daily for 1–2 years; wear splints and garments; moisturise, massage, sunscreen; don't scratch. **Prevention:** raised cooking platform; never refuel a lit stove; tuck the dupatta; stop, drop and roll. **Revised Baux** = age + %TBSA (+ 17 if inhalation injury).

Exam plan

1. Restate: age, job, cause, sites, **%TBSA worked out**, depth, day, inhalation or electrical.
2. **ICF (5–6)** with functional diagnosis and reasons.
3. **Short-term (5):** chest care, oedema, positioning table, splints, hourly ROM, graft care, early walking.
4. **Long-term (5):** stretching, scar care, strength, hand therapy, ADL, work, psychology, prevention.

Draw: rule of nines, skin depth, ICF. **Underline:** anti-deformity, graft immobilisation, pressure garment.

Mastectomy, lymphoedema and cancer surgery

6(f). In 16 of 20 papers; 6 long answers. Full notes p. 48

Asked: a 15-mark case most years (ICF 5 + short-term 5 + long-term 5, or 5 + 7 + 3 with prosthesis). **7 marks:** physiotherapy after MRM, its rationale, complications, lymphoedema in a receptionist. **3 marks:** precautions, three goals, three lymphoedema bandages, prosthesis.

Definitions and types of surgery

Mastectomy: removal of the breast, usually for cancer, often with axillary node removal. **Lymphoedema:** chronic swelling from protein-rich fluid the lymphatics cannot drain (arm, hand, breast, chest wall).

Radical (Halsted)	Breast + pectoralis major and minor + all axillary nodes; rare now; weak shoulder, high lymphoedema risk
Modified radical (MRM)	Breast + axillary nodes, **pectoralis major kept**; commonest in papers
Simple	Breast only
Breast-conserving	Lump + margin, then radiotherapy (fibrosis, stiffness)
Sentinel node vs clearance	Clearance → much higher lymphoedema risk
Reconstruction	Implant, latissimus dorsi or abdominal flap: protect the donor site

Lymphoedema stages (ISL)

0 latent: no visible swelling, may feel heavy. **I**: pitting swelling, goes down with elevation. **II**: elevation rarely helps; pitting, then fibrosis. **III**: elephantiasis: no pitting, thick warty skin, folds. **Diagnosis:** ≥ 2 cm larger circumference or ~10% larger volume than the other arm.

Risk factors: axillary clearance, axillary radiotherapy, infection or seroma, obesity, cellulitis, sudden heavy overuse. Breast cancer: female, age, family history and BRCA, early menarche, late menopause, no children or first after 30, obesity, alcohol.

Surgery to impairments (clinical reasoning)

- Pain and fear → arm held to the chest → **stiff shoulder** (flexion, abduction, ER); scar and radiotherapy fibrosis tether the chest wall and axilla.
- Nodes removed → lymph load exceeds capacity → **lymphoedema** → fibrosis, infection risk.
- **Nerves:** long thoracic (serratus anterior → **winging**), thoracodorsal (latissimus dorsi), intercostobrachial (numb inner arm).
- **Axillary web syndrome:** tight cords from the axilla down the arm in weeks 1–8; limit abduction.
- Raised, protracted shoulder; weight imbalance → neck and back pain. Chemotherapy and radiotherapy: fatigue, low counts, neuropathy, skin and lung changes.

Features: pain and tightness, drains in week 1, ↓ elevation, abduction and ER, arm or hand swelling, numb inner arm, cords, fatigue, low mood, body-image distress. **Early lymphoedema clues (W21):** heavy arm, tight sleeves or rings: measure both arms. Red, hot arm with fever = **cellulitis**: medical treatment first.

Investigations and treatment

Mammography, ultrasound, biopsy; staging scans (bone metastases: avoid heavy loading); blood counts in chemotherapy (low counts: reduce or hold exercise); arm circumference and volume; lymphoscintigraphy, bioimpedance.

Surgery ± reconstruction; chemotherapy (before or after); radiotherapy; hormone therapy (aromatase inhibitors: joint pain, bone loss); trastuzumab (heart).

Complications (7m S20)

Immediate	Pain; chest complications; DVT: breathing, supported cough, early walking, ankle pumps
Weeks 1–2	Seroma, haematoma, wound infection, flap necrosis: shoulder to ~90° with drains; report swelling or redness
Weeks	Stiff or frozen shoulder; axillary cording: graded ROM, sustained stretch
Months–years	Lymphoedema, cellulitis, radiation fibrosis, brachial plexopathy, winging, posture pain, fatigue, low mood

PT assessment

Subjective: surgery, side, nodes removed, drains, reconstruction; chemo and radiotherapy; pain, heaviness, numbness; **hand dominance**; job and household load (cooking, washing, lifting children); goals.

Objective: wound and drains; posture, shoulder height, winging; shoulder ROM both sides (goniometry); scapular strength and grip; **arm circumference every 4 cm from the wrist**, volume, pitting, skin texture; inner-arm sensation; palpate cords.

Outcome measures: DASH, shoulder ROM, arm volume, FACT-B, fatigue scale, 6MWT.

ICF (W20: right-handed teacher, right MRM 3 weeks, on chemotherapy, arm swelling)

- **Body function:** right shoulder pain; ↓ flexion, abduction, ER; early lymphoedema (stage I); scar and axillary tightness, cords; ↓ strength, chemotherapy fatigue; anxiety, body image.
- **Activity:** reaching overhead, writing on the board, combing hair, dressing, lifting, cooking.
- **Participation:** teaching, care of two children, social life.
- **Environment:** family, hospital, a valued job (facilitators); board work, heavy chores, school bags (barriers). **Personal:** 40, motivated, worried about cancer and appearance.

Goals: short-term (4 weeks): less pain, flexion and abduction ~150°, smaller arm, knows skin care, walks 20 min. Long-term (3–6 months): full pain-free range, swelling controlled with a sleeve, exercise 2–3 times a week, prosthesis, back to teaching.

PT management: the timeline

- **Before surgery:** baseline shoulder range and both arm measurements; teach first-week exercises, breathing and posture.
- **Days 1–7 (drains in):** deep breathing, supported cough, early walking; **hand pumping**, wrist and elbow movement, shoulder shrugs, 10 reps several times a day; shoulder flexion and abduction only to **~90°**; arm on a pillow, hand above elbow.
- **Why 90°?** Full-range exercise straight after surgery increases drainage and seroma; waiting ~1 week reduces this and the shoulder still recovers in about a month (Cochrane).
- **Drains out:** full active and assisted range: wand, finger wall-climb, pulley, hands behind head and back; stretches held **20–30 s, 3–5 reps, 2–3 times a day**; scapular setting and posture; serratus work for winging; scar massage once healed; cord stretches; keep stretching through radiotherapy.
- **From ~6 weeks:** progressive resistance (start light, supervised), aerobic walking or cycling, work tasks (board writing, cooking, lifting children).
- **Cancer fatigue (ACSM 2019):** aerobic 3 a week, 30 min, moderate (RPE ~12); resistance 2 a week, ~60% 1RM, 2 × 12–15; at least 12 weeks; cut the dose on bad days rather than stopping.

Complete decongestive therapy (7m S21)

- **1. Intensive, daily 2–4 weeks:** manual lymph drainage (light, slow, skin-stretching; clear the neck, other axilla and trunk first, then upper arm → forearm → hand); **multilayer short-stretch bandaging** day and night; exercises in the bandage; skin care.
- **2. Maintenance, lifelong:** compression sleeve (± glove) by day; self lymph drainage; regular exercise including supervised resistance; skin care; weight control.
- Also: elevation at rest, deep breathing, pneumatic compression only as an add-on; workstation advice (arm supported, breaks, sleeve at work).

Bandages (3m S23): short-stretch (low pressure at rest, high when muscles work, safe at night); **multilayer** (stockinette, finger gauze, padding, short-stretch layers, firmest at the hand); **adjustable Velcro wraps** for self-care. Long-stretch elastic bandages are not used (tourniquet at rest).

Current evidence: slowly progressive, supervised resistance training is safe with or at risk of lymphoedema ("start low, go slow"). Write: "avoid sudden heavy loading; build strength gradually; protect the skin; injections in the other arm where possible."

Breast prosthesis (3m S17)

Temporary soft fibre-filled form in a soft bra soon after surgery. **Permanent silicone** form once healed and swelling settles (~6–8 weeks, or after radiotherapy), matching the other breast's weight, in a pocketed bra. **Why:** symmetry and balance, stops the shoulder rising and spine tilting, clothes fit, confidence, protects the chest wall.

Precautions (3m S21)

Shoulder to ~90° while drains are in; watch seroma and infection; protect the arm from cuts, burns, bites, tight sleeves and bangles; **no heat or vigorous massage** on a swollen arm; build lifting gradually; protect flap donor sites; in chemotherapy hold exercise with fever, very low counts or vomiting; no heavy loading of bones with metastases; no electrotherapy over tumour. **Refer urgently:** red hot painful arm with fever, sudden calf or arm swelling, new bone pain or breathlessness.

Lifelong arm care (S19: why and how)

Why: removed nodes never regrow, so any extra load (infection, burn, injury, heavy unaccustomed work) can start lymphoedema, even years later.

How: clean, moisturised skin; gloves for cooking and washing vessels, oven gloves, long-handled spoons; antiseptic for every cut or bite; no tight bangles or sleeves; build heavy tasks slowly, carry with both hands; stay active, healthy weight; check the arm and report heaviness early; sleeve for travel or heavy work.

Exam plan

1. Restate: age, side and **hand dominance**, surgery and day, chemo or radiotherapy, job and home load.
2. **ICF (5)**: reasons are guarding, nodes removed, chemotherapy.
3. **Short-term (5):** breathing, hand pumping, shoulder to 90° with drains, posture; full ROM once drains are out.
4. **Long-term (5–7):** stretching, scar, strength, aerobic, lymphoedema care, prosthesis, return to work.

Draw: the timeline; an arm with measuring points every 4 cm. **Underline:** 90° until drains out, axillary web syndrome, complete decongestive therapy, lifelong arm care.

Peripheral arterial disease, Buerger's disease and other arterial disorders

6(a). In 18 of 20 papers, mostly ABI and short answers. Full notes p. 61

Asked: in 18 of 20 papers, mostly short answers. **3 marks** (almost every paper): ABI and its grades, three special tests for arterial insufficiency, claudication, evaluation parameters. **7 marks:** Buerger's disease, claudication, PAD assessment and physiotherapy. **Cases:** 25-year-old smoker with claudication (Buerger's); diabetic clerk with claudication (W22).

Definitions and types

- **PAD:** narrowing or block of the limb arteries (usually legs) from **atherosclerosis**.
- **Intermittent claudication:** cramping muscle pain (usually calf) after a fairly **fixed walking distance**, gone within minutes of rest.
- **Buerger's disease** (thromboangiitis obliterans): inflammatory, non-atherosclerotic disease of small and medium arteries and veins of hands and feet in **young smokers**.
- Presentations: asymptomatic; claudication; **chronic limb-threatening ischaemia** (rest pain, ulcer, gangrene); **acute limb ischaemia** (embolus or thrombosis: emergency).
- Others: **Raynaud's** (white → blue → red with cold or stress); aneurysm (aorta, popliteal).

ABI and grades (3m, asked 9 times)

Above 1.40	Non-compressible, calcified (diabetes, kidney disease): use toe–brachial index (≤ 0.70 = PAD)
1.00–1.40	Normal
0.91–0.99	Borderline
≤ 0.90	**PAD**: 0.70–0.90 mild, 0.40–0.69 moderate, **below 0.40 severe** (rest pain, ulcers)

Method: supine, rest 5–10 min; Doppler systolic pressure in both arms (take the higher); at dorsalis pedis and posterior tibial of each ankle (higher of the two); **ABI = ankle ÷ higher arm**, each leg.

Fontaine: I no symptoms; IIa claudication over ~200 m; IIb under ~200 m; III rest pain; IV ulcer or gangrene (III–IV = CLTI).

Special tests for arterial insufficiency (3m W16, S18, W20)

- **Pulses:** femoral, popliteal, dorsalis pedis, posterior tibial; grade 0 absent, 1+ weak, 2+ normal, 3+ full, 4+ bounding.
- **Buerger's test:** raise the leg; angle where the foot blanches (below ~20° = severe); then hang it down: slow dusky red (**dependent rubor**).
- **Capillary refill:** over ~2 s is delayed. **Venous filling time:** over ~20 s after elevation.
- **Treadmill test:** pain-free and maximum walking distance. **Allen's test** for the hand (Buerger's).
- Duplex Doppler (site of block), CT or magnetic resonance angiography before surgery; sugar, HbA1c, lipids, kidney function.

Causes and Buerger's criteria

PAD: smoking (strongest), diabetes (below-knee disease, neuropathy), age over 50–65, hypertension, lipids, kidney disease.

Buerger's: smoker, onset before 45–50, below-knee artery blocks, arm involvement or **migrating superficial phlebitis**, no other atherosclerotic risk factor (Shionoya; Olin). The 25-year-old smoker with claudication and cold sweaty feet (S16, W17) is Buerger's: say so in line one.

Why walking hurts and exercise helps

Plaque narrows the artery → flow enough at rest → walking raises calf O_2 demand, supply cannot rise → ischaemia, metabolites → **pain**; rest lowers demand → pain goes → he walks less → weaker, less fit → distance shrinks → severe narrowing: rest pain, ulcers, gangrene.

Exercise helps: better muscle O_2 use, **collaterals** open, better artery lining, more efficient walking. The block stays, but he walks much further.

Rest pain is worse at night (no help from gravity) and eased by hanging the leg down. **Site:** aorto-iliac → buttock, thigh; femoro-popliteal (commonest) → calf; tibial → foot.

Clinical features

- Calf, thigh or buttock pain on walking, relieved by **standing still**; rest pain in toes at night; cold numb feet.
- Cool, pale, thin, shiny, hairless skin; thick nails; wasted calf; weak or absent pulses; pallor on elevation, rubor on dependency; slow refill; ulcers on toes, heel, pressure points; gangrene.
- **Buerger's:** young male smoker, feet and hands, instep claudication, Raynaud's-like changes, migrating phlebitis, painful fingertip and toe ulcers.
- **Acute ischaemia: 6 P's**: pain, pallor, pulselessness, paraesthesia, paralysis, perishing cold: call the doctor at once.

	Vascular	Neurogenic (canal stenosis)
Onset	Fixed distance	Walking or standing; varies
Relief	Standing still, minutes	Sitting or bending forward, slower
Uphill, cycling	Worse	Often easier
Pulses	Weak or absent	Normal

Medical and surgical management

Stop all smoking (in Buerger's the only proven treatment). Antiplatelet, statin, BP and sugar control (heart attack and stroke risk). Cilostazol for claudication. **Supervised exercise is first-line.** Revascularisation (angioplasty ± stent, femoro-popliteal bypass) if still limiting or limb-threatening. Wound care; amputation if unsalvageable. Acute ischaemia: anticoagulation, embolectomy.

Complications and prevention (W22)

- **Arterial ulcer** → daily foot inspection, footwear, never barefoot, nail care.

- **Infection, gangrene, amputation** → treat wounds early, sugar control, stop smoking, refer early.
- **Heart attack and stroke** (same atherosclerosis) → risk-factor control, exercise.
- **Deconditioning, low mood** → graded walking. **Burns and pressure sores** (numb ischaemic skin) → no hot-water bottles, padded heels. **Neuropathic ulcer** in diabetes → monofilament, offloading.

PT assessment (3m S21: evaluation parameters)

History: claudication site, onset distance or time, stopping time, recovery; rest pain; smoking (pack-years, bidi); diabetes, BP, heart; job, walking demands, stairs. **Inspection:** colour, hair, nails, wasting, ulcers (site, size, base, edge). **Palpation:** temperature both sides, graded pulses. **Tests:** ABI, Buerger's, capillary refill, venous filling. **Sensation:** 10 g monofilament if diabetic. Ankle ROM, calf strength. **Walking:** treadmill or 6MWT (pain-free and maximum distance), claudication scale. **Questionnaires:** Walking Impairment Questionnaire, VascuQoL.

ICF (W22: diabetic head clerk, chain smoker, pain at 5 min, stops at 10)

- **Body function and structure:** calf claudication at 5 min; ↓ walking endurance; calf wasting and weakness; dry thin skin; possible ↓ sensation; ↓ fitness. Reason: narrowed artery cannot meet demand; disuse; starved skin.
- **Activity:** walking over ~10 min, stairs, shopping, walking to the bus. **Participation:** commuting, outings, errands, leisure.
- **Environment:** easy cigarettes, fatty diet (barriers); desk job, family, hospital programme (facilitators). **Personal:** 50, smoker, diabetic.

Goals: short-term (4 weeks): follows the walk–rest programme 3 times a week, daily foot checks, sets a quit date. Long-term (12 weeks+): clearly longer pain-free and maximum walking time, tobacco-free, sugar and BP controlled, commutes without stopping.

Supervised exercise therapy (the core)

30 / 3 / 12

Walk **30–45 min** a session, **≥ 3 times a week**, for **≥ 12 weeks** (ACC/AHA 2024, Class 1A).

- Warm-up 5–10 min with calf stretches.
- Treadmill or level ground: **walk until pain is moderate (grade 2–3 of 4), rest until it has gone, repeat**.
- Progress speed or slope when he walks 8–10 min before moderate pain. Cool-down 5–10 min.
- Pain scale: 1 starts, 2 moderate, 3 intense, 4 unbearable.
- Cannot walk (ulcer, arthritis): arm cranking or cycling. No supervision: home walking with a log book, 30 min, 3–5 times a week.
- Add heel raises and sit-to-stand (2–3 × 10–15, 2–3 times a week), calf stretches 30 s × 3, active ankle and toe ROM.

Buerger's disease (7m S17, S22) and rest pain

- **Absolutely no tobacco** (bidi, chewing, passive). Walking within tolerance; foot and calf exercises.
- **Buerger–Allen (classic):** leg up 45–90° until it blanches (1–2 min) → hang down until red (2–5 min) with ankle movements → flat 5 min; repeat several cycles a few times a day. Weak evidence; supervised walking is the main treatment.
- Keep hands and feet warm with socks and gloves, **never direct heat**; meticulous foot and hand care.
- **Rest pain:** bed head slightly raised or leg slightly down; never elevate an ischaemic leg; pad heels; gentle ankle and toe movement. After bypass or angioplasty: early walking, protect the groin, then the walking programme.

Precautions and stop

No heat on an ischaemic limb (it burns). No compression if ABI is low. Don't elevate in rest pain. Check HR and BP (heart disease is common). Diabetics: sugar and feet before and after. No walking on an open ulcer without offloading. **Stop and refer:** sudden 6 P's, chest pain or breathlessness, new rest pain, new ulcer, black toe.

Education

Foot care: look daily (mirror), lukewarm water tested with the elbow, dry between toes, cut nails straight, soft footwear, never barefoot (even in temples), no heat near feet, report changes. **Lifestyle:** stop all tobacco, walk daily with walk–rest, less fat and salt, take medicines, avoid cold.

Prognosis: few lose the leg, but heart attack and stroke risk is high; smoking predicts worsening and amputation in Buerger's.

Exam plan

1. Name it (PAD or Buerger's in a young smoker) and the Fontaine stage.
2. **ICF or impairments (5–7)** with reasons for pain, wasting and skin changes.
3. **Complications (3)** with prevention, if asked.
4. **Management (7–8):** ABI and Buerger's test; walking dose; strengthening; foot care; stop smoking; no heat; outcome measures.

Draw: ABI number line with Fontaine stages. **Underline:** ABI ≤ 0.90, walk to moderate pain, rest, repeat, collaterals, stop smoking, no heat.

COPD: chronic bronchitis and emphysema

6(b). In 13 of 20 papers. Full notes p. 73

Asked: 15-mark cases (exacerbation; ICU on SIMV) with ICF + short-term + long-term. **7 marks:** energy conservation; ABG, PFT and X-ray in goal setting. **3 marks** (most papers): effects of smoking, X-ray of emphysema, PFT in obstruction, pursed-lip breathing, limb strengthening, hypoxic drive.

Definition and types

COPD: preventable lung disease with persistent, usually progressive airflow limitation (airways and alveoli), causing breathlessness, cough and sputum. **Spirometry: post-bronchodilator FEV1/FVC below 0.70.**

Chronic bronchitis: productive cough on most days for **3 months a year for 2 years** in a row. **Emphysema:** permanent enlargement of air spaces beyond the terminal bronchioles with wall destruction.

GOLD	Severity	FEV1 % pred.
1	Mild	≥ 80
2	Moderate	50–79
3	Severe	30–49
4	Very severe	below 30

Groups: A few symptoms (mMRC 0–1, CAAT under 10); B more symptoms (mMRC ≥ 2, CAAT ≥ 10); **E** any moderate or severe exacerbation in the past year (GOLD 2026).

	Chronic bronchitis	Emphysema
Problem	Airway inflammation, mucus	Alveoli destroyed, recoil lost
Sputum	Marked	Little
Look	Overweight, cyanosed, ankle oedema	Thin, barrel chest, pursed lips, accessory muscles
Gases	Low O_2, high CO_2 early	O_2 near normal until late

Causes; effects of smoking (3m S17, W22)

Causes: tobacco (pack-years = packs a day × years; bidi, hookah); **chulha smoke** (biomass cooking, non-smoking Indian women); occupational dust (sweepers, farmers, miners, textile); outdoor pollution; past TB; alpha-1 antitrypsin deficiency.

Smoking: paralyses and destroys cilia; enlarges mucus glands and goblet cells; inflames and narrows airways; proteases destroy alveolar walls; FEV1 falls faster; CO lowers O_2 carriage; more infections.

Pathophysiology to impairments

- Inflamed, narrowed small airways, big mucus glands, damaged cilia → cough, sputum, infections.
- Lost recoil → airways collapse on breathing out → **air trapping, hyperinflation** → flat diaphragm at a mechanical disadvantage → accessory muscles.
- Faster breathing in exercise → less time to breathe out → **dynamic hyperinflation**: the main cause of exertional breathlessness.
- V/Q mismatch → low O_2, later high CO_2 → hypoxic vasoconstriction → **cor pulmonale**.
- **Breathlessness spiral:** breathless → inactive → weak limb muscles → more breathless. Physio breaks it at deconditioning.

Clinical features

Slowly worsening exertional breathlessness, morning cough, sputum, wheeze, weight loss. **Signs:** barrel chest, accessory muscles, pursed lips, **tripod posture**, ↓ expansion, hyper-resonance, quiet sounds with prolonged expiration, **Hoover's sign** (lower ribs move in), cyanosis, ankle oedema and raised JVP in cor pulmonale.

Exacerbation: sudden worse breathlessness, cough or sputum (more or purulent), usually infection; danger signs: breathless at rest, drowsy or confused (rising CO_2), accessory muscles.

Investigations

- **PFT (3m W20, W19):** FEV1 ↓, FVC normal or slightly ↓, **FEV1/FVC below 0.70**, little bronchodilator reversibility (asthma: large), PEF ↓, scooped flow–volume loop; **RV, TLC, RV/TLC ↑**; DLCO ↓ in emphysema.
- **X-ray emphysema (3m W18, W24):** hyperlucent lungs, fewer peripheral markings; **low flat diaphragms**, over 6 anterior or 10 posterior ribs visible; horizontal ribs, wide spaces; long narrow (tubular) heart; bigger retrosternal space; bullae. Bronchitis: near normal or "dirty chest", tram lines; big hilar arteries in pulmonary hypertension.
- **ABG:** low PaO_2, high $PaCO_2$ in type 2 failure → **target SpO_2 88–92%**. Oximetry on exercise; 6MWT or shuttle; ECG and echo (right heart); sputum, Hb (polycythaemia).

Medical management

- **Stop smoking**, reduce smoke and dust: the only thing that slows decline.
- LAMA and LABA, salbutamol reliever; inhaled steroid in group E with high eosinophils. Vaccines.
- **Long-term oxygen (ATS 2020):** $PaO_2 \leq 55$ mmHg ($SpO_2 \leq 88\%$), or 56–59 (SpO_2 89%) with oedema, polycythaemia or right heart strain; ≥ 15 h a day.
- **Exacerbation:** nebulised bronchodilators, oral steroids, antibiotics if purulent, controlled oxygen, **NIV** for acidotic hypercapnia, intubation if NIV fails. Surgery: bullectomy, volume reduction, transplant.
- **Hypoxic drive (3m W25):** classic: chronic CO_2 retention leaves breathing driven by low O_2. Current (BTS): CO_2 rises mainly from worse V/Q matching and the Haldane effect. Write: "uncontrolled oxygen can cause CO_2 narcosis; give controlled oxygen (Venturi or low-flow cannula) to SpO_2 88–92% and check ABG". Never withhold oxygen.

Complications

Exacerbations; **respiratory failure** (drowsy, flap: position, support NIV, report); cor pulmonale; **pneumothorax from bullae**; muscle wasting; steroid osteoporosis; anxiety, depression.

PT assessment

Subjective: mMRC; cough and sputum (amount, colour, morning load); exacerbations; **pack-years**; oxygen; inhaler technique; sleep position; breathless ADL; home floor, lift, toilet; job and dust; earning role.

Objective: forward-lean posture, chest shape, rate and pattern, accessory muscles, cyanosis, clubbing (think of another cause), oedema; expansion at three levels, trachea; auscultation; SpO_2 at rest and exertion; 6MWT or ISWT with SpO_2 and Borg; quadriceps and grip; PImax.

Outcome measures: mMRC, CAAT, 6MWT, ISWT and ESWT, modified Borg, SGRQ, CRQ.

ICF (W16: 50, smoker 30 years, 2nd floor, no lift; exacerbation; moderate obstruction; hypoxaemia and hypercapnia)

- **Body function:** breathlessness, altered pattern; sputum, poor clearance; airflow obstruction, hyperinflation; low O_2 and high CO_2; ↓ exercise tolerance; weak limb and breathing muscles; anxiety.
- **Activity:** walking, climbing to the 2nd floor, bathing, dressing, carrying. **Participation:** work, home roles, social and religious gatherings.
- **Environment:** 2nd floor without lift, smoke and dust (barriers); family, medicines, rehab (facilitators). **Personal:** 50, 30-year smoker, worried about income.

Goals: short-term: breathing control and relieving positions, clears sputum with ACBT, SpO_2 88–92%, one flight with pacing, correct inhaler use, quit plan. Long-term (6–12 weeks rehab): mMRC better by one grade, better 6MWT, reaches the 2nd floor with one rest, tobacco-free, action plan.

Tests set goals (7m W24): acidotic high CO_2 → acute goals only, no training; compensated high CO_2 → 88–92%, night NIV; desaturation on walking → train with oxygen, intervals; FEV1 grade → realistic goals; ↑ RV → pursed lips, forward lean, slow breathing; flat diaphragm → don't force abdominal breathing; **bullae → no high-pressure techniques**; consolidation → target that lobe.

Acute exacerbation (in hospital)

- **Relieving positions:** high side-lying; sitting leaning forward on a table or knees; standing leaning on a wall. They let the flat diaphragm work better.
- Breathing control; pursed-lip breathing on effort; **ACBT** with huffs; PEP or flutter if helpful; postural drainage only if it clearly helps (head-down can worsen breathlessness; BTS 2009).
- Support NIV (mask tolerance, upright, clear secretions in breaks). Early sitting and walking with oxygen, SpO_2 88–92%.
- **On SIMV (S20, RLL pneumonia):** affected lung uppermost, suction, manual techniques, careful hyperinflation (air trapping, pneumothorax), early sitting, IMT in weaning. Refer to pulmonary rehab.

Pursed-lip breathing (3m S20, W21, W22)

In through the nose; out slowly through loosely pursed lips ("blowing on hot tea"), out about **twice as long** as in, not forced. **Why:** back-pressure splints small airways open → less air trapping and hyperinflation → slower, deeper breathing, better O_2 → less breathlessness and panic on walking and stairs.

Pulmonary rehabilitation (long-term)

Exercise + education + behaviour change, **6–12 weeks, ≥ 2 supervised sessions a week** plus home. Better breathlessness, exercise capacity, quality of life; fewer readmissions after an admission (ATS 2023).

	Endurance	Strength
F	3–5 days a week	2–3 days a week
I	Walk at ~80% of 6MWT or shuttle speed, or cycle 60–80% peak work; **Borg 4–6**	60–70% 1RM, 8–12 reps
T	20–60 min; intervals (1 min on, 1 off) if too breathless	1–3 sets, legs first
T	Walk, treadmill, cycle, arm ergometer	Weights, bands, sit-to-stand, step-ups

Oxygen during training if SpO_2 falls below ~88%. **IMT:** ≥ 30% PImax, ~30 breaths twice a day. Education: inhalers, clearance, energy conservation, action plan, quitting, nutrition. Maintenance at home.

Why strengthen limbs (3m W24): COPD wastes muscle (inactivity, hypoxia, inflammation, steroids, poor food); weak thighs tire early and make lactate, which drives breathing, so many stop from **leg fatigue**; training improves strength, stairs, standing, carrying, breathlessness at the same load, bones and falls, and is less breathless than endurance work.

Energy conservation (7m S19)

Plan heavy tasks across the week and at the best time of day. **Pace:** slow, rest before very breathless, breathe out on effort. **Prioritise** and delegate. **Position:** sit to wash, dress, cook; keep things between waist and shoulder; avoid bending and reaching overhead. **Aids:** trolley, long sponge, shower stool, raised toilet seat. **Stairs:** one step per out-breath, rest on landings, fewer trips. Lowers the O_2 cost of daily tasks, so he stays independent.

Precautions and stop

SpO_2 88–92% if CO_2 is high; no head-down when breathless or after meals; no high PEP or aggressive bagging with bullae; no vigorous percussion on steroid-thinned ribs; no training during an exacerbation. **Stop if:** SpO_2 below ~85–88% despite oxygen, Borg above 7, chest pain, dizziness, confusion, new irregular pulse, BP falls.

Education and prognosis

Stop tobacco; avoid chulha smoke and dust; inhaler technique; daily walking and strength work; daily clearance; pacing; small frequent meals; vaccines; **action plan** for early exacerbation signs. Worse prognosis: low FEV1, frequent exacerbations, low BMI, smoking.

Exam plan

1. Restate: age, job, smoking, home, PFT grade, gases, acute or stable.
2. **ICF (5)** with reasons: hyperinflation, sputum, deconditioning.
3. **Short-term (3–5):** positions, breathing control, ACBT, controlled oxygen, NIV support, early walking.
4. **Long-term (5–7):** pulmonary rehab doses, IMT, energy conservation, stairs, quitting, action plan, outcome measures.

Draw: breathlessness spiral, spirometry curves, forward-lean figures. **Spot:** FEV1/FVC 0.55 with FEV1 60% = GOLD 2.

Lung surgery, thoracotomy, chest drain and chest trauma

6(e). In 12 of 20 papers. Full notes p. 87

Asked: 15-mark cases after lobectomy, segmentectomy or pneumonectomy (functional diagnosis or ICF + short-term + long-term). **7 marks:** complications, intercostal drain and underwater seal, incisions and thoracotomy complications, pneumothorax with drain care, pneumonectomy, lobectomy. **3 marks:** muscles cut in posterolateral thoracotomy, three incisions, indications for a chest drain.

Definitions and operations

Thoracic surgery on lung, pleura, chest wall, oesophagus or mediastinum always lowers lung volume and cough strength and stiffens the shoulder and trunk on that side. **Intercostal drain (ICD):** a tube between the ribs into the pleural space, to an **underwater seal** that lets air and fluid out but not back.

Operations: pneumonectomy (whole lung), **lobectomy** (commonest for cancer), segmentectomy, wedge, sleeve resection, decortication (empyema peel), pleurectomy and pleurodesis, bullectomy, thoracoplasty, transplant.

Indications: lung cancer, localised bronchiectasis, abscess, aspergilloma, TB complications, hydatid cyst (W16), empyema, recurrent pneumothorax, chest trauma.

Incisions (7m S22; 3m W19, W20, S23)

Posterolateral thoracotomy	Below nipple line, round under the scapula tip, up beside the spine; 5th space. Cuts **latissimus dorsi, serratus anterior**, trapezius and rhomboids (if extended), **intercostals**. Most painful. Lobectomy, pneumonectomy, oesophagectomy
Anterolateral	Inframammary: pectoralis major (± minor), serratus anterior, intercostals; trauma
Axillary (muscle-sparing)	Latissimus retracted, serratus split along fibres; upper lobe, pneumothorax
Median sternotomy	Sternum split, no major muscle cut, wired; heart, mediastinum
Clamshell; thoracoabdominal	Both sides of front chest (transplant, trauma); chest + abdomen through the diaphragm
VATS	2–4 ports, little muscle cut: less pain, faster recovery

Surgery to impairments

- Anaesthesia and lying flat → ↓ FRC, basal airway closure.
- Pain from incision, rib spreading and drains → shallow breaths, poor cough → **sputum retention → atelectasis → pneumonia**.
- Lung removed → ↓ TLC and exercise reserve; remaining lung must expand.
- Cut shoulder-girdle muscles and guarding → **stiff shoulder**, trunk bent towards the incision.
- Bed rest and drains → ↓ mobility and fitness, DVT risk.

Features: incision and drain pain worse on breathing and coughing; shallow fast breathing; ↓ expansion and sounds on the operated side; crackles; weak painful cough; drains, oxygen, epidural; stiff shoulder; fatigue.

Tests: pre-op FEV1 and DLCO (predicted post-op function), stair climb, shuttle or CPET (prehab); post-op X-ray (re-expansion, collapse, surgical emphysema); ABG, SpO_2; Hb.

Intercostal drain (7m W20; 3m S25)

- **Indications:** large or breathless pneumothorax; tension (after needle decompression); any pneumothorax on a ventilator; haemothorax; empyema; large or malignant effusion (and before pleurodesis); chylothorax; after chest surgery.
- **Safe triangle:** front edge of latissimus dorsi, side edge of pectoralis major, line at nipple level (5th space), apex below the axilla.
- **Underwater seal:** tube ends ~2 cm under water; air bubbles out, water stops it being sucked back. **Bottle always below the chest.**
- **Swing** = tube patent and in the pleura. **Bubbling** = air leak. **No swing** = kinked, blocked, or lung fully expanded. Low suction may be added.

Medical: epidural, paravertebral or intercostal block, PCA; oxygen; antibiotics; enhanced recovery (walk day 1); chemo or radiotherapy for cancer.

Complications (7m S17, S18)

- **Sputum retention, atelectasis, pneumonia** (↓ sounds, crackles, fever, ↓ SpO_2): deep breathing, ACBT, supported cough, walking.
- Respiratory failure; DVT and PE; arrhythmia (AF): check pulse first.
- **Air leak, bronchopleural fistula** (persistent bubbling; after pneumonectomy coughing thin fluid): report, no high-pressure techniques.
- Haemorrhage (heavy blood in drain, ↓ BP): stop and report. **Surgical emphysema** (crackling under skin): report, check drain.
- Empyema, wound infection; nerve injury (recurrent laryngeal: hoarse, weak cough; phrenic: raised diaphragm).
- **Stiff shoulder, posture, scoliosis**: shoulder and trunk exercises from day 1. Post-thoracotomy pain syndrome: desensitise, TENS.

PT assessment

Pre-op: lung function, exercise tolerance, smoking, sputum, shoulder range, job, hand dominance; teach the plan. **Post-op chart:** operation, incision, drains (number, site, **swing, bubbling**, amount and type), analgesia, oxygen, X-ray. **Respiratory:** pattern, rate, SpO_2, expansion, auscultation, cough. Pain at rest and on coughing (VAS). Shoulder range, posture. Bed mobility, sitting, standing, walking with SpO_2. **Outcome measures:** 6MWT, shoulder ROM, pain, EORTC QoL.

ICF (W16: 40, right-handed labourer, right lower lobectomy for hydatid cyst, posterolateral thoracotomy,

day 2, ICD, productive cough)

- **Body function:** incision and drain pain; ↓ lung volume, shallow breathing, atelectasis risk; retained secretions, weak cough; ↓ right shoulder range and strength; ↓ exercise tolerance. Reasons: pain and anaesthesia, cut latissimus, serratus and intercostals, lung removed.
- **Activity:** bed mobility, walking with the drain, self-care with the right arm, lifting. **Participation:** labour work (heavy, overhead), family income.
- **Environment:** hospital, family (facilitators); heavy job, dust, daily wages (barriers). **Personal:** 40, right-handed, young.

Goals: short-term (~1 week): clear chest, effective supported cough, lung re-expanded, pain controlled, near-full shoulder flexion and abduction, walks and climbs one flight. Long-term (6–8 weeks): normal breathing and chest mobility, full shoulder strength for overhead work, better 6MWT, graded return to labour.

PT management, day by day

- **Before surgery:** explain; teach breathing, ACBT, supported cough, bed mobility, shoulder exercises; stop smoking; **prehabilitation** (aerobic + strength for a few weeks) cuts lung complications and stay (Cochrane).
- **Day 0–1:** sit upright, high side-lying either side (after lobectomy); deep breaths with 3-s hold, thoracic expansion to the operated side, ACBT; **supported huff and cough** (pillow or arm over the incision and drain); ankle pumps; sit out, march or walk with the drain; active-assisted shoulder movement.
- **Day 2–4:** walk 3–4 times a day, stairs before discharge; active shoulder flexion and abduction, trunk side-bending away from the incision, rotation; mirror posture; clearance until the chest is clear.
- **After discharge:** daily walking, shoulder stretch and strength, thoracic mobility, exercise training; heavy work after ~6–8 weeks.
- **Dose:** deep breaths 5–10 every hour; shoulder 10 reps 3–4 times a day. Incentive spirometer is an add-on only (AARC 2011).

Pneumonectomy (7m S23; cases W21, S26)

- **Approach (S26):** posterolateral thoracotomy, 4th or 5th space; skin → latissimus dorsi → serratus anterior → lower trapezius and rhomboids → intercostals → parietal pleura; vessels and main bronchus divided, stump closed.
- The space fills with fluid over weeks; mediastinum shifts towards it.
- **Position:** supine, sitting, or a quarter turn **towards the operated side**. Never fully on the remaining lung (fluid could spill through a leaking stump into the only lung); no head-down.
- One lung does all gas exchange: deep breathing, supported huff, early walking matter even more. Shoulder exercises from day 1; mirror posture (scoliosis risk).
- **Report:** AF, crackles in the remaining lung (fluid overload), fever, coughing thin watery or bloody fluid (worse lying on the good side) or bubbling for days (W21) = **bronchopleural fistula**.

Physio with a chest drain; chest trauma

- Check swing, bubbling, drainage, kinks before every session. Bottle upright and **below chest level**; carry it low when walking.
- **Never clamp a bubbling drain.** Disconnection: chest end under sterile water (or brief clamp per protocol), call the nurse. Drain out: sealed dressing, call for help.
- Support the drain site for coughing; gentle shoulder work near it.
- **Rib fractures (W17 electrician):** pain control (blocks, epidural) is the key; on the ventilator positioning, suction, careful manual techniques away from fractures; then breathing, supported cough, walking. No percussion over fractures.

Precautions and stop

Drain rules above. After pneumonectomy no full side-lying on the remaining lung, no head-down, no vigorous manual techniques. Fresh stump: no aggressive bagging or high PEP. No percussion over fractures, incisions or drains. Epidural: check leg power and BP before standing. **Stop and report:** sudden breathlessness or chest pain, heavy fresh blood in drain or sputum, new irregular pulse or falling BP, spreading surgical emphysema.

Education and discharge

Deep breathing and huffing for weeks at home; daily walking; shoulder and trunk exercises until full; no heavy lifting at first; stop smoking; report fever, breathlessness, wound changes. **Discharge:** drains out, chest clear, walks and climbs stairs, shoulder near full, knows the home programme.

Exam plan

1. Restate: age, job, **hand dominance**, operation, incision, day, drains, sputum.
2. **ICF (5)** with reasons.
3. **Short-term (5):** day by day: breathing, ACBT, supported cough, positioning, drain care, walking, shoulder ROM.
4. **Long-term (5):** exercise training, shoulder and posture, return to work, stop smoking.

Draw: the posterolateral incision with muscles; the underwater seal. **Spots:** drain bottle, portable drain bag; X-ray of a drain, surgical emphysema, post-pneumonectomy space.

Bronchiectasis, including after TB

6(c). In 11 of 20 papers.

Full notes p. 98

Asked: 15-mark cases, usually post-TB (functional diagnosis + short-term + long-term, or reasons for impairments + rehab plan). **7 marks:** right lower lobe physiotherapy, ACBT vs autogenic drainage, management, ergonomic and energy-conservation advice, mucus removal techniques. **3 marks:** clinical features.

Definition and types

Bronchiectasis: permanent, abnormal widening of the bronchi with daily wet cough, sputum and repeated infections. Widened scarred airways cannot clear mucus, so **airway clearance is the core of physio**.

Shape (Reid): cylindrical (mildest), varicose (beaded), **cystic or saccular** (most severe, most sputum, fluid levels). **Extent:** localised (one lobe: after TB, foreign body, pneumonia) or diffuse (CF, immune deficiency, ciliary disorders). Severity: Bronchiectasis Severity Index, FACED.

Causes: post-TB (commonest in India), severe pneumonia, childhood measles or whooping cough; blockage (foreign body, tumour, nodes); cystic fibrosis; primary ciliary dyskinesia (Kartagener's: + sinusitis, reversed organs); immune deficiency; ABPA; rheumatoid arthritis; aspiration; idiopathic.

Vicious cycle (Cole): reasons for impairments (S19)

Infection (e.g. TB) damages the wall → inflammatory enzymes destroy elastic and muscle → **permanent dilatation** → cilia destroyed, mucus cannot clear → **mucus pools**, bacteria colonise → more infection and inflammation → repeat. **Physio breaks the cycle by removing pooled mucus.**

- **Copious productive cough:** pooled mucus. **Foul sputum:** anaerobes in stagnant mucus.
- **Breathlessness, ↓ exercise tolerance:** obstruction by mucus, narrowed airways, deconditioning.
- **Haemoptysis:** enlarged fragile bronchial arteries.
- **Stress incontinence:** repeated coughing raises abdominal pressure on the pelvic floor.
- Fatigue, weight loss, low mood, embarrassment. **Tracheal shift** towards a fibrosed, collapsed lobe (S21: shift to the left = left disease).

Clinical features (3m W19, W20)

Chronic cough with large amounts of mucopurulent sputum, worse in the morning and on changing position; sputum settles in **three layers** (froth, mucus, pus); recurrent infections; haemoptysis; breathlessness, wheeze, fatigue, weight loss, sinus symptoms. **Signs: clubbing**, **coarse crackles** that change after coughing, wheeze, ↓ expansion over a fibrosed lobe, halitosis.

Investigations

- **HRCT (gold standard):** bronchus wider than its artery (**signet ring**), non-tapering airways, cysts: tells which segments to drain.
- **X-ray:** tram lines, ring shadows, cysts with fluid levels; collapse or fibrosis pulling the trachea.
- Sputum culture (Pseudomonas = worse disease; infection control); spirometry (usually obstructive); ABG, SpO_2; tests for cause (immunoglobulins, aspergillus, sweat test, TB).

Medical and surgical management

Airway clearance taught by a physio for every productive patient, and **pulmonary rehab** for reduced exercise capacity (ERS 2025, strong). Antibiotics (14 days in exacerbations; long-term macrolide or inhaled); nebulised hypertonic or normal saline before clearance; bronchodilators; treat the cause; vaccines. Surgery: lobectomy for localised resistant disease or major haemoptysis; bronchial artery embolisation.

Complications

Exacerbations (more clearance, action plan); **haemoptysis** (stop percussion, vibration and head-down; gentle breathing control; report; large bleed = emergency); respiratory failure, cor pulmonale; abscess, empyema, pneumothorax; stress incontinence (pelvic floor, "the knack"); social isolation.

PT assessment

Sputum: daily amount (teaspoons, cups), colour, smell, thickness, blood; best time and position. Cough effectiveness; mMRC; exacerbations a year; current routine; past TB; **ask about incontinence**; job and home. Posture, clubbing, pattern, chest shape; expansion, trachea; **coarse crackles over the lobe, rechecked after treatment**; SpO_2 at rest and exercise; 6MWT or shuttle.

Outcome measures: sputum weight or volume, Leicester Cough Questionnaire, QOL-B, SGRQ, 6MWT, mMRC.

ICF (S19: 50, clerk, post-TB right lower lobe, foul sputum, cough and leak on laughing or talking)

- **Body function:** copious foul sputum, poor clearance; paroxysmal cough; breathlessness, ↓ exercise tolerance; **pelvic floor weakness**; ↓ right lower chest expansion; anxiety about work.
- **Activity:** talking or laughing long, walking and stairs at her pace, housework.
- **Participation:** clerk work with the public; social gatherings (cough, sputum, leakage).
- **Environment:** desk job, TB treated, care access (facilitators); dusty crowded office, no private place to clear sputum (barriers). **Personal:** 50, motivated, worried.

Goals: short-term (2–4 weeks): independent ACBT in the right lower lobe position twice a day; clearer chest; walks 20 min; correct pelvic floor contraction and "the knack". Long-term (3 months+): daily routine kept, fewer exacerbations, better 6MWT, leakage stopped, full workdays and social life.

Airway clearance (the core)

Dose: 10–30 min, **once or twice a day**, more in exacerbations; the technique the patient will actually do. Order: bronchodilator → nebulised saline → clearance → inhaled antibiotic. Combine with the lobe's drainage position if tolerated.

ACBT: breathing control → **3–4 thoracic expansion breaths with a 3-s hold** (air gets behind mucus via collateral channels) → breathing control → **1–2 huffs from mid lung volume** (mouth open, "steaming a mirror") → breathing control; huff from high volume or cough when mucus is central. Repeat until clear or two cycles bring nothing.

Autogenic drainage: breathing at low volume ("unstick"), mid ("collect"), high ("evacuate"); open-mouth breath out; cough suppressed until mucus is central. **PEP, flutter, Acapella:** breathing out against resistance splints airways; oscillation vibrates mucus; good for home and work. **Postural drainage with percussion and vibration:** if helpful and not contraindicated.

(7m W17)	ACBT	Autogenic drainage
Principle	Deep breaths get air behind mucus; huffs move it	Airflow at different lung volumes moves mucus centrally
Learning	Easy, one session	Needs training; older children and adults
Position	With drainage positions	Usually sitting
Cough	Huff, then cough	Suppressed until central: less airway collapse
Best for	Most patients, post-op, exacerbations	Floppy airways, cough-induced bronchospasm, **incontinence**

Postural drainage: affected segment uppermost

Upper apical	Sitting, leaning back ~30°
Upper posterior	Sitting leaning forward over a pillow
Upper anterior	Supine flat
Middle lobe, lingula	Lie on the opposite side, quarter turn back; foot up ~15°
Lower superior	Prone, pillows under hips; flat
Anterior basal	Supine; foot up 20–30°
Lateral basal	Opposite side, quarter turn forward; foot up 20–30°
Posterior basal	Prone, pillow under hips; foot up 20–30°

Right lower lobe (W16, S19): prone foot-up (posterior basal), left side quarter forward foot-up (lateral basal), prone flat (superior); ACBT in each.

No head-down tilt (S18, any 6): haemoptysis; raised ICP or recent head injury; uncontrolled BP, unstable heart, arrhythmia, pulmonary oedema; severe breathlessness or bronchospasm; reflux, hiatus hernia, after meals; aortic aneurysm, recent eye surgery; obesity, distension, late pregnancy; untreated pneumothorax, large effusion, rib fractures. Use flat or sitting positions instead.

Exercise, pelvic floor, ergonomics

- **Pulmonary rehab** as in COPD (6–12 weeks, ≥ 2 supervised sessions a week); clear the chest before exercise; IMT if weak.
- **Pelvic floor:** slow holds up to 10 s × 10 and quick contractions, 3 times a day; **tighten before coughing** ("the knack").
- Breathing control, posture, thoracic mobility; drink enough water.
- **Left lung disease, normal right (7m W22):** rest **on the right side** (diseased lung up drains; healthy lung down gets more blood flow and better gas exchange); clear the chest before work, private huffing break; upright sitting, desk at elbow height; pacing, sit for chores, trolleys; avoid dust and crowds, mask.

Precautions and stop

Haemoptysis: no percussion, vibration or head-down. Bronchodilator before clearance in bronchospasm. Infection control: safe sputum disposal, separate Pseudomonas patients, clean devices daily; confirm TB is not infectious. **Stop and report:** fresh bleeding beyond streaks, falling SpO_2, severe breathlessness or wheeze, sudden chest pain.

Education and prognosis

Clearance every day, even on good days; sputum diary and action plan; stop smoking; daily exercise; clean nebulisers; pelvic floor daily; vaccines. Worse with Pseudomonas, frequent exacerbations, low FEV1, many lobes, low weight.

Exam plan

1. Restate: age, job, cause (TB), **which lobe**, sputum, mMRC, X-ray detail.
2. **ICF or reasons (5–7):** ICF + vicious cycle + reason for each impairment.
3. **Short-term (5):** clearance plan for that lobe (technique, positions, dose), saline, breathing control, pelvic floor.
4. **Long-term (5–8):** rehab doses, home routine, devices, energy conservation, action plan, outcome measures.

Draw: vicious cycle, the lobe's drainage position, ACBT loop. **Clue (W24):** more sputum lying on the right side = left lung disease.

Amputation and stump care

6(h). In 11 of 20 papers. Full notes p. 110

Asked: 15-mark cases after below-knee (and one above-knee) amputation for diabetes or PAD: ICF + management, goals, return to job, suitable prosthesis. **7 marks:** physiotherapy after BKA, stump management. **3 marks:** ideal stump.

Definition, levels, causes

Amputation: surgical removal of all or part of a limb; what remains is the **residual limb (stump)**. Levels: hemipelvectomy, hip disarticulation, **transfemoral** (above knee), knee disarticulation, **transtibial** (below knee), Syme's, partial foot. **Higher level = more energy to walk; keeping the knee is the biggest advantage.**

Causes: PAD and diabetes (commonest in adults), trauma (commonest in the young), infection (gas gangrene, diabetic foot), tumours, burns, frostbite, congenital.

K-levels: K0 cannot walk with a prosthesis; K1 household walker; K2 limited community (kerbs, stairs); K3 community at varying speed, work; K4 high impact.

Ideal stump (3m W18)

- **Length:** transtibial **12–15 cm** below the knee joint line; transfemoral ~**25 cm** from the greater trochanter (room for a prosthetic knee).
- **Conical**, firm, not bulbous; healed, mobile, non-adherent scar away from the weight-bearing end.
- Good padding; smooth bevelled bone ends (fibula slightly shorter than tibia); muscles anchored (myodesis).
- Good circulation and sensation; no neuroma; **full range of the joint above**.

Amputation to impairments

- Loss of ankle and foot (± knee): balance, push-off and shock absorption lost.
- Post-op **oedema**: the stump must shrink and shape before a socket fits.
- Pillow under the stump, long sitting → **contracture**: knee flexion (transtibial); hip flexion, abduction, ER (transfemoral).
- Cut nerves → **phantom sensation and phantom pain**; neuroma → local pain.
- Bed rest and disease → weakness, low fitness; prosthetic walking costs much more energy. The other leg often has the same disease.

Features: stump pain and swelling, drains; phantom sensation and pain (burning, cramping); poor single-leg balance, **falls at night** getting up to the toilet; weak hip extensors, abductors, quadriceps; grief, worry about income, body image.

Tests: pulses, ABI, Doppler of both legs (S21: popliteal 1+ = slow healing); sugar, HbA1c, kidney; Hb, albumin; ECG, echo.

Surgery and prosthesis

Level removes all dead tissue but keeps length, ideally the knee; long posterior flap and myodesis or myoplasty in vascular transtibial. Wound care, pain relief, sugar and BP control, nutrition. Soft dressing with bandaging, rigid removable dressing, or a **shrinker** once healed. Prosthesis when healed and stump volume is stable (several weeks).

Transtibial prosthesis: socket PTB (patellar-tendon-bearing) or total-surface-bearing; suspension by supracondylar cuff, sleeve, pin-lock liner or suction; pylon; **SACH or Jaipur foot** (flexible rubber: squatting, cross-legged sitting, barefoot on uneven ground).

For a diabetic vascular patient (W19): fragile, possibly numb skin → total-surface-bearing socket with gel or silicone liner (or PTB with soft foam liner if cost matters); secure sleeve or pin-lock suspension; light endoskeletal pylon; stable SACH or Jaipur foot (limited community walker; floor-level living); early walking aid and interim prosthesis first; daily checks of the stump and other foot.

Complications and physio role

- **Contractures:** positioning, prone lying, stump board, stretching, strengthening.
- **Oedema, bulbous stump:** correct bandaging or shrinker, exercise.
- Delayed healing, infection: report, protect the wound, sugar control.
- **Phantom pain, neuroma:** desensitisation, mirror therapy, TENS, early prosthesis.
- **Falls** at night; socket skin sores (daily checks, fit review); ulcer or overuse in the other leg (foot care, avoid long hopping); DVT, chest infection, deconditioning; depression (goals, peer amputees).

PT assessment

History: cause, level, date, side, diabetes, BP, heart; **job demands** (driving, door-to-door walking, lifting); home floors, steps, toilet; family, earning role; previous walking level. **Stump:** length from a bony landmark, shape, circumference at fixed points, wound, scar, skin, sensation, temperature, pain, neuroma. Hip and knee range both sides (contracture); strength of both legs, trunk, arms. **Other leg:** pulses, skin, sensation, ulcers. Sitting and standing balance, transfers, wheelchair, hopping with walker; phantom sensation and pain.

Outcome measures: AMPPRO, K-level, Timed Up and Go, 2- or 6-minute walk, Houghton scale, SIGAM grades.

ICF (S21: 48, driver, right BKA for PAD, day 7; right popliteal 1+, left foot pulses 2+)

- **Body structure and function:** loss of right leg below knee, healing wound, weak pulse (slow healing); stump pain and oedema; knee flexion contracture risk; phantom sensation or pain; poor single-leg balance; weak hip and knee extensors; ↓ endurance.
- **Activity:** standing, walking, stairs, toileting, bathing, driving. **Participation:** work as a driver (right foot for pedals), family income, social life.
- **Environment:** prosthetic services (Jaipur foot centres), automatic car or left-foot accelerator, family (facilitators); lost income, Indian toilet, steps (barriers). **Personal:** 48, PAD (smoker or diabetic?), motivated.

Goals: short-term (pre-prosthetic, 4–6 weeks): wound healed, conical stump ready for casting, full knee extension, independent transfers, walks with walker or crutches, phantom pain controlled, self-care at home. Long-term (prosthetic, 3–6 months): good socket fit, full range, community walking without aids (K3), drives an adapted car, back to work.

PT management

- **Before surgery:** explain, meet an amputee, teach transfers, wheelchair, exercises; strengthen arms and other leg; check the other foot.
- **Early post-op (days 1–10):** breathing exercises, ankle pumps of the other leg, sit out early; positioning and prone lying; stump bandaging or shrinker, elevation without bending the knee; isometric then active quadriceps, hip extensors and abductors; full exercise of the other leg, trunk, arms; transfers and wheelchair with a **stump board**; balance in parallel bars; hopping with a walker only for short distances.
- **Prosthetic training:** donning and doffing, daily skin checks; weight shift in bars → standing on the prosthesis → stepping → walking with fewer aids; correct gait deviations (short step, wide base, trunk lean); **stairs: "good goes up, bad goes down"**; ramps, kerbs, uneven ground, getting up from the floor, squatting with a Jaipur foot; build wearing time gradually.

Contracture prevention

No pillow under the stump, under the knee or between the thighs; stump not hanging over the bed edge or resting on a crutch handle; **prone lying 20–30 min, 2–3 times a day**; transtibial: knee straight, stump board; transfemoral: limit sitting, hip straight, not rolled out or abducted.

Stump bandaging (7m W20)

- Elastic bandage **10 cm** (4 in) for transtibial, **15 cm** (6 in) for transfemoral.
- **Diagonal figure-of-eight** turns, no wrinkles, **firmest at the end** and less towards the knee; **no circular turns** (tourniquet).
- Transtibial: anchor above the knee. Transfemoral: include the groin, anchor round the waist (hip spica).
- Re-apply every **4–6 h** and when loose; day and night except bathing and exercise. Shrinker sock once healed (easier, more even).

Pain and phantom pain (S25)

Stump pain: find the cause first: infection, poor blood supply, neuroma, bony end, tight bandage; report infection or ischaemia; treat after analgesia. **Phantom pain** (burning, cramping in the missing part) is common and usually lessens over months: tapping, massage, rubbing with textures several times a day; **mirror therapy**; TENS; shrinker or prosthesis; reassure.

Above-knee amputation (S23: 60, diabetic, lives alone, one floor up)

Contracture: hip flexion, abduction, ER: prone lying, no pillows, limit sitting. Strengthen **hip extensors and adductors**, abductors, trunk, arms, other leg. Hip-spica bandage or shrinker. Prosthesis: ischial-containment or quadrilateral socket, suction or liner, **stance-control or locked knee** for an older low-activity walker, SACH or Jaipur foot. Very energy-costly: build fitness; wheelchair independence first, prosthetic walking if fitness and the other leg allow. Living alone: safe transfers, getting up from the floor, stairs with crutches and rail (or bumping on the bottom), home visit, rails, phone in reach, other-foot checks.

Return to work: driver: automatic car with left-foot accelerator or hand controls; milkman (S19): endurance walking, stairs, carrying, adapted two-wheeler, rest breaks, sugar control.

Precautions and home care

Check the wound before bandaging and exercise; no pressure on an unhealed end; sugar before exercise; monitor HR, BP and breathlessness (walking is costly); avoid long hopping; never sleep in the prosthesis or with an unchecked tight bandage.

Home: inspect the stump daily with a mirror; wash and dry it, wash socks and liner daily; keep the bandage or shrinker on; daily positioning and exercises; **protect the other foot**; control sugar and BP, stop smoking; stable weight (socket fit); walker and light by the bed at night.

Prognosis: most previous walkers with a transtibial amputation walk again; worse with transfemoral level, age, heart and lung disease, poor sugar control.

Exam plan

1. Restate: age, side, level, cause, day, pulses of the other leg, job and family role.
2. **ICF (7)** with one-line diagnosis and reasons.
3. **Management (8):** pre-prosthetic (positioning, bandaging, exercises, transfers, balance, phantom pain); prosthetic (gait, stairs, uneven ground); return to job; other-foot care; outcome measures.

Draw: amputation levels; figure-of-eight bandaging.
Underline: ideal stump, prone lying, stump board, shrinker, Jaipur foot, K-level.

Varicose veins, venous ulcer, DVT and thrombophlebitis

6(a). In 11 of 20 papers. Full notes p. 121

Asked: 15-mark cases (cook with varicose veins; teacher with infected ulcers; air hostess; bus conductor with an ulcer) as ICF + management + do's and don'ts or foot care. **7 marks** (many): venous or varicose ulcer physiotherapy, DVT prevention and physiotherapy, patient education, assessment of venous insufficiency, two special tests.

Definitions

- **Varicose veins:** dilated, twisted superficial leg veins ≥ 3 mm when standing, from faulty valves.
- **Chronic venous insufficiency:** long-term **venous hypertension** → swelling, skin changes, ulcers.
- **Venous ulcer:** skin break of the lower leg, usually **above the inner ankle**, unhealed in ~2 weeks.
- **DVT:** clot in a deep vein (calf, thigh); a piece to the lung = **PE**. Distal (calf) lower PE risk; proximal (popliteal, femoral, iliac) higher.
- **Superficial thrombophlebitis:** red tender cord along a superficial vein. Phlebothrombosis: older term for a deep clot with little inflammation.

CEAP (2020): C0 none; C1 thread veins; C2 varicose; C3 oedema; C4 skin changes (pigmentation, eczema, lipodermatosclerosis, atrophie blanche); C5 healed ulcer; C6 active ulcer. Plus Etiology, Anatomy, Pathophysiology.

Causes

Varicose veins: prolonged standing jobs (cook, teacher, shop worker, traffic police, air hostess, conductor); female, pregnancy, family history, age, obesity; previous DVT; poor calf pump (inactivity, stiff ankle).

DVT, Virchow's triad: stasis (bed rest, surgery, long travel, paralysis, heart failure); **vessel wall injury** (trauma, fracture, surgery, lines, past DVT); **hypercoagulability** (cancer, pregnancy and after delivery, the pill, dehydration, clotting disorders, obesity).

Calf pump to ulcer (S26: pathophysiology)

Calf contracts → squeezes deep veins → blood up; valves stop backflow; relaxation refills deep veins from superficial via perforators. **Valve failure** → reflux on standing → **venous hypertension** even when walking → veins stretch (aching, heaviness worse by evening) → fluid, red cells, protein leak → oedema, **haemosiderin pigmentation**, eczema → inflammation hardens skin (**lipodermatosclerosis**) → a minor knock becomes an ulcer. Stiff ankle or weak calf worsens the pump.

Ulcer theories: fibrin cuff (leaked fibrinogen cuffs capillaries, blocks O_2 and nutrients); **white-cell trapping** (trapped white cells release damaging enzymes).

Standing still: ankle venous pressure = height of the blood column. Walking pumps it down; exercise strengthens the calf and keeps the ankle mobile.

Clinical features

Venous insufficiency: dull ache, heaviness, worse after standing and in the evening, relieved by elevation; evening ankle swelling; night cramps; itching; visible veins standing; pigmentation, eczema, hard skin above the ankle.

Ulcer	Site, look	Pain	Pulses, ABPI
Venous	Gaiter area, above medial malleolus; shallow, irregular, moist, much exudate	Ache; better raised	Present; pigment, oedema; ABPI normal
Arterial	Toes, heel, lateral ankle; punched out, deep, pale, dry	Severe, worse at night and raised	Weak or absent; cold hairless skin; ABPI low
Neuropathic	Sole, metatarsal heads, heel; callus rim	Often painless	Often present; warm dry numb foot

DVT: one-sided calf or thigh swelling, pain, tenderness, warmth, redness; often silent after surgery. **PE:** sudden breathlessness, pleuritic chest pain, fast HR, low SpO_2, haemoptysis, collapse.

Special tests (S16: two tests; 7m W19)

- **Brodie–Trendelenburg:** lying, empty the veins by raising the leg; press the saphenofemoral junction (or thigh tourniquet); stand up. Fill fast on release = **SFJ incompetent**; fill fast with pressure on = **perforators incompetent**.
- **Perthes:** tourniquet below the knee, standing; heel raises or walking. Veins empty = deep veins open; veins swell and hurt = **deep veins blocked** (do not strip superficial veins).
- **Tap test** (impulse felt above = incompetent valves between); **cough impulse** at the SFJ.
- **Duplex ultrasound** = gold standard (reflux, blocks, DVT). **ABPI before any compression.**
- **DVT:** Wells score, D-dimer (normal helps rule out), compression ultrasound; CT pulmonary angiogram for PE. **Homans' sign is unreliable**: no longer used.

Medical management

Varicose veins: stockings, exercise, elevation, weight; **endovenous laser or radiofrequency ablation** (first line); foam sclerotherapy; ligation and stripping; phlebectomy. **Ulcer: compression ~40 mmHg at the ankle** (multilayer or wraps) once ABPI allows; clean, simple dressings; antibiotics only for spreading infection; ablate the vein. **DVT:** heparin then oral anticoagulant ≥ 3 months; thrombolysis or IVC filter in some. **Thrombophlebitis:** NSAIDs, warm compresses, compression, keep walking.

Complications

Ulcer; **bleeding varicose vein** (lie flat, raise leg, press firmly); cellulitis; **PE** (stop, sit up, call for help); post-thrombotic syndrome; ankle stiffness and calf weakness.

PT assessment (7m W19)

History: standing hours, pain pattern through the day, swelling, itch, past DVT, pregnancies, surgery, diabetes, smoking, stairs. **Inspect standing:** veins, oedema, pigmentation, eczema, lipodermatosclerosis, **ulcer** (site, size, depth, base, edge, exudate, smell). **Measure:** calf and ankle girth both sides, ulcer tracing or photo. **Palpate:** temperature, pitting, tenderness, pulses. Trendelenburg, Perthes, **ABPI**. Ankle dorsiflexion; calf strength (single-leg heel raises); gait; monofilament if diabetic.

Outcome measures: Venous Clinical Severity Score, ulcer area, pain scale, CIVIQ.

ICF (S17: 55, diabetic widow, cook, 2nd floor no lift, aching and evening swelling, dilated veins)

- **Body function and structure:** aching, heaviness; evening oedema both legs; dilated veins; ↓ calf pump endurance; ulcer risk (worse with diabetes). Reasons: standing builds pressure, reflux stretches veins, diabetes slows healing.
- **Activity:** standing long to cook, long walks, climbing to the 2nd floor. **Participation:** full working day, household and social roles.
- **Environment:** standing job, hot kitchen, 2nd floor, maybe no family help (barriers); work, care access (facilitators). **Personal:** 55, widow, diabetic, maybe overweight.

Goals: short-term (2–4 weeks): less pain with stockings and elevation, smaller evening ankle girth, hourly ankle pumps, daily foot checks. Long-term (3 months): full shifts with little aching, swelling controlled, 20 single-leg heel raises, no ulcers.

PT management

- **Calf pump:** ankle pumps **10–20 every hour**; heel raises **2–3 × 10–15**; toe walking; walking **30 min** most days; dorsiflexion stretches.
- **Elevation:** legs above heart **30 min, 3–4 times a day**; foot of bed slightly raised; deep diaphragmatic breaths during elevation.
- **Stockings** (after ABPI): on in the morning before getting up, off at night. Weight loss, fitness.
- **After ablation or surgery:** walk the same day, compression as advised, no long standing or sitting for weeks.
- **Venous ulcer (7m W16, W20, W21):** compression ~40 mmHg if ABPI allows + **calf exercise and walking in the bandage** (they work together; progressive resistance helps healing); ankle ROM; elevation; skin care; electrical stimulation, ultrasound, UV (infected) as add-ons.

ABPI	Compression
≥ 0.8	Full (~40 mmHg) is safe
~0.5–0.8	Reduced only, specialist supervision (mixed disease)
below ~0.5	**None**: refer to vascular surgeon

DVT (7m S18, W18, S22)

Prevention: early walking; **ankle pumps 10 every hour**; deep breathing; static quadriceps, gluteal sets, knee bending in bed; hydration; no crossed legs or pillows behind knees; stockings or intermittent pneumatic compression; heparin as prescribed.

After diagnosis: check the anticoagulant and start time; **walk once anticoagulation is therapeutic** (APTA, strong); bed rest does not lower PE risk. Progress walking and leg exercise, elevate at rest, compression for persistent symptoms; watch for PE and bleeding.

Precautions and stop

ABPI before compression (compression on an ischaemic leg → gangrene). No compression in severe heart failure, fragile neuropathic skin, acute cellulitis. No vigorous massage with suspected DVT. No heat on ulcerated or swollen legs. Gentle handling on anticoagulants. **Stop and call:** sudden breathlessness, chest pain, fast HR, haemoptysis (PE); pale, blue, cold or numb toes under a bandage.

Do's, don'ts, foot care (S20, W18, S17)

Do: walk daily; ankle pumps and heel raises when standing or sitting long; raise legs several times a day; stockings morning to night; breaks at work, shift weight, footrest; healthy weight; low heels; moisturise. **Don't:** stand or sit still long, cross legs, tight garters or waistbands, hot-water bags or sit near fire, scratch. **Foot care (diabetes):** inspect daily, lukewarm water, dry, moisturise (not between toes), cut nails straight, never barefoot, sugar control, report wounds early.

Exam plan

1. Restate: age, standing job, sides, swelling and pain pattern, ulcer site and infection, diabetes.
2. **ICF (5):** valve failure, venous hypertension, calf pump.
3. **Management (5–7):** tests and ABPI; calf exercise and walking with doses; elevation; compression; ulcer care; add-ons; post-surgery.
4. **Add-on (3–5):** do's and don'ts or foot care.

Draw: Virchow's triad, ulcer table, gaiter area. **Underline:** calf muscle pump, ABPI before compression, 40 mmHg, mobilise once anticoagulated.

Asthma in adults and children

6(b), 6(d). In 10 of 20 papers. Full notes p. 132

Asked: 15-mark cases (adult farmer in an attack; 7-year-old girl kept from play; 12-year-old missing school) as ICF + short-term + long-term, or counselling + rehab + outcome measures. **7 marks:** rationale of ACBT, paediatric rehabilitation, acute exacerbation, importance of patient education.

Definition and types

Asthma: chronic inflammatory airway disease with **variable** symptoms (wheeze, breathlessness, chest tightness, cough) and **variable airflow limitation**, often reversing with treatment or on its own. **Exacerbation:** worsening beyond usual, needing a change in treatment.

Types: allergic (childhood, eczema, rhinitis, family history); non-allergic (adult onset); **occupational** (flour, grain dust, chemicals: farmers); exercise-induced bronchoconstriction; asthma–COPD overlap in smokers.

Attack (BTS/SIGN)	Any one of
Moderate	Peak flow over 50–75% of best; no severe features
Acute severe	PEF 33–50%; RR ≥ 25; HR ≥ 110; **cannot finish sentences**
Life-threatening	PEF below 33%; **SpO_2 below 92%**; **silent chest**; cyanosis; poor effort; exhaustion; confusion; low BP; arrhythmia; **normal or rising CO_2**

Control: rare daytime symptoms and reliever use, no night waking, no activity limit. Asthma Control Test (ACT), Childhood ACT.

Risk factors and triggers

Risk: family history, atopy, obesity, smoking and passive smoke, prematurity, early viral infections. **Triggers:** infections; dust mites, pollen, mould, animal fur, cockroaches; smoke, **chulha smoke, mosquito coils, incense**; cold air, weather change (winter, rains); exercise; strong smells; stress; aspirin, beta-blockers; work dusts.

Pathophysiology to impairments

- Trigger → mast cells, eosinophils → **three causes of narrowing: bronchospasm, mucosal swelling, mucus plugs**.
- Obstruction mainly on breathing out → **air trapping**, hyperinflation, wheeze, breathlessness.
- Uneven ventilation → low O_2; CO_2 low at first (fast breathing). **Normal or rising CO_2 = tiring: danger.**
- Years of inflammation → **remodelling**: some obstruction becomes fixed.
- Fear of attacks → avoids exercise → deconditioning; some develop fast upper-chest (dysfunctional) breathing.

Features: episodic wheeze, breathlessness, tightness, cough, worse at night and early morning, triggered; **normal between attacks**. In an attack: fast breathing, accessory muscles, prolonged expiration with wheeze, cannot speak, anxiety; silent chest = life-threatening. Children: night cough, cough with play, avoid running, miss school.

	Asthma	COPD
Onset	Childhood, young adult	Over 40, years of smoke
Course	Variable, normal between attacks	Persistent, progressive
Reversibility	**FEV1 up over 12% and 200 mL**	Little; FEV1/FVC stays below 0.70
Allergy	Common	Uncommon

Investigations

Spirometry with bronchodilator (FEV1 rise over 12% and 200 mL); **peak-flow diary** (variability; basis of the action plan); exercise challenge (EIB); FeNO, eosinophils, allergy tests; ABG and SpO_2 in an attack (rising CO_2 → call the team); chest X-ray (normal or hyperinflated; excludes pneumothorax, pneumonia).

Action plan zones: green above 80% of best peak flow; **yellow** 50–80%; **red** below 50% (doctor sets the medicines for each zone).

Medical management

- **Inhaled steroid-containing treatment for everyone**; GINA no longer recommends salbutamol alone at any step. Adults: low-dose ICS–formoterol as reliever (and maintenance at higher steps). Severe: LAMA, leukotriene antagonists, biologics.
- **Inhaler technique with a spacer** (mask for young children).
- **Attack:** repeated reliever by spacer or nebuliser, oral steroids, oxygen to **SpO_2 94–98%**, ipratropium, magnesium; ICU if life-threatening.
- Trigger avoidance, stop smoking, treat rhinitis, written action plan. W22 girl on SOS salbutamol only with monthly attacks → **needs medical review**.

Complications

Life-threatening attack and respiratory failure (sit up, oxygen, call for help); pneumothorax (stop, report); mucus plugging and collapse; deconditioning, obesity; dysfunctional breathing (sighing, light-headed); steroid side effects; missed school or work, anxiety.

PT assessment

History: triggers, attacks and admissions, night symptoms, reliever use, inhaler technique, smoking, job and dust, home, school or work absence, sport and play. **Observe:** pattern, rate, accessory muscles, posture, **speaking in sentences**. Auscultation (wheeze, prolonged expiration). **Peak flow, SpO_2** (ABG if severe). 6MWT or shuttle (children: shuttle or step test) with SpO_2 and peak flow before and after. Nijmegen questionnaire if dysfunctional breathing.

Outcome measures: ACT or Childhood ACT, AQLQ (PAQLQ for children), peak-flow diary, exercise test, school or workdays missed, attacks a year.

ICF (W22: 7-year-old girl, monthly attacks, SOS

salbutamol, kept from play)

- **Body function:** episodic breathlessness and wheeze; ↓ exercise tolerance from inactivity; possible poor breathing pattern; anxiety.
- **Activity:** running and active play, carrying her bag, stairs in attacks.
- **Participation:** missed school; kept from outdoor play and sport; less time with friends.
- **Environment: fearful parents and teachers**, reliever-only medicine (barriers); caring family, school, health services (facilitators). **Personal:** wants to play.

Goals: short-term (4–6 weeks): parents and teacher understand asthma and the plan; correct spacer technique; breathing control and warm-up; carries her own lighter bag. Long-term (3–6 months): plays outdoors and joins sport, better shuttle or 6MWT, fewer missed days and attacks, uses the plan independently.

Acute exacerbation (7m S22)

- **High sitting or forward-lean sitting** with arms supported.
- **Breathing control and relaxation:** slow, gentle lower-chest breathing, relaxed shoulders, calm reassurance; **no deep forced breathing** during bronchospasm.
- Oxygen and nebulised bronchodilator as prescribed; **airway clearance after the bronchodilator**, once spasm eases: ACBT with gentle huffs.
- **No percussion or vibration** in acute bronchospasm. Fluids; mobilise as it settles.
- Before discharge: inhaler technique, action plan, triggers, quitting, follow-up. Discharge when peak flow is over ~75% of best.

Rationale of ACBT (7m S17): breathing control between parts prevents bronchospasm and fatigue; thoracic expansion gets air behind plugs (collateral ventilation); huffs move mucus without the narrowing and coughing fits of forceful coughs; patient controls the pace, so back to breathing control if wheeze rises.

Long-term management

- **Breathing retraining** (Papworth, Buteyko-based): nose breathing, slower rate, relaxed lower chest, less over-breathing; 10–15 min daily; better symptoms and QoL (Cochrane 2020).
- **Aerobic training 2–3 times a week for 8–12 weeks**: fitness and QoL without worsening asthma; **swimming** suits (warm humid air).
- **Exercise-induced symptoms: 10–15 min warm-up** with short bursts, pre-exercise inhaler if prescribed, nose breathing, avoid cold dry air and pollution, gradual cool-down.
- IMT if weak; posture, thoracic mobility, relaxation; self-management: spacer, peak-flow diary, action plan, triggers.
- **Energy conservation (W19 teacher):** pace ("blow as you go"), plan heavy jobs across the week, prioritise, sit for chores, sit for part of each class, limit stair trips, dustless chalk, microphone; damp dusting, mask, no sprays.

Children (7m S18, S25; W22)

Education with the family: triggers, spacer with mask, action plan. **Breathing games:** bubbles, windmills, cotton-ball races, "smell the flower, blow the candle". **Exercise as play:** warm-up games, interval play, skipping, cycling, swimming, 2–3 times a week, progressing; body-weight strength games, posture.

Parent and teacher counselling (W22): asthma can be controlled and well-controlled children play sport; exercise is good, warm-up and reliever prevent attacks; know early signs and plan steps; inhaler and spacer at school, teacher knows how to help; no smoking at home, dust control, no coils or incense; let her carry a light bag; medical review (reliever-only no longer recommended); emergency if she cannot speak, lips blue, reliever not helping.

Outcome measures (3m W22): Childhood ACT, PAQLQ, peak-flow diary, 6MWT or shuttle, school days missed, attacks and reliever use, time in active play.

Precautions and stop

No percussion, vibration, forced breathing or head-down drainage in acute bronchospasm; reliever within reach every session; no exercise in cold dry air, pollution, an attack or a chest infection. **Get help:** cannot finish sentences, very fast breathing, SpO_2 below 92%, silent chest, blue lips, drowsy or confused, reliever not working or needed more than 4-hourly.

Education (7m W21: why it matters)

Most severe attacks and deaths follow poor control: preventers not taken, relievers over-used, poor technique, no plan. Self-management with a **written action plan** and review cuts admissions, emergency visits, night symptoms and missed days. Teach: preventer even when well; spacer, rinse mouth after steroids; peak-flow diary; triggers (hot-wash and sun-dry bedding, no smoking); exercise with warm-up; daily breathing practice; flu vaccine; mask for farmers.

Exam plan

1. Restate: age, job, triggers, severity, smoking, social role.
2. **ICF (5)** with reasons (in children: fear-driven barriers).
3. **Short-term (5):** positioning, breathing control, relaxation, nebuliser timing, gentle ACBT, no percussion.
4. **Long-term (5–7):** breathing retraining, exercise with warm-up, inhaler technique, plan, triggers, quitting, counselling, outcome measures.

Draw: peak-flow zones (80 / 50), forward-lean sitting.
Underline: variable obstruction, reversibility, breathing control, warm-up, action plan, spacer.

Abdominal surgery: pre- and post-operative care

6(f). In 9 of 20 papers, as short answers. Full notes p. 142

Asked: short answers only, but in 6 of the 10 papers since 2021. **7 marks:** importance of pre-op chest physio, physio after gastrectomy, exploratory laparotomy, cholecystectomy day 2 (rationale and procedure), complications with DVT, principles after abdominal surgery. **3 marks:** three incisions, rationale of three pre-op goals.

Definition and incisions (3m S16, W16, S26)

Operations on stomach, bowel, liver, gallbladder, pancreas, spleen, kidney, abdominal wall: open (laparotomy) or laparoscopic. **Post-operative pulmonary complications (PPCs):** atelectasis, pneumonia and other lung problems that delay recovery.

Midline	Linea alba: laparotomy, gastrectomy, bowel, emergencies
Paramedian	2–3 cm from midline (older)
Kocher's	Right subcostal: open gallbladder, liver
Rooftop (Chevron)	Below both rib margins: liver, pancreas, stomach
McBurney's, Lanz	Right iliac fossa: appendix
Pfannenstiel	Low transverse: caesarean, gynaecology, bladder
Inguinal; transverse	Hernia; children, some bowel
Laparoscopic ports	5–12 mm: less pain

Upper abdominal incisions (upper midline, subcostal, rooftop) are near the diaphragm: more pain on breathing and more lung complications.

Risk and pathophysiology

Risk of PPCs: upper abdominal or thoracic site, open surgery, long or emergency operation, nasogastric tube; older age, smoking, COPD or asthma, obesity, poor fitness, poor nutrition, heart failure, sleep apnoea.

Anaesthesia relaxes the diaphragm → ↓ FRC, basal airway closure → wound pain stops deep breaths and coughs; diaphragm weak for days → opioids and lying flat → retained secretions → **atelectasis, low O_2, then pneumonia**. Bed rest → DVT, weakness; cut abdominal muscles weaken trunk support.

Features: wound pain, shallow fast breathing, weak cough, ↓ basal sounds, low SpO_2; drains, catheter, NG tube; ileus; fatigue. **Tests:** pre-op lung function, fitness, Hb, albumin; post-op X-ray (basal collapse, raised diaphragm, effusion), SpO_2 or ABG, blood counts.

Medical: epidural or wound catheters; **ERAS** (pre-op education, short fasting, few drains, early eating, **mobilise day 0–1**, early tube removal); DVT prophylaxis; nutrition. Gastrectomy: upper midline or rooftop, high PPC risk, small frequent meals.

Complications (7m S22)

- **Atelectasis, sputum retention, pneumonia:** pre-op teaching, deep breathing, ACBT, supported cough, upright, early walking.
- **DVT, PE:** early walking, ankle pumps, leg exercises, stockings.
- **Wound infection, dehiscence, incisional hernia:** support the wound on coughing, log-roll, no heavy lifting ~6 weeks.
- Ileus, urinary retention: early walking. Pain, weakness: graded exercise. Pressure sores: turning, sitting out.

Pre-operative physiotherapy (7m W16, W20)

- **Assess** to find high-risk patients; record a baseline.
- **Educate:** what happens after surgery, why breathing and early walking matter, pain relief.
- **Teach and practise:** deep breathing with a 3–5 s hold, ACBT, **supported cough** (hands or pillow on the wound), **log-roll**, ankle pumps, getting out of bed.
- **Prehabilitation** (high risk): aerobic, strength and IMT for a few weeks. Stop smoking early.
- **Evidence:** LIPPSMAck-POP (441 upper abdominal patients): **one 30-min pre-op session roughly halved PPCs** (HR 0.48).

Rationale of goals (3m W21): breathing and cough learned while pain-free and alert are done correctly after surgery; log-roll and early walking taught early → less fear, sooner walking, fewer DVTs; baseline finds high-risk patients; education lowers anxiety and improves co-operation; prehab → better tolerance and faster recovery.

Post-operative physiotherapy

- **Day 0–1:** sit upright or high side-lying; **deep breaths with a hold, 5–10 every hour**; ACBT if sputum; supported huff and cough; ankle pumps, quadriceps and gluteal sets; log-roll, sit over the edge, sit out, walk with help.
- **Day 2 to discharge:** walk 3–4 times a day, stairs if needed; breathing exercises until fully mobile; gentle trunk mobility, posture.
- **After discharge:** daily walking; gentle core activation, progressed after ~6 weeks; **no heavy lifting ~6 weeks**.
- Incentive spirometry only as an adjunct (AARC 2011).

Principles (7m W25): control pain first and support the wound; restore lung volume (upright, deep breaths, walking); clear secretions; **mobilise early** (ERAS); prevent DVT; protect the wound (log-roll, no lifting or sit-ups ~6 weeks); restore posture (patients bend forward) and core; watch for fever, low SpO_2, wound leak, distension; educate and plan discharge.

Exploratory laparotomy (7m W21, S23)

Long midline incision, often emergency (perforation, peritonitis, obstruction, trauma, bleeding): sepsis, ICU, NG tube, drains, catheter, stoma, distended abdomen pushing the diaphragm up: **high PPC risk**. ICU: positioning, clearance, suction, early mobilisation. Ward: upright, deep breaths hourly, ACBT and huff with a pillow on the wound, ankle pumps, log-roll, sit out and walk from day 1 if stable. **Dehiscence and hernia risk:** support the wound, no straining or lifting, binder if the surgeon uses one; support a stoma when coughing. Later: graded walking, posture, gradual core work.

Day 2 after cholecystectomy (7m S24)

Laparoscopic: 4 ports, low PPC risk, home in 1–2 days; gas irritates the diaphragm → **shoulder-tip pain** (phrenic, C3–5), settles in days, walking helps. **Open (Kocher's):** painful breathing, low right basal volume, high PPC risk.

1. **Assess:** pain, SpO_2, temperature, right basal sounds, sputum, drains, mobility.
2. **Time it** with analgesia.
3. **Position:** high sitting, or left side-lying (right side up): more basal volume.
4. **Breathing:** deep breaths with 3-s hold, expansion to the right lower chest, 5–10 hourly: re-opens alveoli.
5. **Clearance:** huff and cough with a pillow on the wound.
6. **Mobilise:** log-roll, walk 3–4 times a day: best for lung volume, DVT, ileus.
7. Ankle pumps 10 an hour; posture, trunk and right shoulder movement; home advice, no heavy lifting.

Precautions, assessment, education

Precautions: always support the wound; check drains and catheter, bags below the wound; epidural: leg power and BP before standing; watch postural drop, bleeding, sudden breathlessness (PE); no percussion near wounds and drains; no heavy lifting or strong abdominal work ~6 weeks.

Assessment: pre-op type and site, risk factors, breathing, cough, mobility; post-op chart, pain on coughing, rate, SpO_2, expansion, auscultation, sputum, bed mobility, walking. Outcomes: chest complications, SpO_2 on air, walking distance, time to independence, length of stay.

Home: walk daily; pillow on the wound for coughing and laughing; roll onto the side to get up; no buckets or gas cylinders ~6 weeks; stop smoking; report fever, wound discharge, calf pain, breathlessness.

Exam plan for a 7-mark answer

Name the incision and its risk → pre-op teaching (with LIPPSMAck-POP) → day-by-day post-op table → wound protection and lifting rule → complications to watch. **Draw:** abdomen with midline, Kocher's, McBurney's and Pfannenstiel incisions. **Underline:** PPCs, supported cough, early mobilisation, ERAS, log-roll, no heavy lifting 6 weeks.

Diabetes, diabetic foot, wound healing, pressure sores and diabetes insipidus

6(h), 6(i)

Full notes p. 149

Asked: 7-mark questions: physiotherapy for glycaemic control, role of physio in diabetes, diabetic foot care (what, how, why), pressure sores, modalities for wound healing, factors affecting wound healing. **3 marks:** electrotherapy for infected wounds and wound healing (with parameters), stages of healing. Diabetes also sits inside 10 long cases.

Definitions and types

- **Diabetes mellitus:** raised blood glucose from lack of insulin, poor response to it, or both. **Type 1:** autoimmune, young, insulin, ketoacidosis. **Type 2:** insulin resistance + failing secretion; obesity, inactivity, family history; commonest, younger in Indians. Gestational; secondary (pancreas, steroids).
- **Diabetic foot:** infection, ulcer or tissue destruction of the foot from **neuropathy and/or PAD**.
- **Diabetes insipidus** (AVP deficiency or resistance): cannot concentrate urine → large volumes (over ~3 L a day) of dilute urine, thirst; **sugar normal**.

ADA	Diabetes	Prediabetes
Fasting glucose	≥ 126 mg/dL	100–125
2 h after 75 g	≥ 200	140–199
HbA1c	≥ 6.5%	5.7–6.4%
Random + symptoms	≥ 200	–

Wagner: 0 at-risk foot; 1 superficial ulcer; 2 deep to tendon, capsule or bone; 3 abscess or osteomyelitis; 4 forefoot gangrene; 5 whole-foot gangrene. **IWGDF risk:** 0 no LOPS or PAD (yearly checks) → 3 LOPS or PAD + past ulcer, amputation or end-stage kidney disease (every 1–3 months).

Why the diabetic foot ulcerates

- **Sensory neuropathy:** no protective feeling: injuries, tight shoes, burns unnoticed.
- **Motor neuropathy:** weak small foot muscles → claw toes, prominent metatarsal heads → high forefoot pressure → callus.
- **Autonomic neuropathy:** no sweating → dry cracked skin → bacteria enter.
- **Ischaemia** slows healing; high sugar weakens immunity, infection spreads.

Pressure under a callus → bleeding and breakdown beneath = **neuropathic ulcer** → + infection + ischaemia → gangrene, amputation.

Other complications: silent MI, stroke, PAD; retinopathy (bleeding with straining, head-down); nephropathy; postural hypotension and fixed HR (autonomic); frozen shoulder, stiff hands, **Charcot foot**.

Features: thirst, polyuria, tiredness, weight loss, blurred vision, slow healing, infections (type 2 often silent). **Neuropathic foot:** numb, burning, warm, dry, callus, claw toes, pulses present, painless ulcer under forefoot or heel. **Ischaemic foot:** cold, pale, no pulses, painful toe and heel ulcers.

Investigations and foot tests

HbA1c (3-month average; target ~7%, individual); glucometer before and after exercise (insulin, sulfonylureas); **10 g monofilament** (loss of protective sensation); 128 Hz tuning fork, ankle reflexes; pulses, ABI (falsely high if calcified), toe pressures; lipids, kidney, ECG, eyes; water deprivation test for insipidus.

Medical management

Lifestyle; metformin; **sulfonylureas (hypoglycaemia)**; DPP-4 inhibitors; SGLT2 inhibitors and GLP-1 agonists (protect heart and kidney); insulin. BP and lipid control; yearly eye, kidney, foot checks. **Foot ulcer: off-loading** (total contact cast), debridement, infection control, revascularisation, sugar control. Insipidus: desmopressin.

Wound healing (3m S19; 7m S25)

Haemostasis (minutes–hours) → **inflammation** (days 1–5) → **proliferation** (day 4–week 3) → **remodelling** (week 3 to 1–2 years). Diabetes slows inflammation and proliferation most.

Local factors: infection, poor blood supply, oedema, dead tissue, pressure and shear, too dry or wet, size and depth, radiotherapy. **General:** age, nutrition (protein, vitamin C, zinc), high sugar, smoking, anaemia, obesity, steroids, chemotherapy, organ failure, immunity, stress. In the diabetic foot: **sugar, ischaemia, unnoticed pressure, infection, smoking**.

Assessment

Type, duration, medicines (insulin, sulfonylureas), HbA1c, hypos, complications; activity, job, diet, smoking, BMI and waist. **Feet:** skin, nails, callus, deformity, footwear, monofilament, vibration, pulses, ulcer (site, size, depth, Wagner). BP lying and standing; resting HR; 6MWT; strength; balance (single-leg stance, TUG). **Outcomes:** HbA1c, weight, waist, 6MWT, ulcer area, balance.

ICF (practice: shopkeeper, 12 years, HbA1c 9%, numb feet, callus, walks barefoot): loss of sensation, callus, ↓ fitness, weight, balance → unsafe walking, long standing → work days at risk; barefoot habit and shop snacks (barriers), family and clinic (facilitators).

Exercise for glycaemic control (7m S16, W18)

Why: contracting muscle takes up glucose **without insulin**; training raises insulin sensitivity for a day or more; lowers weight, BP, lipids, heart risk.

Aerobic	**≥ 150 min a week** moderate–vigorous over 3–7 days; **no more than 2 days in a row without exercise**
Resistance	**2–3 times a week**, non-consecutive; 1–3 sets of 10–15 reps
Flexibility, balance	2–3 days a week; stretches 10–30 s × 2–4; balance with neuropathy

Less sitting	A few minutes of light activity every 30 min

Rule of 15

Sugar below 70 or symptoms (sweating, shaking, hunger, confusion): **15 g fast sugar** (glucose tablets, 3 tsp sugar in water, half glass juice), recheck in **15 min**, repeat, then a snack. Hypos can come hours after exercise.

Goals: short-term: daily foot checks, footwear always, walk 20 min most days with safe readings, knows hypo plan. Long-term: 150 min/week + resistance twice; lower HbA1c and weight; no ulcers; better balance.

Diabetic foot care (7m S19: what, how, why)

Inspect daily	Soles, between toes, mirror or family: numb feet don't feel injury
Wash, dry	Lukewarm water tested with the elbow; dry between toes: burns, fungus
Moisturise	Not between toes: dry skin cracks
Nails	Cut straight; no home surgery on corns
Footwear	Soft, roomy, closed, with socks; check inside; **never barefoot** (home, temple)
No heat	No hot-water bottles or hot surfaces: burns not felt
Sugar, smoking	Better healing and circulation
Seek help early	Any blister, cut, redness, swelling

Active plantar ulcer: **off-load** (cast, boot, crutches); non-weight-bearing exercise (cycling with the other leg, arm ergometer, swimming once healed). Neuropathy without ulcer: moderate walking is safe with good footwear and checks.

Electrotherapy for wounds (3m S16, S22; 7m S24, S25)

High-voltage pulsed current	Twin-peak, ~100 pps, sensory level (100–150 V), 45–60 min daily; **negative pole on an infected wound**, positive to promote healing: attracts repair cells, blood flow, antibacterial
Ultrasound	Pulsed 20%, ~0.5 W/cm^2, 3 MHz, 5–10 min around the wound: moves inflammation on, collagen
UVC	~254 nm, short test-dosed exposures, protect skin and eyes: kills bacteria
Low-level laser	~1–4 J/cm^2 to wound edges: cell stimulation

Parameters vary between books: write a range with the effect. Check circulation first; no heat or electrotherapy over ischaemic or numb skin unchecked.

Pressure sores (7m W19)

Damage from pressure (± shear) over bony points: **sacrum, heels, trochanters, ischial tuberosities**, occiput, elbows. At risk: bed- or chair-bound, numb skin, ICU, poor nutrition, incontinence, elderly.

NPIAP stages: 1 non-blanching redness, intact skin; 2 partial-thickness, shallow pink wound or blister; 3 full-thickness, fat visible; 4 muscle, tendon or bone exposed; unstageable (slough or eschar covers base); deep tissue injury (persistent maroon or purple). **Braden scale** 6–23: **≤ 18 = at risk**.

Physio: reposition on an individual schedule ("at least 2-hourly, or as skin and surface require"); **30° tilted side-lying**; head of bed ≤ 30°; **float heels**; foam or alternating-pressure mattress, chair cushion, weight shifts; early mobility, exercise; lift, don't drag (slide sheets); **electrical stimulation** has the best modality evidence for slow stage 2 and stage 3–4 sores; nutrition, continence; daily skin checks, never lie on a red area.

Insipidus: free access to water and toilet; dehydration signs; night falls; ICU after head injury or pituitary surgery: check fluid balance.

Exercise precautions

Check sugar before and after (insulin, sulfonylureas); carry fast sugar; avoid the peak of insulin action; treat a low first; no exercise with very high sugar and ketones (type 1). **Advanced retinopathy:** no vigorous exercise, heavy lifting, jumping, head-down, breath-holding. Neuropathy: footwear, feet checked after sessions, balance. Autonomic: slow position changes, **RPE not HR**, avoid heat. Silent heart disease: report unusual breathlessness. No heat or ice on numb skin.

Lifestyle: active daily, break sitting every 30 min; fewer refined carbs and sweets; BMI under 23, waist under 90/80 cm; medicines; lifelong foot care; stop tobacco.

Exam plan

Glycaemic control: why exercise works → ADA table → rule of 15 → precautions. **Foot care:** define → three neuropathies → care table with reasons. **Wound modalities:** healing stages → factors → modality table → precautions. **Underline:** insulin sensitivity, 150 min, no more than 2 days off, loss of protective sensation, monofilament, off-loading, daily inspection.

Obesity and metabolic syndrome

6(i)

Full notes p. 158

Asked: 3 marks: grades of obesity (S21, inside a cardiac case), define metabolic syndrome (W24). A risk factor in many long cases; 7 marks if asked alone.

Definitions and grades (3m S21)

Obesity: excess body fat that harms health; screened by **BMI = weight (kg) ÷ height (m)²**; waist shows the riskier abdominal fat.

	WHO	Asian Indian
Underweight	below 18.5	below 18.5
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	**23–24.9**
Obese	I 30–34.9; II 35–39.9; III ≥ 40	**≥ 25**

Abdominal obesity (India): waist **≥ 90 cm men, ≥ 80 cm women**. A staged Indian definition was proposed in 2025; use the cut-offs above.

Metabolic syndrome (3m W24): any 3 of 5

- Waist above cut-off (South Asians 90 / 80 cm)
- Triglycerides ≥ 150 mg/dL
- HDL below 40 (men) or 50 (women) mg/dL
- BP ≥ 130/85
- Fasting glucose ≥ 100 mg/dL

(Or on treatment for any.) Together they raise the risk of heart disease and type 2 diabetes.

Causes and effects

Causes: intake above use (calorie-dense food, sugary drinks, fried snacks, big portions; sitting jobs, screens); genes and family habits (South Asians store central fat at lower BMI); poor sleep, stress, steroids, hypothyroidism, Cushing's, PCOS.

Effects: abdominal fat → fatty acids and inflammation → insulin resistance, high BP, abnormal lipids → diabetes and heart disease; heavy chest wall and abdomen → ↓ lung volumes (worse lying flat), ↑ work of breathing, sleep apnoea, obesity hypoventilation; knee, hip, back load → osteoarthritis; low fitness, heat intolerance, stigma → less activity.

Features: high BMI and waist, exertional breathlessness, snoring, daytime sleepiness, joint pain, **acanthosis nigricans**, stretch marks, low mood. **Complications:** diabetes, hypertension, CHD, stroke, fatty liver, sleep apnoea, post-op chest problems and sternal wound problems, OA, DVT, some cancers.

Tests: BMI, waist, waist-to-hip ratio, skinfolds or bioimpedance; glucose, HbA1c, lipids, liver, thyroid; BP; sleep study; 6MWT. **Medical:** lifestyle first (**5–10% weight loss** already improves BP, sugar, lipids); GLP-1 agonists; bariatric surgery for severe obesity.

Assessment and goals

Weight, height, BMI, waist; BP with a **large cuff**, resting HR; activity, sitting time, diet, sleep, motivation and barriers; joint pain, range, posture, balance; 6MWT with Borg; sit-to-stand. **Outcomes:** weight, waist, 6MWT, sit-to-stand, step count, QoL.

In an ICF (S21 businessman after CABG, BMI 32): class I obesity (obese by Indian cut-off), insulin resistance, low fitness, heavy chest wall (body function); diet, sitting job, stress (personal).

Goals: short-term (4–6 weeks): walk 30 min most days, break sitting every 30 min, activity and food diary. Long-term (6 months): lose 5–10% weight, smaller waist, > 250 min a week + resistance twice, better 6MWT.

Exercise for weight management (ACSM 2009)

Under 150 min/week	Little weight loss
150–250 min	Prevents gain; modest loss
Over 250 min	Clinically important loss, with diet
200–300 min	Keeps weight off

- **Low-impact aerobic** first: brisk walking, cycling, swimming, water exercise (spare knees).
- Moderate intensity (talk test, RPE 12–13); start with **10-min bouts**, add 5 min a week.
- **Resistance 2–3 days a week**: keeps muscle while losing fat, better insulin sensitivity.
- Lifestyle activity (stairs, walking to shops); dietitian; behaviour change (goals, self-monitoring, family).

Precautions and education

Screen for heart disease, diabetes and sleep apnoea before vigorous work; low-impact for painful joints, good shoes; cooler hours, water; large cuff; avoid lying flat long; breathable clothes (chafing). **Education:** lose ~0.5 kg a week; fewer sugary drinks, sweets, fried snacks, maida; more vegetables, pulses, whole grains; smaller portions; sleep 7–8 h; check weight and waist monthly. Fitness helps even if weight changes little.

Tuberculosis, thoracoplasty and post-TB lung disease

6(c). Post-TB disease in 8 long cases. Full notes p. 163

Asked: 15-mark cases: thoracoplasty (W18 nurse), post-TB destroyed lung (S22 farmer, S24 student). TB is also the cause in 4 bronchiectasis cases and 2 lung-surgery cases.

Definitions and types

TB: infection with *Mycobacterium tuberculosis*, spread through the air by coughing; mainly lungs. **Post-TB lung disease:** lasting damage after cure: fibrosis, cavities, bronchiectasis, airway narrowing, pleural thickening, **destroyed lung**. **Thoracoplasty:** removal of parts of several ribs so the chest wall collapses onto a diseased lung or empyema space (now rare).

Types: latent; active primary (children) or post-primary (upper lobe cavities); extrapulmonary (pleura, nodes, spine (Pott's), meninges, abdomen); drug-sensitive or MDR. **Risk:** contact, crowding, poor ventilation, undernutrition, diabetes, HIV, smoking, alcohol, silicosis, steroids, incomplete treatment.

Pathophysiology and features

Granulomas → caseation → **cavities** → healing by **fibrosis**: lobe shrinks, **trachea and mediastinum pulled to that side**, airways distorted (traction bronchiectasis) → vicious cycle of mucus and infection → restriction, obstruction or both → low O_2, right heart strain. After thoracoplasty: chest wall deformity, early paradoxical movement, later scoliosis and shoulder problems.

Active TB: cough over 2 weeks, sputum, haemoptysis, **evening fever, night sweats, weight loss**, anorexia. **Post-TB:** chronic productive cough, breathlessness, recurrent infections, haemoptysis, flat chest and ↓ expansion over the side, tracheal shift towards it.

Tests: CBNAAT/Xpert (also rifampicin resistance), smear, culture; X-ray (upper-lobe cavities, fibrosis, calcification; destroyed lung opaque and shrunken); CT; spirometry (obstructive, restrictive or mixed); SpO_2, 6MWT; HIV and sugar.

Medical (NTEP): ~6 months daily fixed-dose tablets: **2 months HRZE** (isoniazid, rifampicin, pyrazinamide, ethambutol) then **4 months HRE**; longer regimens for resistance; nutrition support (Ni-kshay Poshan Yojana); surgery for destroyed lung, aspergilloma, empyema. **Complications:** massive haemoptysis, pneumothorax, empyema, aspergilloma, cor pulmonale; drug effects (liver, isoniazid neuropathy, ethambutol vision).

Assessment and ICF (S22: 45, farmer, destroyed lung, fever, cough, chest pain, breathless 4 days)

Assess: infectious status (how long on treatment, sputum results), adherence, nutrition, weight; cough, sputum, haemoptysis, mMRC, fever; chest shape, expansion, trachea, auscultation; posture, spine, shoulder (thoracoplasty); SpO_2 rest and exercise, 6MWT; crowding, smoke, earning role. Outcomes: 6MWT, mMRC, SGRQ, weight.

- **Impairments:** breathlessness, retained sputum, pleuritic pain, fever, ↓ lung volume, ↓ exercise tolerance, low SpO_2.
- **Activity:** walking, lifting, field work, self-care. **Participation:** farming, income, social life.
- **Environment:** dust and smoke, distance to hospital (barriers); family, NTEP (facilitators). **Personal:** 45, smoker?, undernourished?

S24 student: right upper crackles, right pleuritic pain, HR 105, RR 28, FEV1 60% and FVC 56% (mainly restrictive): goals are studies, sport, fitness.

Goals: short-term (1–2 weeks): clears sputum alone, breathlessness controlled, walks with stable SpO_2. Long-term (2–3 months): completes rehab, better 6MWT and mMRC, lighter farm work with pacing and mask, fewer exacerbations.

PT management

- **Exacerbation:** ACBT, autogenic drainage, PEP in positions that drain the area (no head-down with haemoptysis); breathing control, relaxation; **better lung down** for gas exchange; supported cough for pleuritic pain; mobilise as fever settles; fluids, food.
- **Pulmonary rehab:** endurance and strength as in COPD, 6–12 weeks, 2–3 supervised sessions a week, oxygen if SpO_2 falls; daily clearance; IMT; thoracic mobility, posture; energy conservation for farm and housework; education.
- **After thoracoplasty (W18):** deep breathing, ACBT, supported cough, early sitting and walking; firm pad early to limit paradoxical movement; **shoulder exercises from day 1**; **mirror posture and trunk work to prevent scoliosis**; later exercise training.

Infection control and stop

Until confirmed non-infectious (often ~2 weeks of effective treatment): staff wear **N95**, patient a surgical mask, ventilated room or outdoors; cough etiquette, covered sputum container with disinfectant; clean shared devices. **Haemoptysis:** stop percussion, vibration, head-down; sit up; report; massive bleed: **lie on the bleeding side**, call for help. Sudden breathlessness or chest pain: pneumothorax.

Education: every dose for the full course; cover coughs, open windows; protein and calories; stop smoking and alcohol; keep breathing exercises and walking after cure; report fever, weight loss, blood in sputum.

Interstitial and occupational lung disease

6(b) Full notes p. 169

Asked: 15-mark case (S18: cotton-mill worker, ILD, grade II dyspnoea: ICF, goals, management); 7 marks: physiotherapy for a farmer with ILD (S21); 3 marks: goals in restrictive disease (S17), three spirometric findings in restriction (S19).

Definitions and types

ILD: inflammation and **scarring of the interstitium** between the air sacs: stiff small lungs, poor O_2 transfer. **Occupational lung disease:** from dusts, fibres, fumes or organic material at work.

Idiopathic	IPF: commonest fibrosing ILD in older adults
Connective tissue	Rheumatoid, systemic sclerosis, myositis
Hypersensitivity	Farmer's lung (mouldy hay), bird fancier's lung
Pneumoconioses	**Silicosis** (stone cutting, mining, sandblasting), asbestosis, coal workers'
Organic dust	**Byssinosis** (cotton, jute): "Monday chest tightness"
Others	Sarcoidosis, drugs (amiodarone, methotrexate), radiation, post-COVID

Pathophysiology and features

Repeated injury → inflammation and scar → **stiff lungs, ↓ volumes and compliance (restriction)** → rapid shallow breathing. Thick alveolar walls → ↓ DLCO; in exercise blood moves faster → **marked exertional desaturation**. Hypoxia → pulmonary hypertension, right heart failure. Breathlessness → inactivity, weakness; troublesome dry cough.

Features: progressive exertional breathlessness, dry cough, fatigue, weight loss; **fine "Velcro" end-inspiratory basal crackles**; clubbing (IPF); rapid shallow breathing; ↓ expansion; SpO_2 falls sharply on walking; signs of the cause.

Investigations (3m S19)

Restriction: FVC ↓ (FEV1 ↓ in proportion); **FEV1/FVC normal or high** (≥ 0.70, often over 0.80); **TLC ↓**, with low DLCO in ILD. (Restriction lowers the plateau; obstruction slows the rise.)

HRCT: reticular lines, **honeycombing**, ground glass, traction bronchiectasis. X-ray: small lungs, reticular-nodular shadows; silicosis: upper-zone nodules, **eggshell calcified nodes**. **6MWT with lowest SpO_2**; ABG; echo; autoimmune tests.

Medical: remove exposure (job change, masks, dust control); antifibrotics (nintedanib, pirfenidone); steroids for inflammatory types; long-term or ambulatory oxygen; **pulmonary rehab (ATS 2023, strong)**; vaccines, quitting, transplant, palliative care. **Complications:** respiratory failure, cor pulmonale, acute exacerbation (often fatal in IPF), pneumothorax, lung cancer, TB with silicosis.

Assessment and ICF (S18: 40, cotton mill 15 years, grade II dyspnoea)

Assess: work history (dust, years, masks), smoking, mMRC, cough; pattern, expansion, crackles, clubbing; **SpO_2 at rest and during 6MWT**, HR, Borg; strength; ADL and work. Outcomes: 6MWT distance and lowest SpO_2, mMRC, SGRQ-I, King's Brief ILD.

- **Impairments:** grade II breathlessness, ↓ volumes, exertional desaturation, dry cough, ↓ tolerance, weakness.
- **Activity:** walking fast, stairs, carrying, housework. **Participation:** mill work, family roles.
- **Environment:** continued cotton dust, need for income (barriers); hospital, possible transfer or masks (facilitators). **Personal:** 40, earning member.

Goals (3m S17): short-term: safe exercise with SpO_2 ~88% or more (oxygen if needed), breathing control and pacing in ADL, cough control. Long-term: better 6MWT and QoL, home programme, energy conservation, no further dust, suitable work.

Pulmonary rehab for ILD (7m S21)

- **Aerobic:** walking or cycling 3–5 days a week, 20–30 min, Borg 3–4; **interval training** if SpO_2 drops quickly.
- **Oxygen during exercise** to keep SpO_2 ~88–90% or more.
- Resistance 2–3 days a week; breathing control and forward-lean recovery; pacing; energy conservation.
- **Dry cough control:** sip water, swallow, nose breathing, suppression.
- Thoracic mobility, posture; IMT may help. Education: oxygen safety (no flames or smoking), when to seek help, workplace protection.

Differs from COPD: airway clearance rarely needed; the limit is O_2 transfer, so **monitor SpO_2 continuously**, use oxygen and intervals, progress gently (pulmonary hypertension).

Precautions and education

Stop or reduce: SpO_2 below ~85% despite oxygen, chest pain, dizziness, severe breathlessness. Pulmonary hypertension: no sudden heavy effort, breath-holding or heavy lifting; watch for fainting. Steroids: fragile bones. **Educate:** avoid the causative dust (transfer, fitted respirator), stop smoking, daily activity within safe SpO_2, vaccines, report fever or sudden worse breathlessness; TB screening in silicosis. IPF progresses over years; some types improve if exposure stops early.

Pleural effusion, pneumothorax, empyema and pleurisy

6(c)

Full notes p. 174

Asked: 15-mark case: empyema with a drain for 15 days (S25: ICF 3 + goals 4 + physio 8). **7 marks:** pneumothorax with precautions and physio with a drain (W22); drain and underwater seal (W20). **3 marks:** X-ray of effusion (S16), drain indications (S25).

Definitions and types

Effusion: fluid in the pleural space. **Pneumothorax:** air, lung collapses. **Hydropneumothorax:** both. **Empyema:** pus. **Haemothorax:** blood. **Pleurisy:** inflamed pleura, sharp pain on breathing.

Light's criteria (exudate if any): fluid/serum protein over 0.5; fluid/serum LDH over 0.6; fluid LDH over ⅔ of upper normal serum LDH. **Transudate** (pressure imbalance): heart failure, cirrhosis, kidney disease, low albumin. **Exudate** (diseased pleura): pneumonia, **TB**, cancer, PE.

Pneumothorax: primary spontaneous (tall thin young men, smokers); secondary (COPD bullae, TB, CF, ILD: more dangerous); traumatic, iatrogenic; **tension**: one-way valve, mediastinum pushed across, venous return stops: emergency.

Pathophysiology and signs

Air or fluid stops the lung expanding → ↓ volume, breathlessness, low O_2. Large collections push the mediastinum away; collapse pulls it towards. Inflamed pleura hurts → shallow breathing. Untreated empyema or haemothorax → thick peel (**fibrothorax**) → trapped lung, stiff chest wall, scoliosis.

	Effusion	Pneumothorax
Expansion	↓ affected side	↓ affected side
Percussion	**Stony dull**	**Hyper-resonant**
Breath sounds	↓ or absent (bronchial above fluid)	↓ or absent
Trachea	Away (large)	Away (tension)
Onset	Gradual	Sudden pain and breathlessness

Pleurisy: pain on breathing and coughing, pleural rub. Empyema: swinging fever, sweats. Tension: severe distress, low BP, fast HR, distended neck veins.

X-ray (3m S16) and management

Effusion: blunted costophrenic angle (≥ ~200 mL on PA); homogeneous basal opacity with a curved upper border rising to the axilla (**meniscus sign**); lost diaphragm outline; large: mediastinum pushed away, white-out. **Pneumothorax:** thin pleural line, no markings beyond; tension: shift away, flat diaphragm. **Hydropneumothorax:** straight **air–fluid level**. Ultrasound, CT, pleural tap.

Effusion: treat cause; aspiration or drain; pleurodesis or indwelling catheter if malignant. **Pneumothorax (BTS 2023):** guided by symptoms, not just size: observe, aspirate, ambulatory device or drain; VATS pleurodesis for recurrence. Tension: needle decompression then drain. **Empyema:** antibiotics, drain, decortication if it will not drain. **Complications:** trapped lung, recurrence, persistent air leak, re-expansion oedema, drain infection.

ICF and goals (S25: 45, driver, tobacco chewer, empyema, drain 15 days)

Impairments: drain-site pain, ↓ expansion (trapped lung risk), sputum with weak painful cough, ↓ tolerance, stiff shoulder and trunk on the drain side, leaning towards it. **Activity:** bathing, dressing, toileting, walking, stairs, lifting. **Participation:** driving work, income. **Environment:** team, family; drain tubing, tobacco. **Personal:** 45, tobacco chewer.

Short-term (2 weeks): drain pain ≤ 3/10 on breathing and coughing, clears sputum with ACBT, better expansion, independent self-care and ward walking, full shoulder movement. **Long-term (6–12 weeks):** full expansion and posture without pleural thickening, fit to sit and drive long hours, tobacco-free.

PT management

- **Check the drain** (swing, bubbling, fluid), temperature, SpO_2; treat after analgesia.
- **Lie on the unaffected side** (affected lung up to expand; healthy lung down for ventilation and perfusion); high sitting; change position.
- **Localised thoracic expansion** to the affected side, hands on the lower ribs, 3-s hold, **5–10 every hour**.
- ACBT and huffing; supported cough over the drain site.
- Walk with the bottle below the chest, daily progress, stairs; shoulder flexion and abduction; **side-bending away** from the drain; mirror posture.
- After drain removal: thoracic mobility, strength, aerobic training; return to driving when trunk and shoulder move painlessly; stop tobacco; pre-op physio if decortication.
- **Pleurisy:** pain relief; lying on the painful side may splint it; hand pressure for deep breaths and cough; gentle expansion; walk.

Pneumothorax precautions (7m W22)

Undrained pneumothorax: no positive pressure (IPPB, NIV, PEP, manual hyperinflation): can cause tension. Drain rules: swing and bubbling checks, bottle below chest, **never clamp a bubbling drain**. No vigorous percussion over drains or wounds; no straining or breath-holding until healed; no flying until cleared; no scuba unless surgery done. **Stop and call:** sudden worse breathlessness, falling BP, tracheal shift; drain out or disconnected; heavy blood. **Educate:** stop smoking (recurrence), finish TB treatment, gradual return to work, know warning signs.

Hypertension

6(a). Rarely asked alone, but part of many cases. Full notes p. 180

Asked: 7 marks: physiotherapy for hypertension (S18). 15 marks: schoolteacher with chest pain and hypertension (S23). "Known hypertensive" appears in most cardiac and vascular cases, so the lifestyle table is reused often.

Definition and grades

Hypertension: persistently raised BP on two or more correct readings: **≥ 140/90** (most guidelines, ISH 2020, India) or ≥ 130/80 (ACC/AHA).

ACC/AHA	Systolic	Diastolic
Normal	below 120	and below 80
Elevated	120–129	and below 80
Stage 1	130–139	or 80–89
Stage 2	≥ 140	or ≥ 90
Crisis	above 180	and/or above 120

ISH: grade 1 140–159/90–99; grade 2 ≥ 160/100. **Primary** (~90%); **secondary**: kidney disease, renal artery stenosis, hyperaldosteronism, thyroid, coarctation, sleep apnoea, pregnancy, steroids, NSAIDs.

Risk factors: age, family history, abdominal obesity, **salt** (pickles, papad, packaged snacks), low potassium, inactivity, alcohol, stress, smoking, diabetes.

Pathophysiology and features

BP = cardiac output × peripheral resistance; mostly stiff narrow small arteries + salt and water retention + sympathetic overactivity. **Target organs:** heart (LVH, failure, CAD), brain (stroke, dementia), kidneys, eyes, arteries (PAD, aneurysm, dissection).

Why exercise lowers BP: better artery lining, relaxed small arteries, less sympathetic activity, weight loss; one session lowers BP for hours (**post-exercise hypotension**).

Usually silent ("silent killer"); headache, dizziness, nosebleed; crisis: severe headache, blurred vision, chest pain, confusion.

Measure correctly: seated, back supported, arm at heart level, right cuff, 5 min rest, two readings; home or 24-h readings. Kidney, sugar, lipids, ECG (LVH), echo, eyes.

Medical management and drugs

ACE inhibitor or ARB, calcium channel blocker, thiazide diuretic (often combined); beta-blocker for specific reasons. Target **below 130/80** (ACC/AHA 2025); ESC 2024 systolic 120–129 if tolerated.

Beta-blockers	Blunted HR, fatigue → use **RPE**
Diuretics	Dehydration, low K^+, cramps → fluids, avoid heat
Vasodilators, CCB, alpha-blockers	BP drop after exercise → **longer cool-down**, rise slowly

Assessment, ICF, goals

Assess: BP both arms (right cuff), HR, drugs and timing, target-organ disease, risk factors; activity, salt, alcohol, smoking, stress, sleep; BMI, waist; 6MWT or step test with BP before and after. Outcomes: home BP, weight, waist, 6MWT, activity minutes.

S23 teacher (chest pain + hypertension, conservative): treat as stable angina with hypertension: confirm the cause with the doctor (ECG, troponin), BP below 180/110, no rest pain; education (BP, angina, GTN, warning signs); walking at RPE 11–13 to 30 min most days, below the angina HR; light resistance breathing out; relaxation; low salt, weight control; home BP. Participation: sit for part of class, plan stairs. See IHD (p. 3).

Goals: short-term (4 weeks): correct home BP, walk 30 min most days, less salt. Long-term (3–6 months): BP at target, ≥ 5% weight loss, regular aerobic, resistance and isometric training, stress habits.

Exercise prescription (7m S18)

Aerobic	Brisk walking, cycling, swimming; moderate (talk test, RPE 12–13); ~30 min most days (ACC/AHA 2025: 75–150 min a week)
Resistance	2–3 days a week; ~50–80% 1RM, ~3 × 10; breathe out on effort
Isometric	**Handgrip 4 × 2 min at 30–40% max**, 1–3 min rest, 3 days a week (or wall squats): largest fall, ~8/4 mmHg (2023 analysis of 270 trials)
Relaxation	Stretching, yoga with slow breathing, relaxation

Warm-up and a **10-min cool-down**; progress duration, then intensity; combine with diet and weight loss.

Precautions and lifestyle

Do not start if resting BP above ~180/110 (recheck, refer) or crisis symptoms. **Stop** if systolic above ~250 or diastolic above ~115, systolic falls as work rises, chest pain, severe breathlessness, dizziness, headache. **No Valsalva**, very heavy lifts or long head-down; keep breathing in isometric holds.

Lifestyle (ACC/AHA 2025): sodium **below 2,300 mg** (ideally 1,500); lose ≥ 5% weight; DASH-style diet (vegetables, fruit, pulses, low-fat dairy); 75–150 min exercise a week; alcohol ≤ 2 (men) or 1 (women) a day; stop smoking; manage stress; treat sleep apnoea. Control cuts stroke and heart disease; lifestyle alone controls many with stage 1.

Valve disease, heart failure, cardiomyopathy, pacemaker and ICD

6(a) Full notes p. 185

Asked: 15 marks: pre- and post-implantation assessment and physio for a pacemaker and ICD in cardiomyopathy (S22). 7 marks: Phase 1 and 2 rehab after valve replacement (W21). 3 marks: NYHA grades (S21). Heart failure inside IHD cases (EF 25%, EF 40%).

Definitions and grades

Valve disease: stenosis (does not open fully) or **regurgitation** (does not close). **Rheumatic heart disease:** valve damage after rheumatic fever (untreated strep throat), mitral valve commonest. **Heart failure:** heart cannot pump enough for the body, or only at high filling pressure. **Cardiomyopathy:** dilated, hypertrophic, restrictive. **Pacemaker** stops the rate going too low; **ICD** detects dangerous fast rhythms and paces or shocks.

EF: reduced ≤ 40%; mildly reduced 41–49%; preserved ≥ 50%. **NYHA (3m S21):** I no limitation; II slight (ordinary activity → symptoms); III marked (less than ordinary → symptoms); IV symptoms at rest.

Mitral stenosis	Mid-diastolic rumble at apex; fast HR shortens filling → breathlessness, AF
Mitral regurgitation	Pansystolic at apex → axilla
Aortic stenosis	Ejection systolic, right 2nd space → neck; **symptomatic severe AS = no exercise** (fainting, sudden death)
Aortic regurgitation	Early diastolic, left sternal edge

"Open valves stenose, closed valves leak": in systole the aortic is open and mitral shut → systolic murmurs = AS or MR; diastolic = MS or AR.

Causes, pathophysiology, features

Causes: rheumatic fever (poverty, crowding), calcification (AS), endocarditis, bicuspid valve; HF from CAD, hypertension, valves, cardiomyopathy, arrhythmia, alcohol. **Jones major:** carditis, migrating polyarthritis, chorea, erythema marginatum, nodules. Pacemaker for heart block, sick sinus; ICD after arrest or EF ≤ 35% with symptoms.

Output cannot rise in exercise → fatigue; back-pressure → breathlessness, **orthopnoea**, night breathlessness, leg oedema, big liver; salt and water retention; **skeletal muscle weakness** (main target of training); arrhythmias.

Tests: echo (EF, valve area), ECG, Holter, X-ray (big heart, congestion, big left atrium in MS), BNP, exercise test or CPET (not in symptomatic severe AS), INR on warfarin.

Treatment: HFrEF four drugs (ACEi/ARB/ARNI, beta-blocker, MRA, SGLT2i) + diuretics, ICD, CRT; RHD: **benzathine penicillin**, balloon valvotomy, repair or replacement (mechanical = lifelong warfarin; tissue; TAVI); device below the left clavicle with venous leads. **Device complications:** lead displacement (arm movement), pocket haematoma, pneumothorax, inappropriate shocks.

Pacemaker and ICD (S22: 34, teacher, cardiomyopathy)

Assess: EF, NYHA, drugs, device type and settings (rate response, **ICD detection rate**), recent shocks; symptoms, daily weight, oedema, orthopnoea; HR, rhythm, BP, SpO_2, crackles; wound and device-side shoulder; 6MWT with Borg. Outcomes: 6MWT, NYHA, MLHFQ or KCCQ, HADS.

ICF: ↓ tolerance, fatigue, breathlessness; wound pain, restricted shoulder; fear of shocks → lifting, reaching overhead (blackboard), stairs, brisk walking → teaching, household, driving rules; device clinic and family (facilitators).

- **Pre-implant:** baseline shoulder range and 6MWT; explain the device, arm precautions, shock plan; breathing and mobility plan.
- **First 4–6 weeks: device-side arm not above shoulder height, no heavy lifting** (leads fix); gentle shoulder movement below shoulder height, elbow, wrist, hand from day 1; walk from day 1; breathing exercises.
- **After 4–6 weeks:** full shoulder range; exercise training.
- **ICD shock:** stop, sit or lie; one shock and well → tell the clinic; more than one or unwell → emergency.

Exercise in heart failure; valve replacement (7m W21)

Training for all stable HF (ESC 2021, Class I): better capacity and QoL, fewer admissions. Aerobic 3–5 days a week, 20–60 min, start RPE 11–13, slow progress, intervals for the unfit; light resistance 2–3 days, 10–15 reps; IMT. **ICD:** train ≥ **10 beats below the detection rate**. Fixed-rate pacemaker or chronotropic incompetence: use RPE and BP.

Valve replacement (sternotomy, as CABG): Phase 1 (5–7 days): pre-op teaching; deep breaths, supported huff, sit out day 0–1; walking room → corridor, stairs; **RPE below 13, HR ≤ rest + 30**; watch AF and heart block (after aortic surgery); sternal precautions. Phase 2 (8–12 weeks): test first, start low; aerobic 3–5 days to 30–60 min at RPE 11–14; leg resistance early, arms after ~6–8 weeks. **Mechanical valve: warfarin** (gentle handling, fall prevention, no contact sport or vigorous percussion); endocarditis prevention (teeth, skin); MS leaves pulmonary hypertension and AF (slower); **TAVI**: no sternal limits, groin care, frailer patients.

Precautions

Devices: no shortwave or microwave diathermy; no TENS, IFT or stimulation near the device or across the chest without device-team advice; no ultrasound over it; no strong magnets. **No exercise if:** decompensated HF (breathless at rest, new orthopnoea, **2 kg gain in 3 days**, more oedema), symptomatic severe AS, uncontrolled arrhythmia, rest chest pain, recent shocks. HCM: avoid dehydration and heavy straining. **Educate:** daily morning weight, salt and fluid advice, never miss penicillin in RHD, carry the device card, phone away from the pocket.

Congenital heart disease: tetralogy of Fallot, ASD, VSD, PDA

6(a) Full notes p. 192

Asked: 3 marks only: clinical features of tetralogy of Fallot (W16, S26) and its X-ray features (S21). 7 marks possible on physiotherapy after paediatric heart surgery.

Definition and types

Congenital heart disease: structural defect of the heart or great vessels present at birth; the commonest birth defect.

Acyanotic, left-to-right shunt	**ASD, VSD, PDA**: too much lung blood → breathlessness, chest infections, poor growth, later pulmonary hypertension
Acyanotic, obstructive	Coarctation, aortic and pulmonary stenosis: pressure overload
Cyanotic (right-to-left)	Blue blood reaches the body

The 5 T's

Tetralogy of Fallot, **T**ransposition of the great arteries, **T**ricuspid atresia, **T**runcus arteriosus, **T**otal anomalous pulmonary venous return.

Causes: mostly none; Down syndrome (AV septal defect), family history; maternal rubella (PDA), diabetes, alcohol, drugs; prematurity (PDA).

Tetralogy of Fallot

Four parts (one fault: outflow septum displaced forwards): **large VSD, overriding aorta, right ventricular outflow obstruction (pulmonary stenosis), right ventricular hypertrophy**. Obstruction pushes blue blood through the VSD into the aorta → cyanosis (worse with worse obstruction).

Squatting (knee–chest) kinks leg arteries, raises systemic resistance, so less blood crosses right-to-left and more goes to the lungs.

Clinical features (3m W16, S26): cyanosis from the first months, worse with crying or effort; **tet (hypercyanotic) spells** after waking, feeding or crying (deep cyanosis, fast breathing, irritability, fainting); **squatting** after exertion; **clubbing**; poor growth, exercise intolerance; harsh ejection systolic murmur at the left upper sternal edge; single S2.

X-ray of TOF (3m S21) and tests

- **Boot-shaped heart** (coeur en sabot): upturned rounded apex (RVH).
- **Concave pulmonary bay** (small pulmonary artery segment).
- **Oligaemic lungs** (dark, few vessel markings).
- Heart size usually normal; right-sided aortic arch in some.

Echo (main test), ECG (RVH), pulse oximetry (newborn screening), Hb (raised in chronic cyanosis), catheter, CT or MRI.

Other lesions

VSD: loud pansystolic murmur, left lower sternal edge; sweating on feeds, poor weight gain. **ASD:** often silent till adulthood; fixed split S2. **PDA:** continuous "machinery" murmur below the left clavicle, bounding pulses. **Coarctation:** weak delayed femoral pulses, high BP in arms. **Eisenmenger:** long-standing left-to-right shunt → pulmonary hypertension → shunt reverses → late cyanosis.

Management and complications

TOF: complete repair in infancy (VSD patch, relieve outflow); Blalock–Taussig shunt in some small infants. **Tet spell:** calm, **knee–chest position**, oxygen, call for help. VSD, ASD, PDA: many close alone; device or surgical closure; drugs for PDA in preterms. Transposition: arterial switch; single ventricle: Glenn, Fontan.

Complications: heart failure, failure to thrive, chest infections; polycythaemia, stroke, brain abscess; endocarditis; Eisenmenger; after repair arrhythmias, leaking pulmonary valve; developmental delay.

Physio after paediatric heart surgery

Assess: operation, day, lines, drains, pacing wires, ventilation, **the child's target SpO_2** (may be lower than normal); rate, recession, expansion, auscultation, secretions; pain (age scale), development, posture, shoulder; parents' worries.

- Head of bed up, side-lying, regular turns; **no head-down**.
- Airway clearance only if secretions: gentle manual techniques, assisted huff, suction; **breathing games** (bubbles, windmills, party blowers).
- Early sitting, standing, walking, play.
- **Sternal care:** lift with a scoop under the bottom and back, **not under the armpits**; no pulling on arms.
- Shoulder and trunk through play; development activities; parent teaching; later normal active play, sport as the cardiologist advises.

Goals: clear chest, sitting and walking within days, confident parents → milestones, active play.

Precautions and parent education

Don't chase normal SpO_2 in a cyanotic heart. Pulmonary hypertension: avoid crying, straining, desaturation; pre-oxygenate before suction. Secure wires and drains; gentle handling on anticoagulants; avoid dehydration (thick blood). **Parents:** lifting after sternotomy; breathing games; recognise a tet spell and use knee–chest; dental hygiene, vaccines, follow-up; avoid over-protection.

Pneumonia, lung abscess and atelectasis

6(c). One ventilated long case (W24).

Full notes p. 198

Asked: 15 marks: ventilated atypical pneumonia on CMV, FiO_2 0.6, PEEP 10 (W24); RLL pneumonia inside the S20 COPD ICU case. Short notes possible on lung abscess and atelectasis.

Pneumonia

Infection of lung tissue: alveoli fill with fluid and cells (**consolidation**). Community- or hospital-acquired (≥ 48 h after admission), ventilator-associated, aspiration; lobar, broncho-, interstitial; pneumococcus commonest.

Features: fever, cough, sputum (rusty in pneumococcal), pleuritic pain, fast breathing, confusion in the elderly; **dull percussion, bronchial breathing**, crackles, ↑ vocal resonance. X-ray: white with **air bronchograms**, **no volume loss**.

CURB-65 (1 point each): new Confusion; Urea over 7 mmol/L; RR ≥ 30; BP systolic below 90 or diastolic ≤ 60; age ≥ 65. 0–1 home, 2 hospital, 3–5 consider ICU.

Antibiotics, oxygen (94–98%, or 88–92% if CO_2 risk), fluids. **Early consolidation is solid exudate**: percussion and drainage don't clear it.

Current (BTS 2009, Cochrane 2022): airway clearance not routine in uncomplicated pneumonia; **sit out ≥ 20 min within 24 h**, then walk. Write: "early mobilisation and positioning for all; ACBT only for retained sputum or underlying lung disease".

- Low O_2: in one-sided pneumonia, **good lung down**.
- Resolving sputum: ACBT 1–2 times a day, affected lung up, humidification. Pleuritic pain: supported breathing and cough, analgesia first.

Ventilated pneumonia (W24: 35, both lungs, CMV, FiO_2 0.6, PEEP 10, 3 days)

Read the settings: FiO_2 0.6 and PEEP 10 are high: stiff, poorly oxygenated lungs, near ARDS; **losing PEEP** (disconnection) collapses alveoli and drops SpO_2.

ICF: poor oxygenation, stiff low-volume lungs, collapse and VAP risk, ICU-acquired weakness, joint stiffness, pressure and DVT risk, sedation → fully dependent → off work; young and fit, ICU team and family (facilitators); tubes and sedation (barriers).

- **Safety check each session:** SpO_2, FiO_2 and PEEP trend, HR, BP, sedation, doctor's plan. At FiO_2 0.6, PEEP 10: upper limit, so **in-bed treatment first**.
- **Head up 30–45°** (aspiration, VAP); regular side-lying; help with prone if ordered.
- Little sputum: suction only when needed with a **closed in-line system** (keeps PEEP); no bagging off the ventilator at this PEEP (ventilator hyperinflation instead).
- Passive → active-assisted limb movement; ankle positioning against foot drop; pressure care.
- As FiO_2 falls and sedation lightens: active exercise, edge of bed, bed cycling, standing and walking on the ventilator. After extubation: deep breathing, ACBT, walking, strengthening, IMT.

Precautions: sepsis (low BP, confusion), worsening hypoxia, fatigue; infection control.

Lung abscess

Localised necrosis forming a **pus-filled cavity**. Usually **aspiration** (alcohol, seizures, drowsiness, poor teeth, swallowing problems): **posterior segment of right upper lobe, superior segment of right lower lobe**; also Staph or Klebsiella pneumonia, blocked bronchus. Swinging fever, **copious foul sputum**, haemoptysis, weight loss, clubbing. X-ray: **thick-walled cavity with an air–fluid level**. Weeks of antibiotics; drainage or surgery if unresolved.

Physio: once it drains into a bronchus: **gravity-assisted drainage of that segment with ACBT, 2–3 times a day**; strong huffs and coughs so pus is cleared, not spilled into other lobes; breathing exercises, walking, nutrition, mouth hygiene. Stop with haemoptysis; suction ready in weak or drowsy patients.

Atelectasis (collapse)

Incomplete expansion of part or all of a lung. **Types:** obstructive or resorption (mucus plug, tumour, foreign body), compressive (effusion, pneumothorax), passive (shallow post-op breathing), adhesive (surfactant loss), cicatrisation (scarring). Small: silent; large: breathless, low SpO_2, ↓ sounds. X-ray: opacity **with volume loss**: trachea pulled towards, raised hemidiaphragm, shifted fissures, crowded ribs.

Physio: deep breaths with **3-s hold** (sustained maximal inspiration), thoracic expansion to the area, **5–10 every hour**; affected area uppermost; humidification, ACBT, suction for a plug; **early mobilisation** (the best deep-breathing exercise); CPAP, manual hyperinflation or IPPB as prescribed in weak or ventilated patients; incentive spirometer as add-on. No positive pressure with an undrained pneumothorax; compressive collapse needs drainage first.

X-ray patterns

	Opacity, volume	Trachea
Consolidation	White, air bronchograms; volume normal	Central
Collapse	White; volume lost	**Pulled towards**
Large effusion	White, meniscus	Pushed away
Tension pneumothorax	Black, no markings	Pushed away

Neonatal and paediatric respiratory conditions

6(d)

Full notes p. 203

Asked: 7 marks: precautions and contraindications for postural drainage in a neonate (W18); paediatric asthma rehab (S18, see Asthma (p. 22)). 3 marks: signs of respiratory distress in children (W24).

Neonatal chest physio: precautions (7m W18)

- **Only when needed** (retained secretions, collapse, blocked tube): never routine.
- **Minimal handling:** cluster care with nurses; short sessions; rest between.
- **Monitor continuously:** SpO_2, HR, breathing, colour, temperature; keep warm.
- **Timing:** before feeds or well after (vomiting, aspiration).
- **No head-down tilt:** flat or head-up; 30° head-up gave less reflux (Cochrane).
- **Gentle techniques:** fragile ribs (osteopenia) and brain vessels; vigorous percussion linked to injury.
- **Suction:** right catheter and pressure, pre-oxygenate, short; watch bradycardia, desaturation.

Contraindications

Unstable baby (low BP, arrhythmia, desaturates with handling); recent or extending **intraventricular haemorrhage**, very preterm in first days; **persistent pulmonary hypertension**; undrained pneumothorax; rib fractures, osteopenia, low platelets (no percussion or vibration); raised ICP; just after feeds; post-op without surgical clearance. **Modified techniques:** positioning, gentle vibration, assisted breathing, suction.

Respiratory distress in children (3m W24)

60, 50, 40

Fast breathing (WHO, full minute, calm child): **≥ 60** under 2 months; **≥ 50** at 2–11 months; **≥ 40** at 1–5 years.

- **Work of breathing:** nasal flaring; **chest in-drawing** (subcostal, intercostal, suprasternal); tracheal tug; neck muscles; head bobbing; see-saw breathing.
- **Noises: grunting** (breathing out against a closed glottis to keep alveoli open), stridor, wheeze.
- **Effects:** cyanosis, low SpO_2, fast HR, cannot feed or talk, sweating.
- **Danger (tiring):** drowsy, apnoea, slowing breathing in an exhausted child, silent chest: urgent help.

Neonatal conditions

- **RDS (premature):** no surfactant → alveoli collapse; grunting, flaring, recession, cyanosis within hours; X-ray **ground glass** + air bronchograms. Antenatal steroids, CPAP, surfactant. **Physio:** not routine (stiff, not wet lungs); positioning, developmental care, minimal handling.
- **Meconium aspiration (term or post-term):** airway block with air trapping and pneumothorax, chemical pneumonitis, surfactant loss, pulmonary hypertension; patchy shadows, over-inflation. Supportive care; routine tracheal suction of non-vigorous babies no longer advised (ILCOR). **Physio:** clearance only when secretions retained; minimal handling.
- **Chronic lung disease of prematurity (BPD):** oxygen need at 36 weeks post-conception; wheeze, infections, poor growth. Clearance during infections, positioning, developmental care (tummy time), home-oxygen education.
- **Paediatric ARDS:** after pneumonia, sepsis, drowning, trauma: positioning (prone as the ICU directs), careful clearance, minimal disconnection, early rehab.

Pneumonia, pneumonitis, bronchiolitis

Pneumonitis: aspiration of milk or stomach contents, **kerosene ingestion** (classic in India), infection. **Pneumonia:** major killer under 5 in India. **Bronchiolitis** (viral, under 2): wheeze, crackles, poor feeding.

Physio: bronchiolitis: conventional chest physio (percussion, vibration) does not help and forced expiration causes vomiting (Cochrane 2023); slow passive expiration may help a little in moderate cases. Pneumonia: no routine clearance; positioning, mobility, play; clear only retained secretions. Aspiration: head-up positioning, feeding advice with speech therapy.

Answer skeleton (W18): indications (not routine) → precautions → contraindications → modified techniques.

Cystic fibrosis

6(b), 6(d)

Full notes p. 208

Asked: not asked 2016–2026; possible 3- or 7-mark short note. Also linked to bronchiectasis and paediatric questions.

Definition and cause

Cystic fibrosis: inherited, **autosomal recessive** (both parents carriers); **CFTR gene** mutation → chloride and water do not move across cell surfaces → **thick sticky secretions** in lungs, pancreas, liver, gut and reproductive tract; very salty sweat.

Features and diagnosis

- **Lungs:** thick mucus, chronic infection (*Staph aureus*, later *Pseudomonas*), progressive bronchiectasis, haemoptysis, pneumothorax, clubbing.
- **Gut, pancreas:** meconium ileus at birth, fatty bulky stools, malabsorption, poor weight gain; CF-related diabetes later.
- **Other:** nasal polyps, sinusitis, liver disease, male infertility, salt loss in heat, low bone density.
- **Diagnosis:** newborn screening; **sweat chloride ≥ 60 mmol/L** (30–59 intermediate); genetic test. Under-diagnosed in India: found late in children with repeated chest infections and poor growth.

Medical points for physio

CFTR modulators (elexacaftor–tezacaftor–ivacaftor) correct the protein in many patients: big gains in lung function and survival where available. **Mucoactives:** dornase alfa, hypertonic saline before clearance. Antibiotics (oral, inhaled, IV), pancreatic enzymes, high-calorie diet, salt supplements, transplant. **Order:** bronchodilator → hypertonic saline or dornase → **airway clearance** → inhaled antibiotic.

ICF: thick secretions, infections, ↓ lung function and exercise tolerance, poor growth, posture and back pain → sport, play, daily treatment burden → school, work, social life (cross-infection rules); CF centre and family (facilitators), cost of drugs (barrier); adolescent adherence.

PT management (lifelong)

- **Infants:** modified postural drainage **without head-down**, gentle percussion or vibration, assisted autogenic drainage, PEP mask, active play (bouncing on a ball, rolling).
- **Children:** PEP mask, flutter or Acapella, blowing games, ACBT once they can huff.
- **Teenagers and adults:** ACBT, autogenic drainage, PEP or oscillating PEP, chest-wall oscillation vest: the one they will actually do.
- **Dose:** once or twice daily, more in infections; review technique regularly.
- **Exercise and sport** at all ages: fitness, mucus clearance, bones, QoL; it does **not replace** airway clearance for most.
- Posture, thoracic mobility, back pain; **pelvic floor** (leakage with cough common in women); inhaler technique; NIV in advanced disease; rehab before and after transplant.

Precautions

Cross-infection: people with CF must be kept **apart from each other** (no shared rooms, gyms or groups); clean equipment; hand hygiene. Haemoptysis: stop percussion and positive-pressure devices, report. Pneumothorax: no positive pressure until drained. Heat: extra fluids and salt. CF diabetes: check sugar around exercise. No head-down in infants or with reflux.

Exam points

Autosomal recessive, CFTR, thick secretions, sweat chloride ≥ 60; lifelong daily airway clearance adapted to age; exercise; strict cross-infection control.

Flail chest, scoliosis and kyphosis

6(j)

Full notes p. 211

Asked: not alone in 2016–2026; rib fractures in the W17 ventilated electrician (burns case). Possible 3- or 7-mark short notes on flail chest, scoliosis or kyphosis.

Flail chest

3 ribs, 2 breaks, 1 floating segment

A chest-wall segment with no bony connection: usually **3 or more neighbouring ribs each broken in 2 or more places**. **Paradoxical movement:** sucked in on inspiration, pushed out on expiration.

Causes: road accidents, falls from height, crush; lesser falls in the elderly. **Features:** severe pain, shallow breathing, poor cough, low O_2; **lung contusion** is the main cause of hypoxia (often worsens over 1–2 days); haemo- or pneumothorax. X-ray/CT: multiple fractures, contusion, pleural air or fluid.

Medical: pain control is the key (epidural, paravertebral, nerve blocks); oxygen; NIV in some; ventilation for failure (positive pressure splints from inside); rib fixation; chest drain.

Flail chest: physio

- **Pain:** treat after analgesia; supported positions; hand or pillow splint for cough.
- **Volumes:** sit upright; deep breaths with 3-s hold, **5–10 every hour**; expansion avoiding pressure on fractures.
- **Secretions:** ACBT with gentle huffs, supported cough, suction if needed.
- **Ventilated:** positioning, suction, careful chest handling, early mobilisation when stable.
- Early sitting and walking; shoulder range on the injured side.

Precautions: no percussion, vibration or shaking over fractures; no positive pressure with an undrained pneumothorax; watch worsening hypoxia, haemoptysis, spine and head injury.

Scoliosis

Sideways spinal curve with vertebral rotation, **Cobb angle ≥ 10°**. **Types:** idiopathic (adolescent commonest, mostly girls), congenital, neuromuscular (CP, muscular dystrophy, SMA, SCI), degenerative, secondary (after thoracoplasty or empyema).

Features: uneven shoulders and waist, prominent scapula, **rib hump on forward bending (Adam's test)**, scoliometer. Large thoracic, early-onset and neuromuscular curves stiffen the chest and limit lung growth: **restrictive pattern**, night hypoventilation.

Medical (SOSORT 2016): observation, scoliosis-specific exercise, bracing for moderate curves in growing children, fusion for large or progressive curves.

Physio: scoliosis-specific exercises (Schroth, SEAS): self-correction, stabilisation, **breathing into the concave side**; thoracic expansion to the concave side, rib mobility, IMT, aerobic fitness; neuromuscular: seating, clearance, assisted cough, night ventilation; after fusion: breathing, early walking, log-roll, spinal precautions. Follow brace schedules; no forceful mobilisation of progressive or fused curves.

Kyphosis

Excess thoracic forward curve (normal ~20–40°). **Postural** (flexible); **Scheuermann's** (rigid, adolescents, ≥ 5° wedging in 3 vertebrae); **osteoporotic** (wedge fractures); **ankylosing spondylitis** (rigid, fused costovertebral joints → very ↓ expansion); **Pott's spine** (angular gibbus); congenital.

Features: rounded back, forward head, back pain; ↓ expansion and vital capacity in severe or rigid curves; exertional breathlessness; falls (osteoporosis).

Physio: spinal extension and back-extensor strengthening, chest stretches, scapular retraction, posture awareness; lateral costal and diaphragmatic breathing; **ankylosing spondylitis: daily chest expansion and spinal mobility**; osteoporosis: weight-bearing, balance, falls prevention, **no loaded flexion**; bracing for Scheuermann's. Careful manual techniques on a rigid, fragile thorax.

Mechanical ventilation: indications, modes, CPAP and PEEP, complications, weaning

In 7 papers, including a ventilated long case (S23).

Full notes p. 216

Asked: an ICU question in every paper since 2023. **3 marks:** respiratory complications, long-term complications, CPAP vs PEEP, three indications. **7 marks:** modes (with ICU physio role), weaning criteria. **15 marks:** GBS on SIMV (S23), with settings to interpret.

What it is

A machine moves gas into the lungs with **positive pressure** when the patient cannot breathe well enough; it buys time, it cures nothing. **Invasive** (ETT, tracheostomy) or **non-invasive** (mask CPAP or BiPAP).

Term	Meaning	Usual adult
FiO_2	Oxygen fraction (0.21 room air, 1.0 = 100%)	Lowest that keeps SpO_2 on target
VT	Tidal volume	**6–8 mL/kg** predicted weight; 6 in ARDS
Rate	Machine breaths/min	12–20
PEEP	Pressure kept at end-expiration	~5 cmH_2O; higher in ARDS
Plateau	Alveolar pressure in a pause	**≤ 30 cmH_2O**
I:E	Inspiration : expiration	~1:2; longer expiration in COPD, asthma
Trigger	Patient effort that starts a breath	Flow or pressure
Pressure support	Added to each patient breath	5–8 near end of weaning

Minute ventilation = VT × rate (~5–8 L/min).

Indications (3m S24)

- **Type 1 failure** (PaO_2 below ~60 despite oxygen): pneumonia, ARDS, pulmonary oedema.
- **Type 2 failure** (high CO_2 with acidosis): COPD when NIV fails, severe asthma, overdose.
- **Pump failure:** GBS, myasthenic crisis, high SCI, flail chest, exhausted muscles.
- **Airway protection:** GCS ≤ 8, head injury, stroke, poisoning, airway burns.
- **Circulation:** cardiac arrest, cardiogenic shock. **Planned:** surgery, heavy sedation, ICP control.

Warning signs: RR above ~35, exhaustion, falling SpO_2 despite oxygen, rising CO_2 with falling pH, drowsiness, falling VC in neuromuscular disease.

Modes (7m S17, W20)

CMV	Set volume or pressure at a set rate; patient cannot add breaths: deeply sedated or paralysed (W24)
Assist-control	Set breath + minimum rate; every triggered breath is a full machine breath: rests tired muscles; fast breathing → air trapping
SIMV	Set machine breaths timed with effort; spontaneous breaths between (usually with PS): partial support (S20, S23, S24)
Pressure control	Set pressure and time; volume varies with stiffness: ARDS
Pressure support	Patient starts every breath; machine adds pressure: comfortable; weaning to 5–8
CPAP	One continuous pressure, no machine breaths: weaning, sleep apnoea, cardiogenic oedema, post-op collapse
BiPAP	IPAP on inspiration, EPAP on expiration by mask: COPD acidosis, post-extubation, neuromuscular at night

Why the mode matters to physio: CMV or A/C at a high rate = muscles resting and wasting → positioning, passive movement, clearance. SIMV, PSV, CPAP = patient breathing → active exercise, breathing exercises, IMT; watch fatigue.

CPAP vs PEEP (3m W21)

	CPAP	PEEP
What	A **mode**: pressure through the whole breath	A **setting**: pressure at end-expiration, added to any mode
Breaths	All spontaneous	Machine or supported breaths on top
How	Mask, nasal, ventilator	Ventilator (PEEP valve on a bag)
Use	Sleep apnoea, cardiogenic oedema, post-op collapse, weaning	Almost every ventilated patient (~5); higher in ARDS

CPAP = PEEP for a patient who breathes alone

Effects: ↑ FRC, alveoli and small airways kept open, better oxygenation and compliance, less collapse; CPAP also ↓ work of breathing and LV load. **High levels:** ↓ venous return and BP, barotrauma, ↑ ICP; mask sores, air swallowing. **Auto-PEEP:** trapped air when expiration is incomplete (COPD, asthma): ↓ BP, harder triggering.

Complications (3m S22, W21)

- **Lungs:** ventilator-induced lung injury (over-stretch, high pressure, repeated opening and closing), pneumothorax, **VAP**, collapse, O_2 toxicity, auto-PEEP, fighting the ventilator → head up, turns, clearance; report sudden high pressure or falling SpO_2.
- **Airway:** tube blocked, displaced or pulled out, right main bronchus intubation; laryngeal and tracheal damage, later **stenosis**; sinusitis → humidification, suction, secure the tube on every turn.
- **Circulation:** ↓ venous return and output (hypotension), fluid retention → check BP.
- **Muscles:** diaphragm wasting, **ICU-acquired weakness**, contractures, pressure injuries, DVT → mobilisation, ROM, positioning, IMT.
- **Other:** stress ulcers, delirium, poor sleep, anxiety, PTSD → orientation, communication, day–night routine.

Long-term (W21): tracheal stenosis or softening, repeated VAP, diaphragm wasting and ventilator dependence, weakness, contractures, pressure injuries, psychological effects. Ventilation over ~1–2 weeks → **tracheostomy** (comfort, less sedation, easier weaning).

Weaning (7m W19)

Daily readiness screen: cause improving; **SpO_2 ≥ 90% on FiO_2 ≤ 0.4–0.5 and PEEP ≤ 5–8** (P/F above ~150–200); stable HR and BP, little or no BP support, no ischaemia; awake, follows commands, can cough; breathing alone, **pH ≥ 7.25**, few secretions, no fever, normal electrolytes.

RSBI (rate ÷ VT in L)	**below 105**
MIP	stronger than −20 to −30 cmH_2O
Vital capacity	over ~10 mL/kg
Minute ventilation	below ~10 L/min

RSBI: a tiring patient breathes fast and shallow: 35 breaths of 0.25 L = 140 (likely fail); 20 of 0.5 L = 40 (likely succeed).

Spontaneous breathing trial: 30–120 min on low pressure support (5–8) or T-piece (ACCP/ATS 2017: low PS). **Stop if:** RR above 35 for over 5 min; SpO_2 below 90%; HR above 140 or change over 20%; systolic above 180 or below 90; agitation, sweating, accessory muscles, see-saw breathing. Passed → extubation (cough and secretion check); NIV after extubation in high risk.

Outcome (2007): simple (first trial); difficult (up to 3 trials or 7 days); prolonged (more than 3 trials or 7 days).

Physio in weaning: keep airways clear (positioning, manual techniques, hyperinflation, suction); **sit upright for trials**; **IMT** for those who fail; early mobilisation; after extubation deep breathing, huffing, supported cough, walking.

Reading settings in a case

Note mode, FiO_2, PEEP, set and actual VT, rate, peak pressure, SpO_2, ABG, and the trend (**rising FiO_2 or PEEP = worse lungs**).

S23 GBS: SIMV, PEEP 6, FiO_2 0.4	Moderate support for weak muscles, lungs fine: active treatment as strength returns
S24 ARDS: SIMV, PEEP 5, FiO_2 0.4	Mild–moderate support: in- and out-of-bed activity if stable
W24 pneumonia: CMV, PEEP 10, FiO_2 0.6	High support, sedated: limit for mobilising; avoid disconnection (closed suction)

Alarms: high pressure = cough, secretions, biting, kinked tube, bronchospasm, fighting, pneumothorax; low pressure or volume = disconnection or cuff leak. If unclear or SpO_2 falls: stop and call; never silence an alarm you don't understand.

Skeletons: modes: definition → sketch traces → table → why it matters to physio. Weaning: definition → readiness → RSBI, MIP → SBT and stop criteria → simple, difficult, prolonged → physio role.

Physiotherapy in the ICU: assessment, safety, equipment and ventilated cases

In 9 papers; 4 ventilated long cases (S20, S23, S24, W24).

Full notes p. 223

Asked: 4 ventilated long cases (S20 COPD + pneumonia, S23 GBS, S24 ARDS, W24 pneumonia). **7 marks:** modes and ICU physio role, short-term goals on a ventilator, physio on a ventilator, ICU equipment, acute ARDS physio, chest physio on SIMV.

ICU assessment (write this first)

Chart: diagnosis, course, days ventilated, trend; ventilator settings; HR, rhythm, BP and **MAP**, SpO_2, RR, temperature, ICP; ABG and X-ray (collapse, consolidation, effusion, pneumothorax, tube position); drugs (sedation, analgesia, BP support, paralysis); Hb, platelets, clotting, lactate, electrolytes, sugar; lines and drains (femoral lines limit hip flexion).

Bedside: GCS; **RASS** (−5 to +4); delirium (CAM-ICU); pattern, ventilator synchrony, chest movement, auscultation, secretions, cough on suction; ROM, tone, contractures, skin, oedema; when awake **MRC sum score** (0–60; **below 48 = ICU-acquired weakness**), bed mobility, sitting balance, **ICU Mobility Scale** (0 lying, 10 walking); pain (behavioural scale).

ICU monitoring

ECG, HR	60–100; suction can slow the heart (vagal); new arrhythmia = stop
Arterial line, MAP	**≥ 65** or as set; turning, sitting, bagging drop BP
SpO_2	92–96% (88–92% if CO_2 retainer); falls with suction, disconnection, effort
RR, graphics	12–20; fast rate or fighting = distress
End-tidal CO_2	35–45; confirms tube in airway
ICP	5–15, treated above ~22; CPP (MAP − ICP) 60–70; suction, coughing, head-down, neck flexion raise it
Urine	≥ 0.5 mL/kg/h
ABG	pH 7.35–7.45, $PaCO_2$ 35–45, PaO_2 80–100

Safety before mobilising (Hodgson 2014)

Traffic lights: **green** go; **yellow** discuss; **red** no active mobilisation without senior agreement.

- **Respiratory green:** tube or trach (not a barrier); **FiO_2 ≤ 0.6, SpO_2 ≥ 90%, RR ≤ 30, PEEP ≤ 10**. Red: SpO_2 below 90%, oscillation, prone.
- **Cardiovascular green:** MAP above target with little support; rate below 120 in a stable tachyarrhythmia. Red: hypertensive emergency drugs, symptomatic low MAP, bradycardia needing drugs, rate above 150, active ischaemia.
- **Neurological green:** RASS −1 to +1, follows commands. Red: RASS above +2 or below −2, raised ICP being treated, uncleared spine, uncontrolled seizures.
- **Other red:** active bleeding; unstable pelvic, spinal or long-bone fracture; big open wound; femoral sheaths.

Near the limit in several areas (W24: FiO_2 0.6, PEEP 10, SpO_2 90%) → decide with a senior ICU doctor. **Stop:** SpO_2 below ~88–90% not recovering; sharp HR change, new arrhythmia, MAP below target; RR above ~35, fighting, distress; tube or line displaced; fall.

ICU equipment (7m W24)

Ventilator	Read settings; don't disconnect at high PEEP or FiO_2
ETT, tracheostomy	Secure on turning; suction; speaking valve later
Monitor	ECG, SpO_2, BP, RR, temperature, $EtCO_2$: before, during, after
Arterial line	Beat-by-beat BP, ABG; no kinking; wrong if transducer level changes
Central line	Drugs, CVP; don't pull on turns
Chest drain	Bottle below chest; swing, bubbling
NG tube, catheter	Pause feeds before head-down; head up 30–45°; bag below bladder
Infusion pumps	High BP-support doses limit mobilisation
Suction, O_2, humidifier, Ambu bag, crash trolley	Check suction pressure; bag at every ventilated bed
ICP monitor, EVD	Clamp as instructed before moving; no head-down
IABP, ECMO, dialysis	Femoral cannulas: in-bed exercise only unless agreed
Mattress, compression, tilt table, bed cycle	Pressure, DVT, early rehab

Physio for the ventilated patient (S17, S21, S26; goals W17)

Short-term goals (W17): clear airways and expanded lungs, better oxygenation; prevent VAP, collapse, aspiration; keep joint range, prevent contractures, pressure injuries, DVT; limit wasting, start active exercise when safe; support weaning; orientation and communication.

- **Check** settings, vitals, ABG, X-ray, sedation, safety; time with analgesia, away from feeds.
- **Position:** head of bed **30–45°**; side-lying, affected area up; turns every **2–4 h**; sit up early.
- **Manual techniques on expiration:** shaking, vibration, percussion for retained secretions (not with fractures, low platelets, instability).
- **Hyperinflation** by bag or ventilator to re-open lung and move secretions, then **suction**.
- Awake on SIMV: deep breaths with a hold on own breaths; **IMT** in weaning.
- **Limbs:** passive ROM when sedated → active-assisted → active; ankle and hand splints.
- **Mobilisation ladder:** rolling → sitting up → edge of bed → standing → marching → chair → walking with the ventilator.
- **Reassess** breath sounds, SpO_2, secretions, pressures; record.

Precautions: high PEEP or FiO_2 → closed suction, ventilator hyperinflation; undrained pneumothorax → no positive pressure; unstable BP → gentle only; raised ICP → head up 30°, neutral neck, short sessions, no head-down; low platelets → no percussion; recent lung surgery → no deep suction or high pressure near the stump.

GBS on SIMV (S23: 45, teacher, PEEP 6, FiO_2 0.4)

Immune attack on peripheral nerves → ascending symmetrical flaccid weakness; ventilated for **weak muscles, not lung disease** (modest settings).

- **Impairments:** weak breathing muscles, low VC (**20/30/40 rule**: VC below 20 mL/kg, MIP weaker than −30, MEP below 40); weak cough, secretions, collapse risk; **swallowing weakness**; **autonomic instability** (suction → bradycardia); flaccid paralysis, areflexia, sensory loss, nerve pain; ankle contracture, pressure, DVT risk; anxiety, cannot communicate.
- **Physio:** 2–4-hourly turns, head up, hyperinflation with vibrations, suction (pre-oxygenate, watch HR); track VC and MIP for weaning; tracheostomy common; weaning with more spontaneous breathing, short non-fatiguing IMT sets, upright; gentle passive ROM, ankles 90°, functional hands; then active work **without fatigue** (overwork weakens recovering muscle); pressure care, tilt table, communication board, reassurance.

ARDS (S24 case; acute phase 7m S25)

Berlin 2012: within 1 week of an insult; bilateral opacities; not explained by heart failure; P/F on PEEP ≥ 5: **mild 201–300, moderate 101–200, severe ≤ 100**. (2024 global definition adds high-flow oxygen and SpO_2/FiO_2.)

Medical: lung-protective ventilation (VT 6 mL/kg, plateau ≤ 30), PEEP, careful fluids, **prone ≥ 16 h a day if P/F below 150** (PROSEVA). **Complications:** lung injury, pneumothorax, VAP, ICU-acquired weakness, contractures, pressure injuries (prone), DVT, delirium, post-intensive care syndrome.

Acute physio: minimal clustered handling; **no disconnection** (closed suction, ventilator hyperinflation, no high pressures); head up, turns; help with safe prone turns (eyes, face, shoulders, knees, lines); passive ROM and positioning. As she improves (FiO_2 0.4, PEEP 5 = green): active exercise, edge of bed, standing and walking on the ventilator, IMT, then graded exercise and follow-up rehab.

Evidence and exam plan

Early mobilisation is safe within criteria: physio during sedation breaks → more independent at discharge (59% vs 35%), less delirium (Schweickert 2009); TEAM 2022: more aggressive early mobilisation did not help survival and caused more adverse events: dose to suit the patient. **ABCDEF bundle:** pain, awakening and breathing trials, sedation choice, delirium, early mobility, family.

Ventilated case skeleton: read the case and settings (low, moderate, high support) → impairments or ICF (chest, muscles and joints, circulation and skin, consciousness, communication) → complications → physio: safety check → positioning → clearance → limbs → mobilisation ladder → weaning → after extubation → precautions.

Suctioning and manual hyperinflation

In 7 papers; hazards of suctioning is a favourite 3-mark question.

Full notes p. 231

Asked: 3 marks (5 papers): hazards or complications of suctioning. 7 marks: suctioning and bagging (manual hyperinflation): indications, procedure, precautions (S22, S24).

Types

Suctioning: removing airway secretions with a catheter on negative pressure when the patient cannot cough them out.

- **Open endotracheal or tracheostomy:** disconnect, sterile catheter; loses PEEP and oxygen.
- **Closed (in-line):** catheter in a sealed sleeve; **keeps PEEP and FiO_2**, less cross-infection: high PEEP, FiO_2, infection.
- **Nasopharyngeal or nasotracheal:** no artificial airway; lubricated catheter or nasal airway. **Oropharyngeal:** Yankauer.
- **Shallow** (to tube length) routinely; **deep** (to resistance, pull back 1 cm) only if shallow fails (AARC 2022).

Indications (only when needed): secretions seen or heard; coarse crackles over trachea; saw-tooth flow graph, rising peak pressure or falling VT; falling SpO_2 from secretions; weak cough after other clearance; aspiration; sputum sample.

Procedure (open)

> **15 s / −200 mmHg / half the tube**

Pass **≤ 15 s**; pressure **no stronger than −200 mmHg** adults (−120 children, newborns); catheter **under half** the tube's inner width (under 70% newborns): 8.0 mm tube → **12 Fr**.

1. Assess and explain (even if sedated); vitals, SpO_2.
2. Hand hygiene, apron, eye protection, sterile glove; check pressure; choose catheter.
3. **Pre-oxygenate** 30–60 s if SpO_2 drops with suction.
4. Insert **without suction** to planned depth.
5. Withdraw **with suction**, rolling the catheter, in under 15 s.
6. Reconnect at once; let SpO_2 and HR recover; usually ≤ 2–3 passes.
7. Mouth last with a separate catheter (never mouth → tube).
8. Reassess and record sounds, SpO_2, secretions, blood, tolerance.

No routine saline down the tube; thick secretions need better humidification and hydration. Closed system: advance, suction on withdrawal, flush.

Hazards (3m W16, S17, S18, W19, S26)

> **Low O_2, slow heart, high head pressure**

- **Hypoxaemia** (oxygen sucked out, PEEP lost): pre-oxygenate, short passes, closed suction.
- **Bradycardia, arrhythmias** (vagal, hypoxia): watch monitor, stop if HR falls.
- **Raised ICP, BP swings** (coughing, distress): sedation, few passes.
- **Mucosal trauma, bleeding:** right pressure, no suction going in, shallow.
- **Atelectasis** (alveolar air sucked out): small catheter, short passes, hyperinflate after.
- **Bronchospasm; infection** (sterile, closed); pain, anxiety, coughing fits; laryngospasm and nasal bleeding (nasal route); rare cardiac arrest.

Care in: raised ICP; high PEEP or FiO_2; low platelets or haemoptysis; unstable rhythm; SCI or GBS (vagal bradycardia); bronchospasm; recent lung resection (no deep suction near stump). **No nasal route** with base-of-skull fracture, CSF leak, facial fractures, nasal bleeding.

Manual hyperinflation (7m S22, S24)

Disconnect and give **larger, slower breaths** with a rebreathing bag, then a **quick release**: imitates a cough. **Aims:** re-open collapsed alveoli (collateral channels), move secretions centrally (fast expiratory flow), short-term better compliance and oxygenation.

- Two people: one bags, one shakes or vibrates. 2-L bag, O_2 ~15 L/min, **manometer**, PEEP valve if needed.
- Side-lying, affected area up (or supine).
- Slow deep breath → **hold 2–3 s** → **quick release** with vibrations on expiration; peak pressure **≤ ~40 cmH_2O**.
- Sets of a few breaths; suction when secretions are central; reconnect, reassess.

Contraindications: undrained pneumothorax (absolute); bullae; fresh bronchial stump; unstable BP or arrhythmia; **PEEP above ~10 or high FiO_2** (disconnection collapses the lung); uncontrolled ICP; severe bronchospasm; haemoptysis; pulmonary oedema.

Evidence (Paulus 2012): short-term gains only; **ventilator hyperinflation** works about as well without disconnection, so preferred with high PEEP or FiO_2.

7-mark skeleton: definitions → indications → suction steps with the three numbers → MHI technique → precautions and hazards.

Therapeutic positioning, V/Q and the zones of the lung

In 6 papers.

Full notes p. 236

Asked: 7 marks: sitting upright as a therapeutic position (W21), positioning in ICU with examples (W22), effects and benefits of ICU positioning (W25). 3 marks: positions for dyspnoea (S22), causes of altered V/Q (S23), zones of ventilation (S25).

V/Q and West's zones (3m S25)

Ventilation ~4 L/min, perfusion ~5 L/min: overall **V/Q ≈ 0.8**. Upright, both are greater at the base, but perfusion rises more steeply (gravity): **apex V/Q ≈ 3, base ≈ 0.6**. Basal alveoli start smaller (less negative pleural pressure), on the steep part of the pressure–volume curve, so they ventilate more.

- **Zone 1** (apex): PA > Pa > Pv: no flow; absent normally, appears with low BP or high PEEP.
- **Zone 2** (middle): Pa > PA > Pv: flow depends on Pa − PA.
- **Zone 3** (base): Pa > Pv > PA: continuous flow.

If "zones" means airways: **conducting** (generations 0–16, no exchange, dead space ~150 mL) and **respiratory** (17–23, gas exchange). Write both if unclear.

Altered V/Q (3m S23)

Low V/Q (shunt if no ventilation): secretions, plugs, bronchospasm, collapse, consolidation, pulmonary oedema, ARDS, lying flat, obesity, pain after surgery. **High V/Q (dead space if no perfusion):** PE, low output or blood loss, emphysema, high PEEP, pulmonary vascular disease. Both lower PaO_2; low V/Q responds to oxygen and positioning; true shunt barely responds until alveoli re-open.

Effects of positions

Upright	Largest FRC, diaphragm moves best: almost everyone
Supine	Diaphragm pushed up, FRC ↓, dorsal collapse, aspiration: avoid long (but tetraplegia breathes better lying; binder when sitting)
Side-lying, spontaneous adult	Lower lung gets more V and Q: **good lung down** for O_2; affected lung up to drain
Side-lying, controlled ventilation	Air to upper lung, blood to lower: matching may worsen; check SpO_2
Infant	Ventilation to the **upper** lung: good lung up
Prone	Even ventilation, re-opens dorsal lung: severe ARDS (P/F below 150), ≥ 16 h (PROSEVA)
Head-down	Drains lower lobes; ↑ work of breathing, reflux, ICP
Head-up 30–45°	Less reflux and VAP; raised ICP
Forward lean	Domes a flat diaphragm; arms fixed for accessory muscles: COPD

Not "good lung down" with a **lung abscess or bleeding** (spill into the good lung) or in infants.

Sitting upright (7m W21)

Respiratory: diaphragm descends → bigger FRC and VT, fewer collapsed basal alveoli, better V/Q and O_2, less work of breathing, deeper breath for a stronger cough, less reflux and aspiration. **Cardiovascular:** keeps BP reflexes trained (prevents postural hypotension); legs down ↓ venous return (orthopnoea, pulmonary oedema). **Neuro and other:** alert, oriented, communicates, safer swallowing; head-up ↓ ICP; first step to standing; off-loads the back. Watch for slumping (squashes abdomen, loads sacrum).

ICU positioning (7m W22, W25)

Benefits: better V/Q and O_2, lung volume, re-opens collapse; gravity drainage; less work of breathing; head-up ↓ aspiration and VAP; turns prevent pressure injuries, contractures, nerve compression; upright keeps reflexes and muscles, prepares for mobilisation; comfort, dignity.

RLL pneumonia, spontaneous	Left side-lying (good lung down; right drains)
LLL collapse	Right side-lying, slight turn, foot raised if allowed
Severe ARDS	Prone ≥ 16 h
Raised ICP	Head up 30°, neck straight, no tight ties, no head-down
Every ventilated patient	Head of bed 30–45°
Pulmonary oedema	High sitting, legs down
Distension, obesity	Sitting or reverse Trendelenburg
Pressure risk	30° tilt, heels floated, scheduled turns
Burns, hemiplegia	Anti-deformity, tone-reducing positions

Every turn: check SpO_2, HR, BP; secure tube, lines, drains; allow settling (SpO_2 can dip for minutes).

Positions for breathlessness (3m S22): high side-lying; forward-lean sitting (forearms on knees or table); relaxed sitting back with arms supported; forward-lean standing on a wall or rail; leaning back on a wall. **Why:** diaphragm domed by the abdomen, arms fixed for accessory muscles, less muscle work and anxiety.

Effects of bed rest and early mobilisation

In 5 papers.

Full notes p. 241

Asked: 7 marks: complications of bed immobility in ICU (W17), effects of bed rest on body systems (S24), cardiovascular and pulmonary effects (S26). 3 marks: one effect on three systems (S19, W21).

Why bed rest harms

Lying flat removes two loads: **gravity along the body** (trains BP reflexes and fluid balance) and **muscle work against gravity** (keeps muscle, bone, fitness). Every system adapts downwards within days; fastest in the old and critically ill.

Numbers: 3 weeks of bed rest cut VO_2max by ~**27%** (Dallas 1968); rectus femoris shrank ~**12.5% in the first week** of critical illness (Puthucheary 2013); strict bed rest loses ~**1–1.5% strength a day**.

Cardiovascular and pulmonary (7m S26)

↑ Resting HR, ↓ stroke volume	Lower blood volume, less filling → sit and stand early, exercise
Orthostatic hypotension	Fluid shifts to the chest → diuresis, ↓ plasma; reflexes unused → sit up in stages, dangle legs, tilt table, ankle pumps, stockings
↓ VO_2max	↓ output; muscles lose capillaries and mitochondria → bed cycling, walking
DVT, PE	Stasis + sticky blood → **ankle pumps 10 an hour**, walking, compression, anticoagulants
↓ Lung volume, basal collapse	Diaphragm pushed up, shallow breathing → upright, deep breaths with hold, turns, walking
Retained secretions, pneumonia	Pooling, weak cough, slow clearance, aspiration → turns 2–4-hourly, head up 30–45°, clearance
Weak breathing muscles	Disuse; diaphragm wastes on a ventilator → spontaneous breathing, IMT

Other systems (S24, W17)

Muscles	Antigravity wasting (quads, glutes, calf, back) → active and resisted exercise, NMES in sedated
Joints	Contractures (ankle plantarflexion, hip and knee flexion, shoulder) → ROM, positioning, splints
Bone	Calcium loss within days, ↑ urine calcium → early weight-bearing, tilt table
Skin	Pressure injuries (sacrum, heels, hips, occiput) within hours → turns, mattress, checks
Metabolic	Insulin resistance, negative nitrogen balance → exercise, nutrition
Kidney, bladder	Stasis, infection, stones, retention → upright, fluids, toileting
Gut	Constipation, poor appetite, reflux → mobility, fibre, sit for meals
Mind and nerves	Poor balance, **delirium**, low mood, poor sleep, sensory deprivation → routine, orientation, family, sitting out

ICU-acquired weakness (W17): generalised symmetrical limb weakness in critical illness with no other cause; **MRC sum score below 48/60**; risks: sepsis, organ failure, long ventilation, immobility, high sugar, steroids, paralysing drugs; slows weaning, affects walking and work for months.

Prevention: early mobilisation

- Safety check (ICU criteria) → turns every 2–4 h, head up, upright sitting.
- ROM: passive → active-assisted → active; bed cycling, bridging, resisted exercise.
- Edge of bed → standing (tilt table if BP falls) → chair → marching → walking (with the ventilator if needed).
- Deep breathing, airway clearance, IMT. Monitor HR, BP, SpO_2, RR, symptoms before, during, after.

Current: bed rest is a treatment with side effects: use only when needed and briefly; early mobilisation is safe and shortens delirium and recovery.

Exam

Heart, lungs, legs

3 marks: heart: postural hypotension; lungs: basal collapse and pneumonia; legs: wasting and DVT (add skin or bone).
7 marks: principle (unloading → adapts down in days) → system table with mechanism → prevention by graded mobilisation, breathing, ROM, DVT and pressure care.

ICU conditions: head injury, tetanus, spinal cord injury and neuromuscular disease

Syllabus 9(e). Not asked on their own in 2016–2026.

Full notes p. 245

Asked: not on its own in 2016–2026 (GBS on SIMV is on the ICU physiotherapy sheet), but ICU cases are rising. Possible: 7 marks on ventilated head injury or C5 tetraplegia; 3 marks on signs of respiratory failure in neuromuscular disease.

Head injury

Primary injury at impact (contusion, diffuse axonal injury, haematomas); **secondary** injury from low O_2, low BP, raised ICP, seizures, fever, sugar swings: the whole ICU plan prevents it. **GCS:** mild 13–15, moderate 9–12, **severe 3–8 ("GCS 8, intubate")**.

ICU picture: sedated, ICP monitor or EVD, craniectomy; decorticate or decerebrate posturing; sympathetic storms; aspiration, VAP, collapse; agitation on waking. **Treat ICP above ~22; CPP (MAP – ICP) 60–70**; normal CO_2; head up 30°; mannitol or hypertonic saline.

Physio: head up 30°, neck straight; side-lying as ICP allows; **vibrations and compressions** on expiration (did not change ICP or CPP in sedated patients); suction only when needed, after sedation and pre-oxygenation, short passes (ICP rose ~20 → 26 and settled in ~10 min); positioning out of abnormal posture, ankle splints, gentle PROM, serial casts later; mobilise once ICP is stable (sit, edge of bed, tilt table); calm consistent handling as he wakes.

Precautions: ICP above target or being treated: no active mobilisation, clustered handling; **no head-down**, neck flexion or rotation, tight ties, high-pressure bagging; craniectomy: no pressure on the bone-free side, helmet; clamp or level EVD; spine not cleared: log-roll.

Tetanus

Clostridium tetani toxin from wounds blocks inhibitory neurones → uncontrolled contraction + autonomic overactivity; onset 3–21 days. **Trismus** (lockjaw), **risus sardonicus**, **opisthotonus**; painful spasms triggered by touch, noise, light; **fully conscious**; laryngospasm, rigid chest → respiratory failure; aspiration; BP and HR swings.

Wound care, immunoglobulin, metronidazole, benzodiazepines, **magnesium**, ventilation and early tracheostomy; **dark quiet room**; vaccinate after recovery (no immunity).

Physio: plan with the team, treat when sedation is adequate; short sessions; no sudden touch or noise; positioning, turns, gentle vibrations, suction with sedation; gentle PROM only when spasms are controlled; recovery: active exercise, **jaw opening**, sitting, walking, strength. Warn before touching; slow firm handling; stop if spasms start; new back pain = vertebral fracture?

Spinal cord injury: breathing

C3, 4, 5 keeps the diaphragm alive

C1–C2: lifelong ventilator or phrenic pacing; **C3–C5:** partial diaphragm, often ventilated then wean; **C6–C8:** diaphragm works, intercostals and abdominals lost: low VC, very weak cough, paradoxical upper chest; **thoracic:** abdominals T6–T12, weaker cough the higher the level.

Spinal shock; **neurogenic shock** (T6 and above: low BP, slow HR); excess secretions (unopposed vagus), collapse and pneumonia; gut distension; VC improves over weeks.

Physio: manually assisted cough (upward and inward push timed with the cough), cough assist, suction; diaphragmatic breathing; **VC is often better lying than sitting** (abdomen supports the diaphragm): **abdominal binder** when sitting; respiratory muscle training (↑ VC, Cochrane 2013); hyperinflation if ventilated; log-roll turns, vibrations; PROM within spinal limits, splints, 2–3-hourly turns; staged sitting, tilt table.

Precautions: unstable spine: log-roll only; suction or turning → severe bradycardia; **autonomic dysreflexia** (T6 and above): very high BP, pounding headache, flushing above the level → sit up, loosen clothing, check catheter and bowel, call for help; postural hypotension; no assisted cough over a fresh abdomen or ileus.

Neuromuscular disease

Myasthenic crisis (infection, surgery, drugs; IVIG or plasma exchange; too much anticholinesterase = cholinergic crisis with secretions); motor neurone disease (NIV improves survival without severe bulbar weakness); Duchenne (scoliosis, night hypoventilation); critical illness polyneuropathy; neurotoxic snakebite.

Warning signs: VC below ~20 mL/kg, weak cough, choking, nasal voice; fast breathing, accessory muscles, abdomen moving in on inspiration, breathless lying flat; morning headache, daytime sleepiness; rising $PaCO_2$ is late. **Peak cough flow below ~270 L/min**: cough may fail in an infection (teach augmentation); **below ~160**: ineffective.

Physio: assisted cough, cough assist, **breath stacking** (bag with one-way valve), suction; support NIV (mask fit); positioning, clearance, swallow referral; ROM, gentle non-fatiguing strengthening, energy conservation; myasthenia: short sessions ~1 h after pyridostigmine, stop at first fatigue. **Avoid fatigue**; respect choices about ventilation in progressive disease.

ICU conditions: pulmonary oedema, shock, organ failure, poisoning, aspiration and drowning

Syllabus 9(e). Not asked on their own in 2016–2026.

Full notes p. 252

Asked: not on its own in 2016–2026 (inhalation burns, ventilated pneumonia and ARDS come closest). Possible: 7 marks on organophosphate poisoning or shock; 3 marks on X-ray of pulmonary oedema or MODS and SOFA.

Pulmonary oedema

Cardiogenic (failing LV, fluid overload: high capillary pressure) or **non-cardiogenic** (leaky capillaries: ARDS, drowning, smoke, altitude, head injury). Severe breathlessness, orthopnoea, night breathlessness, sweating; **pink frothy sputum**; fine bibasal crackles. X-ray: **bat's wing** haze from both hila, **Kerley B lines**, big heart, effusions. Sit up, oxygen, diuretics, nitrates; **CPAP or NIV** relieves breathlessness and cuts intubation (Cochrane 2013).

Physio: high sitting, legs down; breathing control, reassurance; help with the CPAP mask. Froth is oedema, not secretions: **no airway clearance** (unless infection). Later mobilise and heart failure rehab. Never flat or head-down; watch BP (diuretics, nitrates).

Shock

Tissues not getting enough O_2: fast pulse, systolic below ~90 or **MAP below 65**, cold clammy skin (warm in early sepsis), urine below 0.5 mL/kg/h, confusion, **lactate above 2**.

Hypovolaemic	Bleeding, burns, diarrhoea and vomiting
Cardiogenic	Large MI, arrhythmia, end-stage failure
Distributive	Sepsis, anaphylaxis, neurogenic (low BP + slow pulse)
Obstructive	Tension pneumothorax, tamponade, massive PE

Sepsis-3: life-threatening organ dysfunction from a dysregulated response to infection. **Septic shock:** needs vasopressors for MAP ≥ 65 with lactate above 2 despite fluids. Fluids, noradrenaline, antibiotics within 1 h.

Physio: unstable: no active mobilisation; positioning, pressure care, gentle passive movement if BP allows; chest physio only for clear need, short. As support weans: follow the safety traffic lights and the mobilisation ladder. Survivors: months of weakness → strength and endurance training. Turning and sitting can drop BP: stop if MAP falls, pulse races, confusion, clamminess.

Multiple organ failure (MODS)

Progressive failure of **two or more organ systems** needing support; after sepsis, trauma, burns, pancreatitis, major surgery, shock. **SOFA** scores six: breathing (P/F), clotting (platelets), liver (bilirubin), circulation (MAP, pressors), brain (GCS), kidney (creatinine, urine).

Physio by organ: lungs: ARDS plan; circulation: mobilise only with stable MAP on low support; low platelets and liver: no percussion, gentle suction; dialysis: femoral lines don't stop exercise (secure them), watch BP during dialysis; brain: orientation, family, routine; muscles: early ROM, splints, mobilisation. **Post-intensive care syndrome:** weakness, fatigue, memory problems, anxiety, nightmares → follow-up rehab.

Poisoning

Organophosphates block acetylcholinesterase. **Muscarinic:** pinpoint pupils, sweating, salivation, lacrimation, vomiting, diarrhoea, urination, slow heart, **bronchorrhoea**, bronchospasm. **Nicotinic:** fasciculations → paralysis incl. breathing muscles. **Brain:** confusion, seizures, coma. **Intermediate syndrome** at **24–96 h**: neck flexor, proximal, cranial and breathing weakness. Atropine until the chest is dry, HR above 80, systolic above 80, pupils not pinpoint, axillae dry.

Physio: frequent suction and positioning (aspiration pneumonia common), head up; daily **neck flexion test** (lift the head off the pillow): new weakness warns of the intermediate syndrome; then mobilise. Every self-poisoning needs mental health review.

Snakebite (krait, cobra): descending paralysis from ptosis; antivenom, ventilation: treat like neuromuscular weakness. **Paraquat:** lung fibrosis, high O_2 worsens it: oxygen only as prescribed. **Kerosene** (children): aspiration pneumonitis. **Overdose:** aspiration, pressure palsies (foot drop) after long lying.

Aspiration, drowning, smoke

Aspiration pneumonitis (chemical, Mendelson's, hours) vs **pneumonia** (mouth bacteria, days); reduced consciousness, stroke, vomiting, tube feeds flat. Usually the **right lung** (wider, more vertical bronchus): basal segments upright; posterior upper and superior lower segments supine. Physio: head up 30–45°, side-lying, suction and clearance of those segments, swallow referral, oral care.

Drowning (WHO 2002): respiratory impairment from submersion or immersion. Hypoxaemia, surfactant washout → collapse, oedema, ARDS; aspiration; brain injury; hypothermia; neck injury in diving. Physio as ARDS (no high pressures), spinal precautions, later neuro rehab.

Smoke: upper airway swelling (early intubation), lower airway injury; **CO**: oximeter falsely normal, check COHb, high-flow reservoir mask; cyanide. Physio: upright, humidified O_2, deep breathing, ACBT, suction, walking (see Burns (p. 6)). **PE:** sudden breathlessness, pleuritic pain; mobilise once anticoagulated and stable.

Airway clearance techniques: overview and choice

Asked as a 7-mark overview in 4 papers.

Full notes p. 258

Asked: 7 marks: various airway clearance techniques with application, advantages, disadvantages (S19, W24); mucus removal in chronic respiratory disease (W19). 3 marks: indications of bronchial hygiene therapy (W25).

Why and classification

Normal clearance = **mucociliary escalator + cough**. Needed when there is **too much or too thick mucus** (bronchiectasis, CF, chronic bronchitis, infection), **poor cilia** (smoking, CF, ciliary dyskinesia, dry gas via a tube) or a **weak or painful cough** (neuromuscular, post-op, sedation). **Aims:** remove mucus, cut infections and the vicious cycle, improve ventilation and gas exchange, less cough and breathlessness, better QoL.

- **Breathing:** ACBT, FET and huffing, autogenic drainage, directed or assisted cough.
- **Gravity and manual:** postural drainage, percussion, vibration, shaking.
- **PEP devices:** PEP mask, oscillating PEP (Flutter, Acapella), bottle PEP.
- **Mechanical:** oscillation vest, IPV, cough assist, IPPB. **Exercise.**
- **Adjuncts:** hydration, humidification, nebulised saline, mucolytics, bronchodilator first. **Suction.**

Comparison (7m S19, W24)

	Best for	For	Against
ACBT	Most with sputum, post-op, exacerbations	Easy, no equipment, independent, any position	Needs co-operation
Autogenic drainage	Bronchiectasis, CF; wheezy unstable airways	Gentle, no equipment	Hard to learn; 30–45 min
Postural drainage ± percussion	Known lobe; infants; dependent, ventilated	Targets a lobe; carer can do it	Slow; head-down contraindications; needs a helper
PEP mask	CF, bronchiectasis, post-op	Quick, portable, independent	Device, training; no untreated pneumothorax
Oscillating PEP	Bronchiectasis, CF, COPD	Vibration; small; motivating	Cost; Flutter upright only; cleaning
Vest	CF, neuromuscular	Hands-free	Very expensive
Cough assist	Neuromuscular, SCI	Cough when muscles can't	Equipment; care with bullae
Exercise	All mobile patients	Fitness too	Doesn't replace ACT
Suction	Intubated, unconscious, very weak	Direct	Invasive; hazards

Choosing (BTS/ACPRC 2009)

No single best technique: choose by age, condition, ability, preference and what the patient will keep doing. Infants: modified drainage, gentle percussion, play. Older children: PEP, bubble PEP, ACBT. Wheezy or floppy airways: AD, PEP. Weak cough: assisted cough, cough assist, suction. **Order:** bronchodilator → mucoactive (hypertonic saline) → **clearance** → inhaled antibiotic or steroid. Review regularly.

Mucus removal in chronic disease (7m W19)

- **Assess:** daily sputum amount and colour, ease of clearing, which lobes (X-ray, CT), breathlessness, past techniques.
- **Prepare mucus:** fluids; humidified or nebulised saline; bronchodilator first in reactive airways.
- **Daily routine:** one main technique (ACBT, AD or oscillating PEP), in the worst lobe's position if tolerated, **10–30 min, once or twice a day**.
- **Exacerbation:** 2–3 times a day; early medical review.
- **Exercise** (pulmonary rehab); **pelvic floor** (leakage with cough: contract before huffs).
- **Review:** sputum weight, cough or bronchiectasis QoL score, exacerbations.

Indications (3m W25)

Sputum over ~**25–30 mL a day** or hard to clear; mucus plugging with lobar or segmental collapse; weak or painful cough (neuromuscular, post-op, ventilated); chronic suppurative disease (bronchiectasis, CF) as a daily routine.

7-mark skeleton: why (1) → classification (1) → comparison table (4) → choosing (1).

ACBT, forced expiration, huffing and autogenic drainage

In 10 papers, the most-asked technique.

Full notes p. 261

Asked: the most-asked technique (10 papers). **7 marks:** ACBT (W16, S26), its principle, indications and contraindications (S23), rationale in asthma (S17), ACBT vs autogenic drainage (W17), autogenic drainage indications and phases (S20). **3 marks:** huffing vs coughing, FET, ACBT, AD phases, bronchial hygiene indications.

ACBT: three parts

A flexible cycle the patient does **independently**: **breathing control → thoracic expansion → breathing control → forced expiration technique**; breathing control always between active parts.

Breathing control	Gentle relaxed lower-chest tidal breathing, shoulders relaxed, as long as needed. **Rest**: prevents bronchospasm, fatigue, desaturation
Thoracic expansion	**3–4 deep breaths, 3-s hold** (or sniff), relaxed out; hand on the chest for feedback; may add percussion or shaking. Air travels by **collateral channels** (pores of Kohn, canals of Lambert, channels of Martin) behind mucus; hold evens filling; **interdependence** opens neighbours
FET	**1–2 huffs** from mid to low volume (open mouth and glottis, "steaming a mirror"), then breathing control; high-volume huff or cough when mucus is central. Airways narrow downstream of the **equal pressure point**; fast air shears mucus mouthwards

Equal pressure point: in a forced expiration airway pressure falls from alveoli to mouth; where it equals pleural pressure the airway is squeezed and air speeds up. High volume → EPP in central airways; lower volume → EPP moves peripherally. So a **long mid-volume huff reaches further** than a cough from a full breath.

Procedure, dose, indications

Explain; bronchodilator first if prescribed; sitting or the lobe's drainage position; BC until relaxed → 3–4 TEE with hold → BC → 1–2 mid-volume huffs → BC; high-volume huff or one cough when central; repeat. **Stop** when huffs are dry and non-productive in **two consecutive cycles** or the patient tires. **Dose:** 10–30 min, once or twice a day; more in infections; post-op a few cycles every 1–2 h.

Indications: excess or retained secretions (bronchiectasis, CF, COPD, infection, asthma with plugs); collapse; post-op prevention; to make coughing easier; independent home routine.

Contraindications and care: few true ones; not possible if unconscious, confused, very young. Bronchospasm: longer BC, stop if wheeze rises. Haemoptysis: no vigorous huffs or percussion. Untreated pneumothorax, unstable heart, severe post-op pain: modify (supported huff) or avoid forced parts. Fatigue, desaturation: shorter cycles, oxygen.

Asthma rationale (S17): BC prevents bronchospasm and fatigue; TEE gets air behind plugs; huffs avoid coughing fits that narrow sensitive airways; patient sets the pace.

Huff vs cough (3m W16, S20)

	Huff	Cough
Glottis	Open throughout	Closes, pressure builds, opens suddenly
Pressure	Moderate	High
Volume	Mid to low: peripheral airways	High: central airways
Side effects	Less tiring; less collapse, spasm, pain, leakage	Tiring; airway collapse (COPD), spasm, wound pain, stress incontinence
Best for	Moving mucus from the periphery; post-op; unstable airways	Clearing mucus at trachea and main bronchi

Teach a huff: medium breath, then out fast but not violently through an "O" mouth: a breathy rush, not a cough; a tube keeps the glottis open; support the wound.

Autogenic drainage (7m S20; 3m W19)

Controlled breathing at different lung volumes (Chevaillier) to create the highest airflow without airway collapse, moving mucus from small to large airways.

1. **Unstick:** small breaths at **low lung volume** (expiratory reserve): loosens peripheral mucus.
2. **Collect:** breaths at **mid volume** (tidal): gathers mucus in middle airways.
3. **Evacuate:** deeper breaths at **high volume**, then a gentle huff.

Upright; slow nasal breath in, **hold 2–3 s**, sigh out through open mouth and throat, fast but no wheeze; feel or hear crackles to change phase; **suppress the cough** until mucus is central; 30–45 min. **Indications:** bronchiectasis, CF (from ~8–12 years), COPD with sputum, collapsing or wheezy airways, cough-induced incontinence. **For:** independent, no equipment, gentle. **Against:** hard to learn, needs concentration, slow, not for young children (assisted form for infants).

7-mark skeleton (S23): definition + cycle drawing → each part with its principle → indications → contraindications → dose, end point, advantages.

Postural drainage, percussion and vibration

In 2 papers, and part of every airway-clearance answer.

Full notes p. 267

Asked: 3 marks: any six contraindications to postural drainage (S18); 7 marks: precautions and contraindications in a neonate (W18). Drainage positions are part of every bronchiectasis long answer.

What it is

Postural drainage (gravity-assisted positioning): the affected segment's bronchus points down towards the trachea so gravity helps secretions reach the large airways, where they are huffed out. Always combined with a breathing technique (ACBT), ± percussion, vibration, shaking. **Classic:** 11–12 positions, many head-down. **Modified:** same side-lying and prone positions **without head-down tilt** (infants, reflux, breathlessness).

Dose: ~10 min per productive position, session 20–40 min, once or twice a day; more in infection; not after meals.

Positions

Bad side up, bronchus down

Segment	Position	Bed
Upper apical	Sitting, leaning back ~30°	Flat or sitting
Upper posterior	Sitting leaning forward over a pillow; or opposite side, quarter turn to prone	Flat
Upper anterior	Supine	Flat
Lingula (left)	Right side, quarter turn back, pillow behind	Foot up ~15°
Middle lobe (right)	Left side, quarter turn back, pillow behind	Foot up ~15°
Lower superior	Prone, two pillows under hips	Flat
Anterior basal	Supine (or opposite side turned slightly back)	Foot up 20–30°
Lateral basal	Opposite side, quarter turn forward	Foot up 20–30°
Posterior basal	Prone, pillow under hips	Foot up 20–30°

Lie on the healthy side for lateral segments; turn towards prone for posterior, towards supine for anterior; foot up for lower lobes, middle lobe and lingula. Angles vary between books (foot raised ~35–45 cm); **direction matters most**.

Indications: large sputum (bronchiectasis, CF, abscess); retained secretions or collapse of a known lobe; patients too weak or young for active techniques (modified positions).

Contraindications to head-down tilt (3m S18)

- Recent or active **haemoptysis**
- **Raised ICP**, recent head injury or neurosurgery
- Uncontrolled hypertension, unstable heart, arrhythmia, pulmonary oedema, heart failure
- Severe breathlessness or bronchospasm
- **Reflux**, hiatus hernia, recent oesophageal surgery, just after meals
- Aortic aneurysm, recent eye surgery
- Gross obesity, abdominal distension, late pregnancy
- Untreated pneumothorax, large effusion or empyema, rib fractures, painful recent surgery

Use modified positions instead. **Neonates (W18):** no head-down (reflux, IVH in preterms), minimal handling: see Neonatal (p. 36).

To percussion and vibration: rib fractures, flail chest, osteoporosis, bone metastases, recent spinal surgery; haemoptysis, low platelets; unstable BP or rhythm; bronchospasm; burns, fresh grafts, wounds, new pacemaker; surgical emphysema; intolerable pain.

Percussion, vibration, shaking

- **Percussion:** rhythmic clapping with **cupped hands**, relaxed wrists, over a thin towel, 30 s to a few minutes, alternating with BC or TEE: air cushion sends an energy wave that loosens mucus.
- **Vibration:** fine rapid shaking from tensed arms with gentle compression, **on expiration only**: ↑ expiratory flow.
- **Shaking:** coarser rhythmic bouncing compression on expiration after a deep breath: thick mucus.
- Adjuncts only, if secretions and helpful. Infants: fingertips or soft cup, no tilt. Ventilated: on expiration, often after hyperinflation.

Current (BTS/ACPRC 2009): active techniques preferred; head-down only when helpful and safe; percussion not routine. Write: "gravity-assisted positions for the affected lobe with ACBT; modified positions where contraindicated; percussion and vibration as adjuncts".

PEP, flutter, Acapella and other airway-clearance devices

In 7 papers.

Full notes p. 271

Asked: 3 marks (5 papers): PEP mask, Acapella and Flutter, two PEP devices, principle with two devices, three uses. 7 marks: PEP principle, classification and types (S21); assistive devices for airway clearance (W21).

PEP principle

Breathing out against a resistance, so airway pressure stays above atmospheric through expiration; PEP mask target **10–20 cmH_2O**.

- **Splints airways open**: back-pressure stops early collapse of floppy small airways (like pursed lips).
- **Air behind mucus**: slight volume rise, collateral channels; on expiration that air pushes mucus centrally.
- **Oscillating PEP** adds vibration: pressure flutters many times a second, shearing mucus off and making it less sticky.

Then clear with huffs and a cough.

Types (7m S21)

PEP mask (low pressure)	One-way valve, fixed resistor, **manometer**; set by the physio; sitting or drainage positions; nebuliser can be added
Flutter	Pipe with a **steel ball in a cone**; breath lifts the ball, it drops back repeatedly; must be held at the right angle (**sitting only**); small, cheap
Acapella	**Counterweighted lever and magnet**; dial sets resistance and frequency; **any position**; suits low flows
Others	RC-Cornet, Aerobika: position-independent
Bottle (bubble) PEP	Tube into water: **depth in cm ≈ cmH_2O** (10 cm ≈ 10); bubbles add oscillation; very cheap, child-friendly; change water daily
High-pressure PEP	Forced expiration through the mask at high pressure: specialist CF use

Technique (PEP mask)

1. Sit, elbows on a table (or drainage position); mask firmly over nose and mouth.
2. Breathe in slightly deeper than normal; out actively, not forcefully, manometer **10–20 cmH_2O**.
3. **8–10 breaths**, remove the mask, **2–3 huffs** and a cough, rest with breathing control.
4. Repeat ~**15–20 min**, once or twice a day (more in infection).

Flutter: tilt until vibration is felt most in the chest. Clean after each use; never share (CF cross-infection).

Uses (3m S26): independent, portable clearance (bronchiectasis, CF, COPD); unstable collapsible airways (COPD, tracheomalacia); post-op lung expansion; better aerosol delivery with a nebuliser; play-based practice in children (bubble PEP).

Contraindications: untreated pneumothorax (main); active haemoptysis; raised ICP; haemodynamic instability; recent face, mouth, skull or oesophageal surgery; sinusitis, nosebleed, ear problems or perforated eardrum; exhaustion; bullae (low pressure, care).

Other assistive devices (7m W21)

Incentive spirometer	Visual feedback for slow deep breaths with hold: post-op add-on
IPPB (Bird)	Positive pressure boosts each patient breath: weak or tired; not with undrained pneumothorax
Oscillation vest	Rapid chest-wall squeezes: CF, neuromuscular; hands-free, expensive
IPV	Rapid small bursts with saline: thick secretions
Cough assist (MI-E)	Positive breath in, quick switch to negative, ~**±40 cmH_2O**: weak cough (neuromuscular, SCI)
Manually assisted cough	Push in and up on abdomen or lower ribs with the cough: SCI (not over a fresh wound)
Suction	For those who cannot cough secretions out

7-mark skeleton: principle (2) → types, one line each (2.5) → technique cycle (1.5) → indications and contraindications (1).

Breathing exercises, lung expansion and breathing retraining

In 7 papers, counting pursed-lip breathing. Full notes p. 275

Asked: 3 marks: thoracic mobility exercises (W17), diaphragmatic breathing (W18), pursed-lip breathing effects and rationale (S20, W21, W22). 7 marks: diaphragm recruitment and strengthening (S21), breathing retraining (S23).

Types and aims

Breathing control: relaxed tidal breathing, calms panic. **Diaphragmatic:** more diaphragm, less accessory work, basal ventilation. **Pursed-lip:** slower breathing, less air trapping (COPD). **Thoracic expansion:** re-expand an area, loosen secretions. **SMI and incentive spirometry:** post-op expansion. **Thoracic mobility:** free a stiff chest and spine. **Breathing retraining:** correct dysfunctional breathing (asthma, anxiety).

Diaphragmatic breathing (3m W18; 7m S21)

Technique: half-lying, head and knees supported (diaphragm domed, good length); later sitting, standing, walking. Shoulders relaxed; hand on the upper abdomen below the ribs; breathe in slowly through the nose, **hand rises**, upper chest quiet; out gently, not forced; **5–10 breaths** several times a day; watch dizziness.

Recruit and strengthen: quick stretch in and up on the upper abdomen at end-expiration, then light resistance on inspiration; a **sniff** recruits strongly; progress supine → sitting → standing (binder in tetraplegia); manual resistance, classic small weights on the abdomen, best evidence **threshold IMT ≥ 30% MIP**; use in walking, stairs, speech.

Indications: post-op, neuromuscular disease and SCI (diaphragm may be the main muscle left), weaning, upper-chest breathing. **Severe COPD:** with a flat diaphragm, forced abdominal breathing can raise work and breathlessness (Gosselink 1995): prefer pursed lips, positions of ease; stop if worse.

Pursed-lip breathing (3m S20, W21, W22)

In through the nose; out slowly through loosely pursed lips ("hot tea"), **out ~twice as long**, not forced. Back-pressure **splints small airways** → less air trapping → slower deeper breathing, better VT and SpO_2 → less breathlessness and anxiety on walking and stairs. See COPD (p. 12).

Thoracic expansion; lung expansion

Localised expansion: hand over the area (lateral costal, posterior basal, apical; uni- or bilateral); gentle pressure at end-expiration, "breathe into my hand", **3-s hold**, relaxed out; 3–4 breaths then breathing control. Post-op, effusion, empyema, collapse, stiff chest, scoliosis (concave side).

SMI: slow deep breath to the top, **hold 3–5 s**, relaxed out (collateral channels). **Incentive spirometer:** visual feedback; **volume-oriented** better than flow-oriented (balls tempt fast breaths); 5–10 breaths hourly + walking. Not if cannot be taught, will not co-operate, or VC below ~10 mL/kg (then IPPB or CPAP). **AARC 2011:** not alone or routine; deep breathing, cough and walking are the core.

Thoracic mobility (3m W17)

- **Trunk rotation** with breathing out, back to centre breathing in, 10 each way.
- **Side-bending with arm overhead** while breathing in (thoracotomy, pleural disease, scoliosis).
- **Bilateral arm elevation** up on inspiration, down on expiration.
- **Pectoral stretch** in a doorway with deep breaths, hold 20–30 s.
- **Thoracic extension** over a rolled towel or the chair back.

Ankylosing spondylitis, kyphosis, scoliosis, after chest surgery and chest burns, COPD posture.

Breathing retraining (7m S23)

For **dysfunctional breathing**: fast, upper-chest, mouth breathing, sighing, over-breathing; symptoms (breathlessness, tightness, dizziness, tingling) out of proportion; screen with the **Nijmegen questionnaire**; common in asthma and anxiety.

- **Education:** over-breathing lowers CO_2 and causes symptoms; reassure.
- **Nose breathing** at rest and light activity; **lower-chest** breathing, relaxed shoulders.
- **Slower, smaller, gentler** breaths with a pause after expiration.
- Relaxation; control sighs and yawns; speech phrasing.
- Practice **10–15 min once or twice a day**, then in daily activity (Papworth; Buteyko adds reduced breathing).

Evidence: better asthma QoL and symptoms (Cochrane 2020); UK trial 2018 with a DVD programme; does not replace inhalers.

Neurophysiological facilitation and PNF of respiration

In 3 papers, all since 2022.

Full notes p. 279

Asked: rising, all since 2022. 7 marks: neurophysiological facilitation of respiration (W22); three PNF techniques for respiration (W24); PNF to improve lung expansion (S25).

NPF: what and who

Neurophysiological facilitation: external touch and pressure (proprioceptive and tactile) producing **reflex changes in breathing**: deeper breaths, more movement in a chosen area, rate changes (Bethune 1975, unconscious adults).

For patients who cannot follow commands: unconscious or sedated (incl. ventilated with own breathing), head injury, stroke, confused, young children. **Aims:** deeper breathing, lung expansion, basal and segmental ventilation, secretion movement, oxygenation. **Dose:** hold through several breaths (30 s – 1 min), repeat while it works; watch breathing, SpO_2, HR.

NPF techniques (7m W22)

Technique	How	Response
Perioral pressure	Firm maintained finger pressure on the upper lip	More upper-abdominal movement, deeper breaths, sighs; mouth closure, swallowing (sucking reflex)
Intercostal stretch	Push the upper border of a rib down to widen the space above; hold	More expansion there (intercostal stretch reflex)
Vertebral pressure	Firm pressure on spinous processes: upper T2–T5 or lower T7–T10	Upper: more abdominal movement, deeper breaths. Lower: more apical movement
Co-contraction of abdomen	Lower ribs and pelvis on one side drawn towards each other	Upper-abdominal movement, posterior basal expansion, firmer abdominal wall
Anterior stretch basal lift	Supine; hands under lower posterior ribs lift gently and hold	Posterior basal expansion, abdominal movement
Maintained manual pressure	Firm steady hand on an area of chest wall	More movement under the hand on inspiration

Precautions: no pressure over fractured ribs, unstable spine, osteoporosis, painful wounds or burns; stop if agitated, HR or BP unstable, SpO_2 falls; raised ICP: short, gentle. **Evidence:** small studies, effects last minutes: combine with positioning and other techniques.

Skeleton: definition and who (1) → six techniques table (4) → aims and dose (1) → precautions (1).

PNF for respiration (7m W24, S25)

Limb PNF principles (manual contacts, stretch, resistance, commands, **irradiation** from strong patterns) applied to breathing muscles in patients who **can co-operate**.

Repeated quick stretch with resistance	Hands on lower lateral ribs (or upper abdomen); quick stretch in and down at end-expiration, light resistance through inspiration, re-stretch during it ("push my hands out"): stretch reflex, more expansion
Bilateral D2 flexion with inspiration	Arms up and out on breathing in, D2 extension on breathing out, can be resisted: irradiation, upper chest expands
Trunk lift and chop	Lift (diagonal up, extension, rotation) on inspiration; chop (diagonal down, flexion) on expiration and cough
Scapular posterior elevation	Side-lying, resisted on inspiration: upper and lateral expansion
Rhythmic initiation	Passive → active-assisted → active → resisted breathing rhythm: teaches a coordinated pattern
Approximation; timing for emphasis	Rib compression at end-expiration; hold a strong pattern to direct expansion to a weak area

For "three techniques" (W24): repeated quick stretch with resistance, bilateral D2 flexion with inspiration, trunk lift and chop, each with method and effect.

Indications: weak or uncoordinated breathing (stroke, SCI, neuromuscular), post-op and post-ICU low expansion, basal collapse. **Precautions:** rib fractures, wounds and drains under the hands, pain, unstable cardiovascular status.

Respiratory muscle training

In 3 papers.

Full notes p. 282

Asked: 3 marks: two types of inspiratory muscle training (S18). 7 marks: exercise programme to strengthen respiratory muscles (W22); diaphragm recruitment (S21, see Breathing exercises (p. 52)).

Why and who

Breathing muscles may be **weak** or **overloaded** (stiff lungs, obstruction, hyperinflation) → breathlessness, poor exercise tolerance, weak cough, weaning failure. Like limb muscles they respond to **progressive overload**.

Measure: MIP (PImax) and MEP with a mouth pressure meter. MIP stronger than ~**−80 cmH_2O** rules out important weakness; in COPD MIP weaker than ~**60** marks those who gain most.

Who: COPD with weak inspiratory muscles, heart failure, before high-risk heart, lung or abdominal surgery, ICU weaning failure, early neuromuscular disease and SCI, dysfunctional breathing, athletes.

Types (3m S18)

Threshold loading	**Spring-loaded valve** opens only at a set pressure; load independent of flow; easy to set as % MIP; most used
Inspiratory resistive	Breathing in through a **small opening**; load depends on flow, so a set pattern or target device is needed
Normocapnic hyperpnoea	High ventilation for 10–30 min with a rebreathing bag (normal CO_2): endurance; needs equipment
Tapered-flow resistive	Electronic device varies resistance through the breath: heavier loads, costly
Expiratory training	Threshold device on expiration: cough, voice, swallowing (neuromuscular, Parkinson's, MS)

Classic: small weights on the upper abdomen in supine; manual diaphragmatic resistance.

Dose (threshold IMT)

- **Intensity:** start ~**30% MIP**, progress to 50–60%+; re-measure MIP every 1–2 weeks.
- **Volume:** ~**30 breaths (or 15 min), twice a day**.
- **Frequency:** 5–7 days a week, **≥ 6–8 weeks**, then maintenance.
- **Technique:** upright, nose clip if needed, quick strong breath in through the device, slow relaxed breath out, rest between sets.

Evidence: alone, IMT improves strength, endurance, breathlessness and walking, most in weak patients (2011); added to full pulmonary rehab it raised strength but not exercise capacity (2018); helps ICU weaning failure (2015). An add-on, not a replacement for whole-body exercise.

Write: "IMT at ≥ 30% MIP, ~30 breaths twice a day for ≥ 8 weeks, for inspiratory weakness, alongside general exercise".

Programme to strengthen respiratory muscles (7m W22)

1. **Assess:** MIP, MEP, pattern, expansion, peak cough flow, Borg, mMRC, posture.
2. **Diaphragmatic breathing** with facilitation and manual resistance, half-lying → sitting → standing.
3. **Threshold IMT** at the dose above.
4. **Expiratory training** and huffing if cough is weak; abdominal strengthening (bracing, curl-ups as allowed).
5. **Accessory and postural muscles:** scapular and upper-limb exercise (arm ergometer, resisted patterns), thoracic mobility.
6. **Whole-body aerobic training**: the ventilatory load also trains breathing muscles.
7. **Endurance:** normocapnic hyperpnoea; singing or wind-instrument games for children.
8. Review every 1–2 weeks; re-measure MIP; adjust.

Precautions

Recent or untreated pneumothorax; unhealed chest surgery; acute COPD or asthma exacerbation; unstable heart failure, recent MI; perforated eardrum; raised ICP; active haemoptysis. Stop with fatigue (harmful in neuromuscular disease), dizziness, breathlessness.

Dyspnoea, relieving positions, energy conservation and pacing

In 8 papers.

Full notes p. 285

Asked: 8 papers. **7 marks:** energy conservation (S19, S22), ergonomic advice in left lung bronchiectasis (W22), dyspnoea definition, perception and treatment (W25). **3 marks:** energy conservation, pacing, positions for dyspnoea.

Definition (ATS 2012)

Dyspnoea: "a subjective experience of breathing discomfort that consists of qualitatively distinct sensations that vary in intensity": measured by the patient's own rating. Sensations: **air hunger**, **work or effort**, **chest tightness**; intensity (sensory) and distress (emotional). Measure: modified Borg 0–10 in activity; mMRC 0–4 in daily life. **Orthopnoea** (lying flat: heart failure), **PND** (wakes breathless), **platypnoea** (sitting or standing, relieved lying).

How it is perceived (W25)

- **Motor drive and effort:** a copy of the motor command (**corollary discharge**) reaches the sensory cortex; weak or overloaded muscles need a bigger command = effort.
- **Chemoreceptors:** central (CO_2, acidity, medulla), peripheral (low O_2, carotid and aortic bodies) → more drive, air hunger.
- **Mechanoreceptors:** airway stretch and irritant receptors, **J receptors** (stiff wet lungs), chest wall spindles and tendon organs report movement.
- **Mismatch (neuromechanical dissociation):** high drive, lungs can't respond (hyperinflation, stiffness, weakness) → main source.
- **Emotion:** limbic system adds fear; anxiety speeds breathing → more hyperinflation → more breathlessness.

Causes: ↑ demand (exercise, fever, acidosis, hypoxia), ↑ load (obstruction, stiffness, hyperinflation), weak muscles, anxiety.

Treatment (W25)

Treat cause	Bronchodilators, heart failure, infection, anaemia; O_2 if hypoxaemic
↓ Work of breathing	Positions of ease, pursed lips, breathing control, clearance, NIV
↓ Demand	Energy conservation, pacing, weight loss, aids
↑ Capacity	Exercise training (desensitises, less ventilation per load); IMT if weak
Calm the brain	Education, relaxation, breathlessness plan, **handheld fan** on the face, anxiety management
Palliation	Low-dose opioids in advanced disease

Positions of ease (3m S22)

High side-lying (pillows, top arm supported); **forward-lean sitting** (forearms on thighs or pillows on a table); relaxed sitting back with arms on armrests; **forward-lean standing** on a wall, sill, rail or frame; relaxed standing against a wall. **Why:** abdomen pushes a flat diaphragm into a dome (better contraction); fixed arms let accessory muscles pull on ribs; supported positions cut muscle work and anxiety; upright ↑ lung volume.

Energy conservation (S19, S22, W22; 3m W17, S25)

The 4 Ps: Plan, Pace, Prioritise, Position

- **Plan:** spread heavy jobs over the week; best time of day (after clearance and medicines); gather things first.
- **Pace:** slow and steady; rest before very breathless; **"blow as you go"** on the effort.
- **Prioritise:** drop or share non-essential jobs.
- **Position:** sit to bathe, dress, iron, cook; things between waist and shoulder; avoid bending and overhead work; trolley, long sponge, shower stool, raised toilet seat.
- Pursed lips when walking; stairs one step per out-breath, rest on landings; avoid smoke, dust, heat, cold; loose clothes; fan.

One-sided disease (W22, left bronchiectasis): lie on the **right** (healthy lung down: more air and blood; diseased lung up: drains); desk at elbow height; huffing breaks; avoid dust.

Activity pacing (3m S20, W21)

Activity in planned manageable amounts with rests, staying below the breathlessness that forces a stop: **no "boom and bust"**. Break tasks into steps (bed in stages); alternate heavy and light; **stop, rest, recover, restart** at moderate breathlessness (Borg 3–4); stairs one step per out-breath; interval walking at a slower pace; sit to wash and dress, rest between, dressing gown instead of towelling.

W25 skeleton: definition (1) → mechanism with diagram (3) → treatment table (3).

Oxygen therapy: devices, targets and hazards

In 4 papers.

Full notes p. 290

Asked: 7 marks: low-flow oxygen therapy (S20). 3 marks: methods of delivery (S23), hazards (S25), hypoxic drive (W25, see COPD (p. 12)).

Oxygen is a drug

Given to correct or prevent **hypoxaemia**, prescribed with a **target SpO_2**; it does not treat breathlessness itself.

Most acutely ill adults	**94–98%**
CO_2 risk (COPD, obesity, chest wall, neuromuscular, CF)	**88–92%**, guided by ABG
Critical, peri-arrest, CO poisoning	Reservoir mask 15 L/min, then titrate

Indications: acute hypoxaemia (pneumonia, asthma, heart failure, PE, post-op), critical illness, CO poisoning, exercise desaturation, long-term oxygen (see COPD). **Not** for breathless patients with normal SpO_2 or MI with $SpO_2 \geq 90\%$.

Methods of delivery (3m S23)

Low-flow (variable performance): less than the patient's inspiratory flow, room air mixes in, FiO_2 varies. **High-flow (fixed performance):** supplies all of it, FiO_2 fixed.

Device	Flow	FiO_2	Points
Nasal cannula	1–4 (to 6) L/min	~24–40% (~4% per L)	Eat, talk; falls with mouth breathing
Simple mask	5–10	~35–60%	**Never below 5 L** (CO_2 rebreathing); not for CO_2 retainers
Reservoir (non-rebreather)	15, bag inflated	~60–90%	Emergencies, CO
Venturi	Printed on valve	Fixed 24, 28, 31, 35, 40, 60%	**Best for CO_2 risk** (start 24–28%); flow +50% if RR above 30
High-flow nasal	to ~60 L/min, heated, humid	21–100%	Some PEEP, washes out CO_2; type 1 failure, post-extubation
Trach mask, T-piece	Humidified	Set	Tracheostomy, tube
Head box, incubator	Set	Analyser	Newborns

Sources: wall supply, cylinders, concentrators.

Low-flow oxygen therapy (7m S20)

- **Definition:** flow lower than the patient's inspiratory flow → mixed with room air → FiO_2 varies.
- **Devices:** nasal cannula, simple mask, reservoir mask, tracheostomy mask (oxygen catheter in older books).
- **Uses:** mild–moderate hypoxaemia in stable patients; home and ambulatory oxygen; exercise desaturation.
- **Advantages:** simple, cheap, comfortable; can eat, talk, walk.
- **Disadvantages:** FiO_2 not exact (higher with slow shallow breathing, lower with fast), so less suitable when CO_2 must be controlled; nasal dryness.
- **Humidification:** not needed at low flow; consider for high flow over ~24 h, tracheostomy, dryness.
- **Monitor:** SpO_2 against target, RR, drowsiness (rising CO_2), ABG if at risk.

Hazards (3m S25)

- **CO_2 retention** in at-risk patients (V/Q and Haldane effect; drive plays a small part): drowsy, confused, acidotic.
- **Absorption atelectasis** (nitrogen washed out, oxygen absorbed, poorly ventilated alveoli collapse).
- **Oxygen toxicity** (FiO_2 above ~0.6 for long). **Retinopathy of prematurity.**
- **Fire**: no smoking, flames, chulha; no grease on fittings; face burns in smokers.
- **Masking deterioration**: good SpO_2 hides worsening breathing; watch the rate.
- Dry nose, nosebleeds, ear and nose pressure sores, mask claustrophobia, aspiration if vomiting; tubing trips, empty or heavy cylinders.

Physio role

Check prescription and target; never change oxygen without it. SpO_2 during walking; oxygen to keep SpO_2 above ~88–90% in training; portable cylinders or concentrators. Keep oxygen on during clearance and positioning; watch SpO_2 after turns and suction. Teach LTOT users: **≥ 15 h a day**, no smoking, clean cannula, pacing. **Oximeter errors:** cold fingers, poor circulation, movement, nail polish, **CO (falsely normal)**, over-reading in darker skin.

Humidification, nebulisers and inhalers

In 7 papers.

Full notes p. 294

Asked: 7 marks: principle of jet and ultrasonic nebulisers (S20); indications, contraindications and types of nebulisation (S24). 3 marks: humidification and uses (S17), why intubated patients need it (W17), three nebuliser conditions (S19), definitions (S21), MDI (W25).

Definitions (3m S21)

Humidification: adding water **vapour** (a gas) to inspired gas, to warm and moisten it. **Nebulisation:** turning a liquid (drug) into an **aerosol** of fine droplets to breathe in. **Aerosol:** tiny liquid or solid particles in a gas.

Humidification (3m S17, W17)

Normally the nose warms and humidifies air so that a few cm below the carina it is **37 °C, 100% RH, ~44 mg H_2O/L** (isothermic saturation boundary).

Intubated (W17): the tube bypasses the nose → cold dry gas → thick sticky secretions, slow cilia, damaged lining, **blocked tube**, collapse, heat and water loss. AARC 2012: humidify every invasively ventilated patient.

HME ("artificial nose")	Passive: traps heat and moisture from expiration; **≥ 30 mg/L**; not with thick, copious or bloody secretions, big leak, hypothermia, nebulisation, or low VT (dead space)
Heated humidifier	Active, heated water chamber, heated-wire circuit; **33–44 mg/L at 34–41 °C**: long ventilation, thick secretions, high-flow cannula, NIV
Bubble humidifier	Oxygen through room-temperature water: little real humidity
Saline nebulisation	Loosens sputum; tracheostomy care

Uses: intubated and tracheostomy patients, long high-flow oxygen, thick secretions, NIV, dry airway. **Hazards:** overheating and airway burns, water in tubing (washed in, infection, false triggers), contaminated water, HME resistance, dead space, blockage.

Nebulisers (7m S20, S24)

Droplet size: over 10 µm mouth and throat; 5–10 µm large airways; **1–5 µm small airways and alveoli (respirable)**; under 1 µm breathed out.

Jet	Gas 6–8 L/min through a narrow opening: low pressure (**Bernoulli, Venturi**) draws liquid up and shatters it; **baffle** removes big drops. Cheap, any drug, suspensions; noisy, slow (10–15 min), residual drug
Ultrasonic	**Piezoelectric crystal** vibrates; surface waves throw off droplets. Quiet, fast; warms some drugs, poor for suspensions (budesonide), costly
Vibrating mesh	Plate pushes liquid through a mesh of tiny holes. Efficient, little waste, quiet, portable, in-line on ventilators; expensive, careful cleaning

Indications (S24; S19): acute asthma or COPD (bronchodilators); CF and bronchiectasis (hypertonic saline or dornase before clearance; inhaled antibiotics); ventilated patients (in-line); croup; sputum induction; patients who cannot use inhalers.

Contraindications and care: drug allergy; hypertonic saline can cause bronchospasm (bronchodilator first, check peak flow); **COPD with CO_2 risk: drive with air, not oxygen**, oxygen by cannula (BTS); tachycardia, tremor, low K^+; aerosol infection risk: clean, dry, don't share.

Technique: sit up; 2–5 mL; mouthpiece (mask for children, very breathless); normal mouth breathing with occasional deep breaths; until it splutters; rinse mouth after steroids; wash and dry.

Inhalers (3m W25)

MDI: pressurised canister + propellant; metering valve gives a fixed dose (e.g. salbutamol 100 µg). Cap off, **shake**; breathe out away; seal lips; **start a slow deep breath (3–5 s) and press once at the start**; **hold up to 10 s**; wait 30–60 s before a second puff; rinse after steroids. For: small, quick, cheap. Against: needs coordination, throat deposition, cold spray.

Spacer: puff into the chamber, breathe over a few breaths: less coordination, less throat drug; mask for young children; in acute asthma **MDI + spacer = nebuliser** (Cochrane 2013).

DPI (Rotahaler, Accuhaler, Turbuhaler): released by a **fast deep breath**; no propellant or coordination; not for young children or severe attacks; don't breathe into it.

Electrotherapy and therapeutic exercise in heart, lung and vascular conditions

Wound modalities in 4 papers.

Full notes p. 298

Asked: 3 marks: modalities for an infected wound (S16), modalities with parameters for wound healing (S22). 7 marks: modalities for wound healing (S24), factors and modalities (S25): tables on Diabetes (p. 26). Exercise for circulation sits inside PAD, venous and mastectomy answers.

Pain

TENS after chest surgery: electrodes **either side of and parallel to** the thoracotomy, sternotomy or abdominal wound (not on it); conventional ~**80–100 Hz**, short pulses, strong comfortable tingling, during breathing exercises and coughing. Better than standard analgesia alone (review of 11 trials); an add-on. Also: pillow-supported cough, wound-up side-lying, relaxed sitting; cold for acute chest-wall pain; careful heat for neck and shoulder spasm; relaxation.

Swelling

Elevation + muscle pump	Post-op, venous, lymphatic; **not** in severe arterial disease
Compression	Venous disease, ulcers, lymphoedema, after DVT, burns scars; **ABPI first**: full ≥ ~0.8, reduced 0.5–0.8 with specialist, none below ~0.5
Intermittent pneumatic compression	DVT prevention, lymphoedema add-on; not over untreated DVT, infection, uncontrolled heart failure
MLD, decongestive therapy	Lymphoedema (see Mastectomy (p. 8))
Cold	Acute traumatic swelling; not on numb or ischaemic skin

Wounds and weak muscles

Wound modalities: high-voltage pulsed current (best evidence for slow pressure ulcers), ultrasound, UVC, low-level laser: parameters on the Diabetes sheet.

NMES of quadriceps: strong contractions **without breathlessness**: severe COPD, heart failure, ICU, after major surgery. Mixed evidence (COPD gains in 2016 meta-analysis; 2014 review no clear benefit; small gain after surgery 2023): a **bridge to active exercise**. Electrodes over quadriceps, smooth strong contraction, on–off cycles, 20–30 min several times a week.

Precautions

Pacemaker or ICD: no shortwave or microwave; TENS only away from chest and device with advice. **Ischaemic limb:** no heat (burns); check sensation and circulation. Anticoagulants: gentle. **DVT:** no massage or heat on that limb. **Cancer:** nothing over a tumour. Never over the anterior neck (carotid sinus), pregnant abdomen, broken or numb skin, or in a patient who cannot report discomfort.

Therapeutic exercise

Pain	Gentle AROM, relaxation, breathing control, posture, graded activity
Mobility	Shoulder after thoracotomy, mastectomy, CABG (within precautions); thoracic mobility; burns scar stretch
Posture	COPD rounded shoulders, post-thoracotomy lean, post-mastectomy raised shoulder, kyphosis: thoracic extension, scapular retraction, pectoral stretch, mirror
Arterial	Walk to moderate claudication, rest, repeat: **30–45 min, 3 a week, ≥ 12 weeks** (see PAD (p. 10))
Venous	Ankle pumps, heel raises, walking, elevation, ankle range, with compression
Lymphatic	Rhythmic exercise in compression, deep breathing, slowly progressive resistance

Buerger–Allen (classic): legs up ≥ 45° until pale (1–3 min) → hang down until pink (2–5 min) moving ankles and toes → flat, covered, ~5 min; repeat several cycles, a few times a day. Weak evidence; walking is first-line; avoid in rest pain, ulcer, gangrene.

Applied yoga in cardio-respiratory conditions

Syllabus 4(l). Not asked on its own in 2016–2026.

Full notes p. 302

Asked: not on its own in 2016–2026; possible 3- or 7-mark note or viva. Fits one or two lines in the long-term plan of COPD, asthma and cardiac answers.

What yoga includes

- **Asanas:** postures for stretch, strength, balance.
- **Pranayama:** slow deep breathing, **anulom-vilom** (alternate nostril), **bhramari** (humming), ujjayi; forceful **kapalbhati, bhastrika**; breath retention (**kumbhaka**).
- **Relaxation, meditation:** shavasana, yoga nidra.
- Ministry of AYUSH **Common Yoga Protocol**.

How it may help: slower deeper breathing and control; less anxiety and sympathetic drive; flexibility and posture; culturally familiar activity.

Evidence

After MI	**Yoga-CaRe** (India, 3,959, 2020): better self-rated health and return to activity; **no significant fall in major events**; good where usual rehab is unavailable
Asthma	Cochrane 2016: small gains in QoL and symptoms; no clear lung function change
COPD	2019 meta-analysis (11 trials, 586): ~**25 m more 6MWT**, slightly better FEV1, mainly breathing-focused yoga
Hypertension	Modest BP falls with regular breathing and relaxation: an add-on
Heart failure	Small trials: better QoL and tolerance with gentle supervised yoga

Claim vs evidence: yoga does not cure asthma or reverse heart disease or replace medicines. Write: "yoga as an adjunct to standard rehab: breathing-focused pranayama, gentle asanas and relaxation, avoiding breath-holding, straining and head-down postures".

Sample supervised session (stable COPD, cardiac Phase II onwards)

1. **Loosening** 5–10 min: neck, shoulders, wrists, ankles, gentle trunk.
2. **Asanas** 15–20 min, breathing normally: tadasana, vrikshasana near a wall, ardha kati chakrasana, supported trikonasana, vajrasana or sitting forward bends, bhujangasana if allowed; rests.
3. **Pranayama** 10 min: slow breathing with longer exhalation, anulom-vilom **without retention**, bhramari.
4. **Relaxation** 10 min: shavasana or yoga nidra.

3–5 times a week, as part of, not instead of, aerobic and strength training.

Avoid or adapt

- **Breath-holding and straining** (Valsalva): heart disease, hypertension, aneurysm, after eye or brain surgery.
- **Kapalbhati, bhastrika:** uncontrolled BP, heart disease, recent chest or abdominal surgery, hernia, pregnancy, epilepsy, asthma or COPD flare.
- **Inversions** (sirsasana, sarvangasana, halasana): hypertension, heart failure, glaucoma, reflux, neck problems, breathless patients.
- **After sternotomy:** no arm weight-bearing, prone on the chest or strong back bends for ~6–8 weeks.
- **Steroid osteoporosis:** no loaded flexion or twisting.
- **No session:** unstable angina, uncontrolled arrhythmia, exacerbation, fever.

Continue all medicines; stop with chest pain, dizziness, marked breathlessness; learn from a trained teacher in a supervised group.

Handling skills and exercise equipment: stockings, bandaging, taping, splints, monitors

Practical skills list; short-case demonstrations.

Full notes p. 305

Asked: practical exam only: short-case demonstrations and viva on stockings, bandaging, dressings, taping, splints and orthoses, exercise and monitoring equipment, and monitoring adherence (practical list 3, 5, 7).

Compression stockings

Anti-embolism stockings (bed-bound, DVT prevention) vs **graduated compression** (walking patients with venous disease or lymphoedema; highest at the ankle). Classes differ by country: follow the prescription.

Check first: ABPI (none below ~0.5; specialist 0.5–0.8), skin, sensation, uncontrolled heart failure. **Measure in the morning:** narrowest ankle, widest calf, heel to below the knee crease.

Donning: morning before getting up (or after 15–20 min elevation); turn inside out to the heel; foot in, heel centred; ease up a little at a time (rubber gloves, frame aid); smooth wrinkles; top band two fingers below the knee crease, **never rolled down** (tourniquet). **Off:** peel down inside out; at night unless prescribed; check skin daily; wash; replace every few months.

Bandaging and dressings

- **Venous ulcer:** after ABPI; ankle 90°; pad bony points; toes to below the knee incl. heel, spiral or figure-of-eight, ~50% overlap, firmest at the ankle; check toes after.
- **Lymphoedema:** multilayer short-stretch (see Mastectomy (p. 8)). **Stump:** figure-of-eight, firmest at the end, no circular turns (see Amputation (p. 18)).
- **Always loosen and report:** pain, numbness, tingling, pale or blue digits, swelling above the bandage.
- **Dressings:** work around them safely: no shear on dressings or grafts, support the wound on coughing, keep dry; burns exercise at dressing changes. **Report:** soaking, loose or wet dressing, spreading redness, heat, pus, smell, fever, wound opening. Types: gauze, film, hydrocolloid, foam (heavy exudate), alginate (very wet), silver (infected).
- **Taping:** kinesiology tape for mild lymphoedema, posture, chest-wall pain; limited evidence; clean intact skin, test patch; not over wounds, drains, grafts, irradiated or numb skin; remove if itching or blisters, peel along the hair.

Splints, binders, orthoses

Burns splints	Anti-deformity positions; check fit and pressure daily (swelling changes fit)
Ankle–foot splint	Foot drop, heel sores (ICU, GBS, head and spinal injury); heel off the mattress, skin checks each shift
Abdominal binder	Tetraplegia (breathing and BP sitting); post-op support if advised; over lower ribs and abdomen, not the chest
Sternal support vest	Some post-sternotomy patients; does not replace precautions
Collar, thoracolumbar brace	Spinal stability as ordered; log-roll; skin under the collar
Pressure garments	Burns scars, lymphoedema; ~23 h a day; replace as they loosen

Exercise and monitoring equipment

Treadmill	Show emergency stop and safety clip; start slow and flat; light hand support; small steps up; slow before stopping
Cycle, arm ergometer	Knee slightly bent at the bottom; low resistance first; arm crank at or below shoulder height (after sternotomy only when allowed)
HR monitor	Chest strap beats wrist; check against pulse; unreliable on beta-blockers and in AF
Pulse oximeter	Warm finger, no nail polish, steady trace; ear probe or recheck at rest when walking
Sphygmomanometer	Right cuff, arm at heart level, seated and rested; recheck high readings
Weights, bands	Load from 1RM or reps; breathe out on effort; progress with "2 for 2"
Pressure biofeedback	Pad under the lower back (~40 mmHg) or abdomen; keep or lower pressure while drawing in: core and posture
Balance board, Swiss ball	Balance and trunk training in deconditioned and post-ICU patients; near a rail, supervised

Monitoring adherence

How: exercise and symptom diaries, step counts, attendance, check inhaler and breathing technique each visit, open questions ("How many days did you walk last week?"). **Barriers:** cost, travel, work hours, fear of breathlessness or chest pain, low mood, poor understanding, family beliefs ("rest is best"), other illnesses. **Helps:** goals the patient chose; short picture programme in his language; link to routines; family; phone or WhatsApp reminders, video reviews; praise; adjust the plan.

Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests

In 9 papers.

Full notes p. 308

Asked: 9 papers. **7 marks:** absolute and relative contraindications (W18, W20, S26); exercise tests with advantages and disadvantages (S19). **3 marks:** Bruce protocol, cycle ergometer pros and cons, 6MWT vs shuttle, absolute contraindications.

Why test; types

Why: diagnosis (ischaemia, arrhythmia, desaturation, bronchospasm); prognosis and risk stratification; functional capacity (METs, distance); prescription (safe HR, walking speed, oxygen need); outcome. **Types:** laboratory (treadmill, cycle with ECG; CPET with gas analysis) and field (6MWT, shuttle, step tests).

Contraindications

Acute heart, acute clot, tight valve

Absolute: MI within 2 days; ongoing unstable angina; uncontrolled symptomatic arrhythmia; active endocarditis; **symptomatic severe aortic stenosis**; decompensated heart failure; acute PE, pulmonary infarction or DVT; acute myocarditis or pericarditis; acute aortic dissection; disability preventing safe testing.

Relative: left main stenosis; moderate–severe AS with unclear symptoms; tachyarrhythmia with uncontrolled rate; complete or advanced heart block; HOCM with severe gradient; recent stroke or TIA; mental impairment; **resting BP above 200/110**; uncorrected anaemia, electrolytes, hyperthyroidism.

When to stop

Absolute	Relative
ST elevation over 1 mm (no old Q; not aVR, aVL, V1)	ST depression over 2 mm, horizontal or downsloping
Systolic falls over 10 mmHg with rising work + ischaemia	Same fall without ischaemia
Moderate–severe angina	Increasing chest pain
Dizziness, near fainting, unsteadiness	Fatigue, breathlessness, wheeze, cramps, claudication
Pallor, cyanosis	**Systolic over 250 or diastolic over 115**
Sustained VT	Other arrhythmias, new bundle branch block
Monitoring failure; patient asks	–

Bruce protocol (3m S16)

Symptom-limited treadmill test; **speed and gradient rise every 3 min**; ECG, BP, symptoms; result = time, stage (METs), ischaemia, arrhythmia, BP response.

1	1.7 mph (2.7 km/h), 10%	~5 METs
2	2.5 mph (4.0), 12%	~7
3	3.4 mph (5.5), 14%	~10
4	4.2 mph (6.8), 16%	~13
5	5.0 mph (8.0), 18%	~15

Modified Bruce (elderly, deconditioned, early post-MI): two extra starting stages at 1.7 mph, **0% then 5%**, avoiding the big first jump.

Cycle ergometer (3m S22): for: exact workload in watts, independent of weight; still upper body (easy ECG, BP); safer with poor balance, obesity, orthopaedic problems; cheaper, quieter. Against: thighs tire first, so peak VO_2 ~10–20% lower than treadmill; unfamiliar; less walking-specific.

Field tests (3m W22)

6MWT: flat straight course ≥ 30 m; as far as possible in 6 min at own pace, rests allowed, standard encouragement each minute; record distance, SpO_2, HR, Borg before and after, stops; practice test. **Don't test:** unstable angina or MI in the past month; caution resting HR above 120, BP above 180/100. **Stop:** chest pain, intolerable breathlessness, cramps, staggering, sweating, pallor. **MCID ~25–33 m** (often 30 m) in chronic lung disease.

ISWT: 10 m course round two cones, paced by beeps, speed up every minute through 12 levels; ends when the cone is missed: maximal, relates to peak VO_2. **ESWT:** constant speed from ISWT; endurance time; sensitive to rehab.

	6MWT	ISWT
Pacing	Self	Beeps
Intensity	Submaximal, daily life	Incremental to maximum
Course	≥ 30 m	10 m
Result	Distance in 6 min	Distance or level
Best use	Function, desaturation, outcome	Peak capacity, training intensity

Harvard step test: 20-inch (~50 cm) step, **30 steps/min for 5 min**; 30-s pulse counts at 1–1.5, 2–2.5, 3–3.5 min. **Index = (seconds × 100) ÷ (2 × sum of counts)**: below 55 poor, 55–64 low average, 65–79 high average, 80–89 good, ≥ 90 excellent. Example: 300 s, counts 70 + 60 + 50 → 30000 ÷ 360 = **83 (good)**. Healthy people only.

Exercise physiology and prescription: METs, energy systems, intensity, training

In 6 papers.

Full notes p. 313

Asked: 3 marks: define MET with value and two activities (W22, S24, S25), aerobic vs anaerobic (W22), Karvonen (W25). 7 marks: ageing and a fitness programme (S16), methods of measuring intensity (S23), energy systems with examples (S24).

MET (3m W22, S24, S25)

1 MET = energy used sitting quietly at rest = **3.5 mL O_2/kg/min** (~1 kcal/kg/h). 4 METs = four times resting.

Eating, sitting bath	1.5	Toilet	2.3
Showering, grooming	2.0	Dressing	2.8
Walking ~2 mph	2.8	Walking ~3 mph	3.8
Stairs slowly	4.5	Brisk 3.5–3.9 mph	4.8
Stairs, general	6.8		

(2024 Compendium.) **Light** below 3, **moderate** 3–5.9, **vigorous** ≥ 6 METs. Phase I up to ~3 METs; good capacity ≥ 7 METs; sex when 3–5 METs tolerated.

Energy systems (7m S24)

ATP–PC	Creatine phosphate remakes ATP, no O_2: maximal efforts **up to ~10 s**: sprint, jump, single heavy lift
Anaerobic glycolysis	Glucose → **lactate** without O_2, fast, little ATP: **10 s–2 min**: 400 m, fast stair flights
Aerobic	Carbohydrate and fat with O_2 in mitochondria, slow, large yield: **over 2 min** and rest: walking, cycling, marathon

Fat dominates at rest and low intensity; carbohydrate as intensity rises. RQ: carbohydrate 1.0, fat 0.7.

(3m W22)	Aerobic	Anaerobic
O_2	With	Mainly without
Intensity, time	Low–moderate, long	High, seconds to 2 min
Fuel, by-product	Fat, carbohydrate; CO_2, water	CP, glucose; lactate
Fibres	Type I	Type II
Training	Capillaries, mitochondria, VO_2max	Power, speed, strength, lactate tolerance

Measuring intensity (7m S23)

	%HRR	%HRmax	RPE	METs
Light	30–39	57–63	9–11	2–2.9
Moderate	40–59	64–76	12–13	3–5.9
Vigorous	60–89	77–95	14–17	≥ 6

- **Karvonen (W25):** target HR = rest + intensity × (max – rest); max = 220 – age (or 208 – 0.7 × age). Age 60, rest 70: max 160, HRR 90, 50% → **115**.
- % HRmax; % VO_2max or VO_2 reserve (CPET = gold standard); METs; **RPE** 6–20 or CR10 (3–5 in pulmonary rehab); **talk test** (talk not sing = moderate); ~80% of 6MWT or shuttle speed; ventilatory or lactate threshold; **≥ 10 beats below the ischaemic threshold**; % 1RM for strength.
- Use RPE and talk test with **beta-blockers, AF, pacemakers, heart transplant**.

Training principles and adaptations

FITT-VP; overload, specificity, progression, individuality, **reversibility** (gains fade in weeks), warm-up and cool-down.

Acute: ↑ HR, stroke volume, output, systolic BP (diastolic similar), ventilation, muscle blood flow and O_2 extraction. **Chronic:** ↓ resting and submaximal HR, ↑ stroke volume and max output, ↓ resting BP; less ventilation per load; more capillaries and mitochondria, ↑ VO_2max; strength and size with resistance; better insulin sensitivity, HDL, weight.

Ageing and fitness (7m S16)

Heart: ↓ max HR, stiffer arteries (↑ systolic), slower HR response, ↓ VO_2max. **Lungs:** ↓ recoil, stiff chest wall, ↑ RV, ↓ VC, earlier airway closure, weaker muscles, slightly ↓ PaO_2. **Muscle, bone:** sarcopenia (fast fibres), bone loss, stiff joints, poor balance.

Programme (WHO 2020): 150–300 min moderate (or 75–150 vigorous) a week; strength ≥ 2 days; balance and strength ≥ 3 days (falls); start low, go slow; longer warm-up and cool-down; check drugs and heart first.

Investigations: chest X-ray, PFT, ECG and bedside tests

3-mark questions in most papers; four of the five practical spots.

Full notes p. 317

Asked: 3 marks in most papers: X-ray of effusion, emphysema, COPD, TOF; spirometry in restriction, obstruction, COPD; ECG of MI; ABI. 7 marks: ABG, PFT and X-ray in COPD goal setting (W24). **Practical spots:** X-ray, ABG, PFT, ECG, devices.

Chest X-ray: read in order

1. **Details and quality:** name, date, PA or AP (AP enlarges the heart), rotation (spinous processes midway between clavicle ends), inspiration (~6 anterior or 10 posterior ribs), exposure.
2. **A** airway (trachea central?); **B** lungs and pleura zone by zone, costophrenic angles; **C** cardiac (**CTR below 0.5** on PA), borders, mediastinum; **D** diaphragm (right slightly higher; flat in hyperinflation; free air); **E** everything else (bones, sternal wires, surgical emphysema, lines and tubes).

Silhouette sign: a white area blurring a border lies next to it: lost right heart border = right middle lobe; lost hemidiaphragm = lower lobe.

Consolidation	White, air bronchograms, volume kept → positioning, mobility; clearance only if sputum retained
Collapse	White, smaller lobe; fissure, trachea, mediastinum pulled towards; raised diaphragm → target that lobe
Effusion	Dense base, meniscus, blunted angle; large pushes away → good side down, expansion; drain first if large
Pneumothorax	Black, no markings beyond a pleural line; tension pushes away → no positive pressure until drained
Hyperinflation	Dark lungs, flat diaphragms, over 6 anterior ribs, narrow heart → forward lean, pursed lips; bullae = no high pressure
Pulmonary oedema	Bat's wing, Kerley B, big heart, effusions → sit up; no clearance for oedema
ILD	Fine reticular basal shadows, small lungs → oxygen in training
Bronchiectasis	Tram lines, ring shadows → daily clearance
Lung abscess	Cavity with air–fluid level → abscess uppermost
Tubes	ETT tip ~5 cm above carina (right bronchus → left lung collapse); NG below diaphragm; drain in pleura

Pulmonary function tests

FVC total forced exhalation; **FEV1** first second; **FEV1/FVC** normally ~0.75–0.80; **PEF** fastest flow (asthma diary); **TLC, RV, FRC** (body box or dilution): high in hyperinflation, low in restriction; **DLCO** low in emphysema, ILD, vascular disease, normal in asthma.

	FEV1	FVC	Ratio	TLC	Examples
Obstructive	↓	N or ↓	↓ **(below 0.70)**	N or ↑	COPD, asthma, bronchiectasis
Restrictive	↓	↓	**N or ↑**	↓	ILD, pleural thickening, kyphoscoliosis, obesity, neuromuscular, post-resection
Mixed	↓	↓	↓	↓	Post-TB, CF

Bronchodilator response: FEV1 or FVC rise ≥ 12% and 200 mL (ATS/ERS 2005); 2022 standard: **over 10% of predicted**. Large = asthma. **Flow–volume loop:** scooped expiration in obstruction; small narrow loop in restriction; flattened in upper airway obstruction. **S25:** FVC 60%, FEV1 82%, ratio 94% predicted = **restrictive** (old TB scarring).

ECG

25 mm/s: small square **0.04 s**, large **0.2 s**; 10 mm = 1 mV. P atria, QRS ventricles depolarise, T ventricles repolarise. **PR 0.12–0.20 s** (3–5 small); **QRS below 0.12 s**; QTc below ~0.44 (men), 0.46 (women). **Rate:** 300 ÷ large squares between R waves (regular), or QRS in 10 s × 6 (irregular). **Order:** rate, rhythm, P before every QRS, PR, QRS width, ST and T.

Sinus tachycardia	Above 100: find cause; resting above 120 → postpone
Sinus bradycardia	Below 60: beta-blockers, fit people; stop if dizzy or BP falls
AF	**No P, irregularly irregular**: exercise if rate controlled, use RPE; anticoagulant bleeding
Ectopics	Early beats (ventricular wide): frequent, multiform or runs in exercise → stop
VT	Fast wide regular: emergency; no pulse → CPR
VF	Chaotic, no QRS: arrest → CPR, defibrillate
Heart block	1st: PR above 0.20; Mobitz I: PR lengthens then drops; Mobitz II: sudden drops; 3rd: P and QRS unrelated. Mobitz II, 3rd: no exercise until paced
ST change	Elevation in neighbouring leads = acute MI; depression = ischaemia → stop, report (see IHD (p. 3))
Paced	Spikes before P or QRS: device precautions

Bedside cardiovascular tests

- **Pulses:** radial, brachial, carotid (one side), femoral, popliteal, dorsalis pedis, posterior tibial; 0 absent, 1+ weak, 2+ normal, 3+ bounding.
- **Postural hypotension:** BP after 5 min lying, then 1 and 3 min standing: **fall ≥ 20 systolic or ≥ 10 diastolic within 3 min** (2011). Bed rest, diuretics, antihypertensives, diabetes, SCI.
- **ABI:** highest ankle ÷ highest arm per leg: 0.91–1.40 normal, ≤ 0.90 PAD, below 0.40 severe, above 1.40 calcified (see PAD (p. 10)).
- **Claudication distance:** treadmill ~3.2 km/h at 12% (or graded, or 6MWT): pain-free and maximal walking distance.
- **Capillary refill:** press nail bed 5 s, colour back within ~2–3 s.
- Auscultation: see Chest signs (p. 66); ABG: see ABG (p. 65).

Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires

In 5 papers.

Full notes p. 322

Asked: 3 marks: Borg CR10 (S16, S18), modified Borg (S26), NYHA grades (S21), one QoL questionnaire each for COPD, heart failure and PAD (W24). mMRC is quoted inside cases.

Borg RPE (6–20)

Overall exercise effort; ≈ HR ÷ 10 in healthy young adults (13 ≈ 130).

6 No exertion	7 Extremely light	9 Very light
11 Light	**13 Somewhat hard**	15 Hard
17 Very hard	19 Extremely hard	20 Maximal

Even numbers have no words. Phase I **below 13**; Phase II 11–16 (start 11–13); moderate 12–13.

Modified Borg / CR10 (3m S16, S18, S26)

3 moderate, 5 severe, 7 very severe, 10 maximal

0 Nothing at all	0.5 Very, very slight	1 Very slight
2 Slight	**3 Moderate**	4 Somewhat severe
5 Severe	6 –	**7 Very severe**
8 –	9 Very, very severe	**10 Maximal**

Borg's own CR10 words: 0.5 extremely weak, 2 weak, 5 strong, 7 very strong, 10 extremely strong; ratings above 10 allowed. **Category–ratio:** word anchors (categories) spaced so 4 feels about twice 2 (ratio). Uses: before and after 6MWT, pulmonary rehab training at 3–5, stop above 7, pacing.

mMRC dyspnoea scale

0 only with strenuous exercise; **1** hurrying on the level or a slight hill; **2** slower than peers or stops at own pace on the level; **3** stops after ~100 m or a few minutes; **4** too breathless to leave the house, or when dressing. Daily-life breathlessness; GOLD "more symptoms" = mMRC ≥ 2.

NYHA (3m S21) and CCS

	NYHA (heart failure)	CCS (angina)
I	No limitation	Angina only with strenuous, rapid or prolonged exertion
II	Slight; ordinary activity → symptoms	Slight: walking or stairs rapidly, uphill, after meals, cold, stress; over 2 blocks or 1 flight
III	Marked; less than ordinary activity → symptoms, comfortable at rest	Marked: 1–2 blocks or 1 flight at normal pace
IV	Symptoms at rest	Any activity; may occur at rest

Quality-of-life questionnaires (3m W24)

COPD	**SGRQ** (0–100, higher worse, 4 points matters); CRQ (0.5 per item); CAT (8 items, 0–40; ≥ 10 high impact)
Heart failure	**MLHFQ** (21 items, 0–105, higher worse); KCCQ (0–100, higher better)
PAD	**WIQ** (distance, speed, stairs; 0–100, higher better); VascuQoL (25 items)
Generic	SF-36, EQ-5D, WHOQOL-BREF

Generic vs disease-specific (syllabus 10a): generic compares across diseases but is less sensitive to change; disease-specific detects small changes after rehab: use one of each, and compare the patient's own rating with your assessment (10b).

ABG interpretation and why patients desaturate

In 3 papers.

Full notes p. 326

Asked: 3 marks: reasons for desaturation with exercise (S21), causes of respiratory acidosis (S24). Inside COPD and ICU cases; ABG, PFT and X-ray in COPD goal setting (W24, 7 marks); a practical spot.

Normal values

pH	**7.35–7.45**; below = acidaemia, above = alkalaemia
$PaCO_2$	**35–45 mmHg** (4.7–6.0 kPa): respiratory part, set by ventilation
HCO_3^-	**22–26 mmol/L**: metabolic part
PaO_2	**80–100 mmHg** (10.7–13.3 kPa), lower in the elderly; **below 60 = respiratory failure**
SaO_2	95–98%
Base excess	–2 to +2: negative in metabolic acidosis

Four steps

1. **pH:** acidaemia or alkalaemia?
2. **Cause:** $PaCO_2$ opposite to pH = respiratory; HCO_3^- same direction as pH = metabolic.
3. **Compensation:** the other system moves the same way as the cause; partial (pH still abnormal) or full (pH normal, never over-corrected).
4. **Oxygenation:** PaO_2 against FiO_2 (P/F ratio).

ROME

Respiratory **O**pposite, **M**etabolic **E**qual.

Acute resp. acidosis	HCO_3^- +1 per 10 mmHg $PaCO_2$ rise
Chronic resp. acidosis	HCO_3^- +3.5–4 per 10
Acute resp. alkalosis	HCO_3^- –2 per 10 fall
Metabolic acidosis	Expected $PaCO_2$ = 1.5 × HCO_3^- + 8 (± 2) (Winter's)
Metabolic alkalosis	$PaCO_2$ +0.7 per 1 mmol/L HCO_3^-

Worked examples

- **pH 7.28, $PaCO_2$ 62, HCO_3^- 28:** respiratory acidosis; expected HCO_3^- ~26 acute, ~32 chronic; 28 between → **acute-on-chronic** (COPD exacerbation).
- **pH 7.48, $PaCO_2$ 30, HCO_3^- 22:** acute respiratory alkalosis (anxiety, early asthma, PE, pain).
- **pH 7.25, $PaCO_2$ 28, HCO_3^- 12:** metabolic acidosis; expected $PaCO_2$ 1.5 × 12 + 8 = 26 ± 2 → appropriate (DKA, Kussmaul).

Causes (3m S24: respiratory acidosis)

Respiratory acidosis (hypoventilation)	**Brain:** sedatives, opioids, head injury, stroke. **Nerves and muscles:** GBS, myasthenia, high SCI, dystrophy, exhausted muscles. **Chest wall:** flail chest, kyphoscoliosis, obesity hypoventilation. **Lungs:** COPD exacerbation, late severe asthma, pneumonia, oedema, obstruction. Ventilator too low
Respiratory alkalosis	Anxiety, pain, fever, hypoxaemia (pneumonia, PE, early asthma), altitude, pregnancy, ventilator too high
Metabolic acidosis	DKA, lactic acidosis (shock, sepsis), kidney failure, diarrhoea, poisons
Metabolic alkalosis	Vomiting, NG suction, diuretics, low K^+, antacids

Respiratory failure

Type 1 (hypoxaemic): PaO_2 below 60, $PaCO_2$ normal or low: gas exchange (pneumonia, oedema, ARDS, PE, ILD). **Type 2** (hypercapnic): PaO_2 below 60 + $PaCO_2$ above ~45–50: pump failure or high load (COPD, neuromuscular, chest wall, sedation). **P/F** normal ~400–500 (95 ÷ 0.21 ≈ 450); ARDS 201–300 mild, 101–200 moderate, ≤ 100 severe. Low pH = acute (NIV); normal pH + high HCO_3^- = chronic.

Exercise desaturation (3m S21) and physio meaning

- **Worse V/Q** as output rises; **diffusion limitation** (faster flow, less transit time; ILD, emphysema with low DLCO); **hypoventilation** (dynamic hyperinflation, tired muscles); **low mixed venous O_2** (more extraction); pulmonary hypertension, shunt through a PFO; **measurement error** (cold fingers, movement: use ear probe).
- SpO_2 in the 6MWT; **oxygen in training if below ~88%**; intervals; ILD desaturates most.
- **Acute type 2:** gentle chest care, support NIV, no training, report drowsiness. **Chronic CO_2 retention:** SpO_2 88–92%. **Kussmaul breathing** is compensation, not a breathing problem: inform the team. Recheck after big changes.

Abnormal breathing patterns and chest signs

In 2 papers.

Full notes p. 330

Asked: 3 marks: Cheyne–Stokes respiration (S16); three abnormal breathing patterns (S22). Draw the shape and write one cause: a full answer for each pattern.

Breathing patterns

Tachypnoea	Above ~20/min: fever, pneumonia, oedema, PE, anxiety, pain, ILD
Bradypnoea	Below ~12/min: opioids, sedatives, raised ICP, hypothermia, hypothyroidism
Kussmaul	Deep, fast, laboured, regular: **metabolic acidosis** (DKA, kidney failure) blowing off CO_2
Cheyne–Stokes (S16)	**Breaths gradually deepen then become shallow, then an apnoea of seconds; repeats**: heart failure (slow circulation delays chemoreceptor feedback), stroke, brain injury, altitude, opioids; normal in sleeping infants and elderly
Biot's (ataxic)	Groups of quick equal breaths with irregular pauses (ataxic: completely irregular): medulla damage (meningitis, raised ICP, brainstem stroke, opioids): grave sign
Apneustic	Deep gasping inspiration held, short expiration: pons damage
Central neurogenic hyperventilation	Sustained deep rapid breathing: midbrain, upper pons
Paradoxical	Part moves the wrong way: flail chest (in on inspiration); diaphragm paralysis or fatigue (abdomen in on inspiration); obstruction
Prolonged expiration	Pursed lips: COPD, asthma
Sighing, upper chest	Dysfunctional breathing, hyperventilation

Orthopnoea (lying flat), PND (wakes breathless), platypnoea (upright, better lying).

Chest signs

- **Barrel chest** (↑ AP diameter): hyperinflation (emphysema, chronic asthma).
- Pectus excavatum or carinatum, kyphoscoliosis: restriction when severe.
- Accessory muscles, intercostal indrawing, tracheal tug: ↑ work of breathing.
- **↓ Expansion one side** = disease on that side.
- **Trachea** pulled towards collapse or fibrosis; pushed away by large effusion or tension pneumothorax.
- **Clubbing:** bronchiectasis, CF, abscess, empyema, cancer, ILD, cyanotic heart disease (**not COPD alone**).
- **Central cyanosis** (tongue, lips) = low arterial O_2; peripheral only = poor circulation or cold.

	Percussion	Breath sounds
Normal	Resonant	Vesicular
Consolidation	Dull	**Bronchial**, crackles, ↑ vocal resonance
Collapse	Dull	↓ or absent
Effusion	**Stony dull**	↓ or absent, ↓ vocal resonance
Pneumothorax	**Hyper-resonant**	↓ or absent
Emphysema	Hyper-resonant	Quiet, prolonged expiration

Added sounds

- **Wheeze:** musical, narrowed airways (asthma, COPD); fixed single wheeze = local obstruction.
- **Coarse crackles:** secretions in larger airways (bronchiectasis, infection); **change after a cough**: a sign for airway clearance.
- **Fine crackles:** late inspiratory "Velcro" (fibrosis); fluid (pulmonary oedema).
- **Pleural rub:** creaking with breathing (pleurisy).
- **Stridor:** harsh inspiratory noise, upper airway narrowing: emergency.

Cardiac rehabilitation: definition, indications, contraindications, phases, outcome measures

Part of the most-asked long answer; 7-mark questions in 5 papers. Full notes p. 334

Asked: inside 16 IHD long cases; 7 marks: Phase I after MI (W16, W17), risk stratification (S21), Phase 1–2 after valve replacement (W21), Phase 2 after PTCA for sport (W24); 3 marks: Phase I (S20), Karvonen (W25). Doses and examples are on IHD (p. 3).

Definition, aims, indications

Cardiac rehabilitation: a coordinated, multi-part programme that restores physical, psychological and social function and slows the disease through **exercise, education and risk-factor control** (AHA/AACVPR). WHO: all activities so patients, by their own efforts, regain as normal a place as possible in the community.

Aims: safe return to daily life, work, leisure; ↑ exercise capacity; risk-factor control; fewer events and admissions; better QoL and mood; lifelong self-management.

Indications: MI or ACS; after CABG or PCI; stable angina; stable HFrEF; valve repair or replacement (incl. TAVI); heart transplant; also pacemaker or ICD, cardiomyopathy, adult congenital disease, PAD (walking programme), high-risk people. Age, low EF, ICD and diabetes are reasons to **stratify and supervise**, not to exclude.

Contraindications (ACSM)

Unstable angina; resting BP above 180/110; symptomatic drop over 20 mmHg on standing; significant aortic stenosis; uncontrolled arrhythmia or sinus rate over 120; decompensated heart failure; 3rd-degree block without pacemaker; active peri- or myocarditis; recent embolism, acute thrombophlebitis, aortic dissection; fever; uncontrolled diabetes or potassium; severe orthopaedic or psychological problems.

Core components (AHA/AACVPR 2024)

Patient assessment (ECG, exercise test or 6MWT, risk stratification, goals); nutrition (salt below 5 g); weight (Indian cut-offs); risk factors (BP, lipids, diabetes, tobacco, medicines); psychosocial (depression, stress, family, work); aerobic training (FITT-VP); strength training 2–3 days; activity counselling (≥ 150 min/week, less sitting); quality improvement.

Phases I–IV

I inpatient	Stable day to discharge: prevent bed-rest effects, breathing exercises after surgery, graded self-care and walking, stairs, education, GTN, home plan, referral. **~3 METs; RPE below 13; HR ≤ rest + 20 (MI) or + 30 (surgery)**
II early outpatient	Within weeks of discharge, **8–12 weeks**, centre, home or hybrid: exercise test, risk stratification, monitored aerobic and strength training, risk factors, psychology, return to work, driving, sex. **40–80% HRR or RPE 11–16**, start low; 3–5 days; resistance 2–3 days
III community	Months: less supervision, self-monitoring with pulse and RPE
IV lifelong	Independent: **≥ 150 min/week moderate + 2 days strength**; periodic review

Across conditions

- **MI, CABG:** Phase I limits; sternal precautions 6–8 weeks; legs first.
- **PCI:** no sternal limits; protect the puncture site ~1 week; faster progress.
- **Heart failure:** start RPE 11–13, intervals, long warm-up and cool-down; watch weight gain; IMT.
- **Valve surgery:** sternal precautions; AF; anticoagulation (bleeding, falls).
- **Pacemaker, ICD:** arm below shoulder 4–6 weeks; HR ≥ 10 below the detection rate.
- **Transplant:** denervated heart, HR rises slowly: RPE, long warm-up. **AF, beta-blockers:** RPE and talk test. **PAD:** walk to moderate claudication.

EF 50 / 40, METs 7 / 5

Risk (AACVPR): low = EF ≥ 50% and ≥ 7 METs; high = EF below 40%, symptoms below 5 METs, complex arrhythmias or complicated course; sets ECG monitoring and progression.

Outcome measures and evidence

Capacity: exercise test (METs, VO_2), 6MWT, ISWT, sit-to-stand. **Risk factors:** BP, lipids, HbA1c, weight, waist, smoking. **Symptoms:** CCS, NYHA, Borg. **QoL:** MacNew, Seattle Angina, MLHFQ or KCCQ, SF-36, EQ-5D. **Mood:** HADS, PHQ-9. **Behaviour:** steps, adherence, return to work. **Events:** readmissions, MI, revascularisation.

Evidence: MI ↓ ~28%, admissions ↓ ~35%, probably fewer deaths, better QoL (Cochrane 2026: 107 trials, 26,886 people); home and digital programmes as good as centre-based; Yoga-CaRe better health, not fewer events. Few eligible join (especially in India and women): refer at discharge.

7-mark skeleton: definition and aims → indications → contraindications (any six) → phases table → outcomes and benefits.

Pulmonary rehabilitation: definition, indications, contraindications, components, outcome measures

Long-term part of every chronic lung disease case. Full notes p. 339

Asked: the long-term part (5–8 marks) of every COPD, bronchiectasis, ILD and post-TB case: write pulmonary rehab with doses. 7 marks: paediatric asthma rehab (S18), energy conservation (S19), ILD (S21), ABG, PFT, X-ray in COPD goals (W24).

Definition and aims

ATS/ERS 2013: "a comprehensive intervention based on a thorough patient assessment followed by patient-tailored therapies that include, but are not limited to, **exercise training, education, and behavior change**, designed to improve the physical and psychological condition of people with chronic respiratory disease and to promote the long-term adherence to health-enhancing behaviors."

Aims: less breathlessness and fatigue; ↑ capacity and function; better QoL and mood; fewer exacerbations and admissions; lasting self-management. **Why it works:** it treats what can change: weak muscles, poor breathing pattern, fear, inactivity, poor self-management (the breathlessness spiral).

Indications and contraindications

Indications: COPD limited by breathlessness (mMRC ≥ 2) and **soon after an admission** (ATS 2023 strong); **ILD** (strong); bronchiectasis, CF, post-TB; asthma with persistent breathlessness; pulmonary hypertension (conditional, supervised); lung cancer, before and after resection, LVRS, transplant; chest wall and neuromuscular disease; post-ICU, post-COVID. Age, smoking and low O_2 do not exclude.

Contraindications: unstable heart disease (unstable angina, recent MI, uncontrolled arrhythmia, severe AS, decompensated failure); locomotor or neurological problems preventing exercise; severe cognitive or psychiatric illness; **acute exacerbation** (treat first); severe pulmonary hypertension with exertional syncope, uncontrolled BP (specialist only).

Assessment

History, comorbidities (heart, diabetes, osteoporosis, anxiety, depression), drugs and inhaler technique, smoking; spirometry; mMRC; CAT, CRQ or SGRQ; HADS; **6MWT or ISWT with SpO_2** (training speed; oxygen if below ~88%); quadriceps, grip, sit-to-stand, balance; the patient's own goal ("reach my 2nd-floor flat without stopping").

Exercise training (the core)

	Endurance	Strength
F	3–5 days a week	2–3 days a week
I	Walk at **~80% of 6MWT or shuttle speed**, or cycle 60–80% peak; **Borg 4–6**	**60–70% 1RM**, 8–12 reps
T	20–60 min; **intervals** (1 min on, 1 off) if too breathless	1–3 sets, legs first, then arms, trunk
T	Walking, treadmill, cycle; arm ergometer	Weights, bands, sit-to-stand, step-ups; water bottles, sand bags at home

Add-ons: IMT if weak; NMES for very weak or bed-bound; flexibility, posture, balance; **oxygen if SpO_2 below ~88%**.

Education, support, structure

Education: the disease; inhaler and spacer; **written action plan**; pursed lips, breathing control, positions of ease; airway clearance if sputum; energy conservation and pacing; quitting; vaccines; nutrition; travel and oxygen; panic management; advance care planning. **Psychosocial and nutrition:** screen and treat anxiety and depression, groups, diet, family.

Structure: 6–12 weeks, ≥ 2 supervised sessions a week + home exercise; centre, home or telerehab (ATS 2023: offer the choice); then a maintenance plan (gains fade within months).

India: few centres; home programme with stairs, chair, water-bottle weights, walking, breathing exercises and pranayama, reviewed by phone or video.

Outcome measures and evidence

6MWT	MCID ~25–33 m (30)
ISWT	~47.5 m; ESWT, cycle test
Breathlessness	mMRC (1 grade), Borg, CRQ (0.5 per item)
Health status	CAT (2 points), SGRQ (4 points)
Other	HADS; sit-to-stand, quadriceps, grip, balance; steps, exacerbations, admissions

Evidence: COPD gains above the MCID in breathlessness, fatigue, QoL, capacity (Cochrane 2015); after an admission ↓ readmissions (2016); telerehab similar (2021). **Old answer** "breathing exercises and postural drainage" is wrong: **exercise training is the core**.

7-mark skeleton: definition and aims → indications and contraindications → assessment → components (exercise table, education, psychosocial, length) → outcomes.

Respiratory anatomy, physiology and mechanics of the thorax

Revision; spirogram and oxygen curve to draw.

Full notes p. 343

Asked: not on its own; underpins every technique answer. Related 3-mark questions: zones of the lung and altered V/Q (see Positioning (p. 44)), spirometry patterns (see Investigations (p. 63)).

Airways and lobes

Nose warms, humidifies, filters; carina at the sternal angle (T4–T5); **right main bronchus wider, shorter, more vertical** (aspiration goes right). ~23 generations: **0–16 conducting** (dead space ~150 mL), **17–23 respiratory**. Type I alveolar cells exchange gas; **type II make surfactant** (stops small alveoli collapsing); macrophages clear particles. Mucociliary escalator slowed by smoking, dry gas, infection.

Right lung 3 lobes (oblique + horizontal fissures), **left 2** (upper with lingula, lower). **Segments:** right upper apical, posterior, anterior; middle lateral, medial; lower superior + 4 basal (medial, anterior, lateral, posterior). Left upper apicoposterior, anterior; lingula superior, inferior; lower superior, anteromedial, lateral, posterior basal.

Surface markings: oblique fissure from T3 spine round to the 6th costal cartilage (lower lobes mostly behind); horizontal fissure along the 4th costal cartilage; lung border 6th rib (MCL), 8th (MAL), 10th (back); pleura two ribs lower. **Nerves:** vagus constricts, β2 dilates (salbutamol), **phrenic C3–C5**. Pulmonary artery ~25/10 mmHg.

Mechanics

Pump handle	Upper ribs 1–6, sternum forward and up: ↑ AP diameter
Bucket handle	Lower ribs 7–10 swing up and out: ↑ transverse
Caliper	Floating ribs 11–12 outwards
Diaphragm descent	↑ vertical: biggest share (~¾ of quiet VT)

Zone of apposition: diaphragm against the lower ribs lifts and widens them. Hyperinflation flattens it → Hoover's sign; forward lean re-domes it.

Muscles: quiet inspiration: diaphragm, external and parasternal intercostals, scalenes. Forced inspiration: SCM, scalenes, upper trapezius, pectorals (arms fixed), serratus, back extensors. **Quiet expiration: passive recoil.** Forced expiration and cough: abdominals, internal intercostals, latissimus.

Pressures: pleural −5 at rest, −7.5 end-inspiration; alveolar −1 in, +1 out (cmH_2O). **Compliance:** lungs ~200, lungs + chest ~110 mL/cmH_2O; low in fibrosis, oedema, obesity, deformity; high in emphysema. Resistance mainly in medium bronchi. Work of breathing 3–5% of energy at rest.

Volumes (young adult man)

TV 0.5 L	IRV 3.0 L	ERV 1.1 L	RV 1.2 L
IC = TV + IRV 3.5	**FRC = ERV + RV 2.3**	VC = IRV + TV + ERV 4.6	TLC 5.8

RV, FRC, TLC need body box or dilution. **FRC** falls after surgery, with pain, obesity, lying flat → basal airway closure → collapse; restore with upright positions, deep breaths, walking. Minute ventilation = VT × rate (~6 L/min).

Gas exchange, transport, control

Diffusion slows with a thick (fibrosis, oedema) or small (emphysema, resection) membrane. O_2: **98.5% on Hb**, 1.5% dissolved; 1 g Hb carries 1.34 mL (anaemia ↓ delivery despite normal SpO_2). CO_2: bicarbonate ~70%, carbamino ~23%, dissolved ~7%. **Bohr:** CO_2 and acid release O_2 in tissues. **Haldane:** O_2 binding releases CO_2 in lungs (why high O_2 raises CO_2 in COPD). **Dissociation curve:** flat top; SpO_2 stays above 90% until PaO_2 ~60, then falls fast: why **60 mmHg** defines failure.

Control: medulla (rhythm); pons (pneumotaxic fine-tuning); **central chemoreceptors** (CO_2 via CSF acidity: main drive); **peripheral** (carotid, aortic: low O_2, strongly below ~60); stretch (Hering–Breuer), irritant, J receptors; cortex (voluntary) and limbic (anxiety).

Physiology behind techniques

Collateral ventilation (pores of Kohn, canals of Lambert, channels of Martin) → deep breaths with hold, SMI, PEP. **Equal pressure point** → mid-volume huffs. **Gravity** → positioning, drainage. **Zone of apposition** → forward lean, binder. **FRC** → upright, walking. **Surfactant** → regular deep breaths. **Mucociliary clearance** → humidification, hydration, quitting. **Chemoreceptors, Haldane** → 88–92% in COPD. **Cortical control** → breathing control, pursed lips, retraining.

Cardiac anatomy and physiology for physiotherapists

Revision; coronary territories and the conduction system.

Full notes p. 348

Asked: not on its own; the coronary territories table is used in MI cases (II, III, aVF = inferior, RCA). Revision for the cardiac pages.

Anatomy and coronary arteries

Middle mediastinum; apex beat left 5th space, mid-clavicular line. Right side: RA → **tricuspid** → RV → **pulmonary** valve → lungs (low pressure). Left: LA → **mitral** → LV → **aortic** valve → body (high pressure, thickest wall). Pericardium: effusion or tamponade restricts filling.

Artery	Supplies	Blocked: leads
LAD	Anterior LV wall, front ⅔ septum, apex	V1–V4
Circumflex	Lateral and posterior LV, LA	I, aVL, V5–V6
RCA	RA, RV, inferior LV, back of septum; **SA node and AV node** in most	II, III, aVF: watch bradycardia and block

Veins → coronary sinus. Sympathetic β1 speeds and strengthens; vagus slows.

Conduction

SA node (right atrium, natural pacemaker **60–100/min**) → atria contract (**P wave**) → **AV node** (delay ~0.1 s lets ventricles fill: **PR interval**; backup **40–60/min**) → **bundle of His** → right and left bundle branches → **Purkinje fibres** → ventricles contract (**QRS**; backup **20–40/min**). T wave = ventricular repolarisation. Lower pacemakers are slower: why complete heart block gives a slow pulse.

Cardiac cycle and sounds

Systole: isovolumetric contraction, ejection. **Diastole:** isovolumetric relaxation, rapid filling, atrial top-up. LV 120/0–10, aorta 120/80; RV 25/0–5, PA 25/10 mmHg.

S1 mitral and tricuspid close (start of systole); **S2** aortic and pulmonary close, splits on inspiration; **S3** rapid filling into a failing ventricle (normal in the young); **S4** atrial kick into a stiff ventricle (hypertension, ischaemia).

LV perfusion happens mainly in diastole (~⅔ of each beat at rest); tachycardia shortens diastole while demand rises: why fast rates bring angina and beta-blockers help.

Output and regulation

Cardiac output	HR × SV: ~5 L/min rest; 20–25 in maximal exercise (fit young)
Stroke volume	EDV − ESV: ~70 mL
Ejection fraction	SV ÷ EDV: **55–70%**; ≤ 40% reduced
Preload	Filling stretch: ↑ by muscle pump, breathing, lying
Afterload	Resistance: ↑ by high BP, aortic stenosis
Contractility	↑ sympathetic; ↓ damaged muscle

Frank–Starling: more filling → stronger contraction (to a limit); the failing heart is on a flatter curve. **BP = CO × peripheral resistance**; **MAP ≈ diastolic + ⅓ pulse pressure** (120/80 → ~93). **Baroreceptors** (carotid sinus, aortic arch) correct a fall on standing; fail after bed rest, drugs, autonomic damage → postural hypotension. **O_2 demand** rises with HR, BP, wall tension, contractility: **rate–pressure product** = HR × systolic.

Physiology behind physio decisions

- Diastolic coronary flow, RPP → stay below the angina HR; long warm-up; no sudden bursts.
- **Valsalva** swings BP → breathe out on effort; no breath-holding when lifting.
- RCA territory → inferior MI: watch slow HR and block when mobilising.
- Frank–Starling, failing heart → lower intensity, intervals, watch weight gain.
- Baroreceptors → sit up in stages after bed rest; postural BP.
- Muscle pump → ankle pumps; cool down slowly (pooling).
- Autonomic HR control → beta-blockers, pacemakers, transplant: use RPE.

Cardiac and respiratory drugs: what physiotherapists need to know

How each drug changes assessment and exercise.

Full notes p. 352

Asked: not on its own; needed in every case answer and viva. One line on a drug's effect on your plan (for example "on a beta-blocker, so intensity by RPE 11–13") shows clinical reasoning.

Heart and BP drugs

Beta-blockers (metoprolol, bisoprolol)	↓ HR, BP, angina, arrhythmia → **Karvonen overestimates: RPE, talk test**; fatigue; non-selective → bronchospasm in asthma
ACEi, ARBs (ramipril, losartan); sacubitril–valsartan	↓ BP, protect heart and kidney → postural and post-exercise hypotension, longer cool-down; ACEi dry cough
Diuretics (furosemide, spironolactone)	Dehydration, postural dizziness, **low K^+** (cramps, arrhythmia); treat after the morning dose has worked
Nitrates (GTN, isosorbide)	Relieve angina → carry GTN, take it **sitting**; headache; before a known trigger
CCBs (amlodipine; diltiazem, verapamil)	Ankle swelling, hypotension; slower HR response (diltiazem, verapamil)
Antiplatelets (aspirin, clopidogrel, ticagrelor)	Bruising, bleeding; **never advise stopping**
Anticoagulants (heparin, warfarin, apixaban)	Gentle handling, care with percussion, suction, falls; **INR** often 2–3 (higher for mechanical valves)
Thrombolytics (streptokinase, tenecteplase)	High bleeding risk day 1: no vigorous chest physio
Statins	Report new muscle aches or weakness
Digoxin	Toxicity: nausea, confusion, visual change, slow irregular pulse (worse with low K^+)
Amiodarone	Slow HR; lung fibrosis, thyroid problems (new cough, breathlessness)
Inotropes, vasopressors	High or rising doses: no active mobilisation

Respiratory drugs

SABA (salbutamol)	Reliever: give **15–30 min before** clearance or exercise; keep at hand; tremor, tachycardia
LABA (salmeterol, formoterol)	12–24 h; not for sudden relief (except formoterol combinations in asthma)
Antimuscarinics (ipratropium; tiotropium)	Bronchodilation, less mucus (COPD); dry mouth
Inhaled steroids (budesonide, fluticasone)	**Rinse mouth** (thrush, hoarse voice); spacer
Oral steroids (prednisolone)	Long courses: **osteoporosis** (care with percussion, falls), weakness, high sugar, thin skin, slow healing
Theophylline, doxofylline	Narrow margin: palpitations, nausea, tremor, seizures
Montelukast	Asthma add-on; mood or sleep changes
Mucoactives (NAC, carbocisteine, hypertonic saline, dornase)	Clearance after them; hypertonic saline → bronchodilator first
Antibiotics	Inhaled ones **after** clearance

Order (CF, bronchiectasis): bronchodilator → mucoactive → clearance → inhaled antibiotic or steroid.

Other drugs

- **Insulin, sulfonylureas:** hypos during and hours after exercise: check sugar, carry glucose. Metformin, gliptins, SGLT2 inhibitors: low hypo risk; SGLT2 dehydration.
- **Anti-TB:** isoniazid neuropathy (pyridoxine); rifampicin orange urine and tears, interactions; ethambutol blurred vision; liver toxicity: report.
- **Opioids, analgesia:** treat 30–60 min after oral doses; drowsiness, slow breathing; epidurals: leg weakness, low BP sitting up.
- **ICU sedatives, paralysis:** no movement or cough; active work in sedation breaks. **Pyridostigmine:** treat ~1 h after the dose.

Timing physio around medicines

Bronchodilator 15–30 min before clearance or exercise; analgesia before post-op chest physio; exercise 1–2 h after a meal with insulin, sugar before and after; GTN before a known trigger; after the morning diuretic has worked; coordinate sedation (tetanus, ICU) with the nurse.

Basic life support (CPR) for physiotherapists

Skill and viva; 2025 CPR updates.

Full notes p. 356

Asked: not in the 2016–2026 theory papers; a practical skill demonstration and viva favourite. Possible 3- or 7-mark note.

Recognising arrest; chain of survival

Cardiac arrest: unresponsive with **no normal breathing** (none, or occasional gasps: **gasping is not breathing**; start CPR). Each minute without CPR lowers survival.

Chain (adult, outside hospital): call early → high-quality CPR → early defibrillation → advanced care → post-arrest care → recovery. In hospital: recognise deterioration early.

Adult BLS steps

Check, Call, Compress, Shock

1. **Safety** of the scene.
2. **Response:** tap shoulders, shout "Are you all right?"; don't shake the neck.
3. **Call:** in hospital the code team; outside **108 or 112** on speaker; send for an **AED**.
4. **Breathing** (± carotid pulse if trained) for **≤ 10 s**.
5. No normal breathing → **compressions at once, 30:2**.
6. **AED:** switch on, follow prompts; restart compressions right after each shock.
7. Continue until normal breathing, the team takes over, or exhaustion; **swap every 2 min**.

High-quality CPR

Hands	Heel on the centre of the chest (lower half of sternum), other on top, arms straight, shoulders over hands
Rate	**100–120/min**
Depth	**5–6 cm**
Recoil	Full; don't lean
Pauses	**Under 10 s**
Ratio	**30:2** without an advanced airway
Breaths	Head tilt–chin lift (jaw thrust if neck injury); 1 s each, just visible chest rise; no over-ventilation
Advanced airway	Continuous compressions, 1 breath every 6 s
Surface	Start at once even on a bed (2025); backboard helps but don't delay

Compression-only CPR for untrained or unwilling rescuers: hard, fast, continuous; India's **COLS** guideline for lay people.

AED

Switch on, follow prompts; bare, dry chest; pads **below the right clavicle** and **left side below the armpit** (as pictured), not over a pacemaker; **stand clear** for analysis; shock if advised ("Stand clear", check, press); **restart compressions immediately for 2 min**, no pulse check; reanalysis every 2 min. Shockable: **VF and pulseless VT**.

Children, choking, recovery position

	Child (1 to puberty)	Infant (under 1)
Depth	⅓ chest, ~5 cm	⅓ chest, ~4 cm
Technique	One or two hands	Two thumbs encircling, or heel of one hand (two-finger method dropped, 2025)
Ratio	30:2 one rescuer, **15:2 two rescuers**; rate 100–120	

Children arrest mostly from breathing problems: breaths matter more.

Choking: mild (can cough, speak): encourage coughing. **Severe** adult or child: **5 back blows then 5 abdominal thrusts**, alternate (AHA 2025); pregnant or obese: chest thrusts. **Infant:** 5 back blows face-down on the forearm, then 5 chest thrusts; no abdominal thrusts. Unresponsive: floor, call, CPR; look in the mouth before breaths, no blind finger sweeps.

Recovery position (breathing normally, no spinal injury): on the side, upper hip and knee bent in front, hand under cheek, head slightly back; keep checking breathing.

In physio settings

Know the AED, trolley, emergency number and trained staff; phone at hand in rehab. **Warning signs:** chest pain, sudden breathlessness, dizziness, pallor, sweating, new irregular pulse, falling BP → stop, sit or lie down, vitals, call early. VF on a monitor + unresponsive = arrest (a loose lead also looks chaotic: check the patient). After heart surgery follow the unit protocol but never withhold compressions for fear of the sternum. Check DNR wishes in advance. BLS certificates last ~2 years.

Answer writing: the paper, ICF, goals and documentation

How to write 3-, 7- and 15-mark answers.

Full notes p. 361

Asked: ICF and goals in nearly every long answer; structure decides marks in every section. Syllabus Chapter 5 (ICF, goals, documentation).

The paper and a time plan

Read and choose	Plan the long answers: 5 min
Q1: 5 of 6 × 3	15 marks: ~6 min each (30 min)
Q2: 5 of 6 × 7	35 marks: ~14 min each (70 min)
Q3, Q4: 1 of 2 each × 15	30 marks, patient cases: ~30 min each (60 min)
Revise	Underline keywords, add doses, precautions: 15 min

~2 minutes a mark. Slow writer: long answers first. Choose the case you know best.

Short answers

3 marks (half a page): one-line definition → 3–5 points, a small table or labelled sketch → one line of physio relevance. *Example (ABI):* definition; highest ankle ÷ highest brachial; 0.91–1.40 normal, ≤ 0.90 PAD, below 0.40 severe, above 1.40 calcified; use before compression.

7 marks (1.5–2 pages): definition (2 lines) → principle, classification or why (flowchart) → 6–8 headed points or a table with doses → labelled diagram → precautions, contraindications → one concluding line (evidence or outcome measure).

The 15-mark case

1. Read the split (5 + 5 + 5, 3 + 4 + 8) and give space in proportion.
2. **Underline clues:** age, job, earning role, home (stairs, lives alone), post-op day, drains, settings, tests, what the patient wants.
3. **Functional diagnosis** in one sentence, then the **ICF table** with a clinical reasoning column.
4. Problem list by priority → **SMART goals** with time frames.
5. Management by phase or impairment with **technique, dose, reason**; precautions and stop criteria; education, home programme.
6. Outcome measures; discharge or progression criteria.

Doses with units (reps, sets, times a day, minutes, RPE, % HRR): a plan without doses loses marks.

Applying ICF (Ch 5)

Health condition	Diagnosis and stage: "CABG day 3", "post-Koch's bronchiectasis", "40% burns"
Body function, structure	Pain VAS 6, breathlessness, sputum, ↓ expansion, weakness, stiff shoulder, oedema, low SpO_2
Activity	Cannot sit, sleep, climb stairs, walk to the field, self-care
Participation	Cannot work as driver, teacher, cook; only earner; misses school; avoids people
Environment	2nd floor no lift, lives alone, family, cost, dusty job, distance
Personal	Age, smoker, motivation, fear, "wants to resume job", "ignored advice"

Template: "[Age]-year-old [job] with [condition, day or stage], presenting with [3–4 impairments], leading to [activity limitations], restricting [participation], with [facilitator] and [barrier]." **Mistakes:** symptoms as participation; no contextual factors; describing disease not function; impairments not linked to treatment.

Goals and documentation

SMART: specific, measurable, achievable, relevant, time-bound; written as what the patient will do. **Short-term:** days to ~2 weeks ("clears sputum with ACBT within 3 days"). **Long-term:** ~6–12 weeks ("climbs to the 2nd floor with one rest by 8 weeks"). Patient-specific: bank manager drives 1 h to work by ~4 weeks; tiffin cook cooks for ten clients by 8–12 weeks; farmer walks to the field; milkman delivers door to door; schoolchild attends daily and plays. Link each goal to an impairment and each treatment to a goal ("because…").

SOAP: S what the patient reports; **O** measured findings and treatment given; **A** your interpretation and progress; **P** next steps. Also consent, precautions, adverse events, dated outcome measures.

Presentation: headings, numbered points, tables; underline keywords; labelled diagrams (ICF, ACBT cycle, drainage positions, CR phases, rule of nines, spirogram); always precautions; keep pathology short; leave lines for additions. **Before handing in:** every part answered? doses with units? time frames? precautions? diagram labelled? question numbers right?

Practical exam pack: long case, short case, spots and journal

Follows the official 80-mark scheme.

Full notes p. 365

Practical exam: 80 marks + 20 internal assessment. Long case 45, short case 2 × 10, spots 5 × 2, journal 5. (College terminal and prelim: spots 5 × 3 = 15, journal no marks.)

Marking scheme

Long case 45	(a) subjective and physical examination 10; (b) evaluation and PT diagnosis (ICF) 10; (c) plan of care and goals 10; (d) **demonstrate one test and one treatment** on the patient 15. Cognitive, psychomotor and affective domains
Short case 20	Two physiotherapy skills, 10 each
Spots 10	Chest or cardiac X-ray, ABG, PFT, ECG, devices: 5 × 2
Journal 5	Assessment, evaluation, diagnosis, prognosis, intervention with ICF

Long case proforma

Subjective: name, age, sex, **occupation demands**, hand dominance, address (stairs, distance); chief complaints in the patient's words with duration; HPI: breathlessness (onset, triggers, mMRC or NYHA, orthopnoea, PND), cough and sputum (amount, colour, smell, blood), chest pain, palpitations, fainting, swollen feet, claudication distance, fever, weight loss; past (TB, diabetes, BP, heart, surgery, ventilation); **pack-years** (cigarettes a day ÷ 20 × years), bidi, tobacco, alcohol, diet, sleep, bladder and bowel; drugs (inhaler technique, beta-blockers, home oxygen); family, support, finances, home, work; **the patient's goals**.

Objective: HR and rhythm, BP, RR, SpO_2 (air or O_2), temperature, pain; build, BMI, posture, cyanosis, clubbing, pallor, oedema, lines, drains, scars; chest shape, pattern, accessory muscles; trachea, **tape expansion at axilla, nipple line, xiphisternum**; percussion, auscultation, cough, sputum; pulses incl. feet, capillary refill, pitting grade, postural BP, ABI; shoulder and thoracic range, posture, MRC strength, bed mobility, transfers, gait, balance; 6MWT or shuttle with SpO_2 and Borg, or sit-to-stand; mMRC, Borg, NYHA, CCS, CAT, SGRQ, MLHFQ; X-ray, ABG, PFT, ECG, echo, bloods.

Then: ICF and functional diagnosis → problems → short- and long-term goals → plan with doses and precautions → demonstrate one test (chest expansion, 6MWT set-up, ABI, auscultation, postural BP) and one treatment (ACBT, drainage position, incentive spirometry, monitored Phase I walking).

Presenting (3–5 min)

"Mr X, 55, a bank manager from Pune, day 3 after CABG for triple vessel disease. Main complaints: wound pain, breathlessness on walking. Smoker, 20 pack-years, hypertensive. HR 92 regular, BP 130/84, SpO_2 95% on air, ↓ sounds at the left base, weak cough, walks 40 m at RPE 12. My functional diagnosis is ...; short-term goals ...; long-term goals ...; plan ... . I will now demonstrate a 6MWT set-up and ACBT."

Viva: why this goal; SpO_2 falls?; stop criteria; progression; what the X-ray or ECG shows; why this dose; drugs and exercise; outcome measures; home programme.

Short case: 8 steps for any skill

1. Introduce, check identity, explain, **consent**.
2. Hand hygiene; privacy, draping; mask in suspected TB.
3. **Vital signs and contraindications** for this skill.
4. Position patient and yourself.
5. Explain simply → demonstrate → patient does it → correct.
6. Watch breathing, SpO_2, pain, colour; stop if needed.
7. State the **dose** and home programme.
8. Reassess, thank, document.

Likely skills: ACBT, huffing, supported cough, AD; drainage positions, percussion, vibration; PEP, Flutter, Acapella, MDI + spacer, nebuliser; diaphragmatic and pursed-lip breathing, thoracic expansion, SMI, mobility; NPF, PNF; positions of ease, pacing, relaxation; IMT; 6MWT, shuttle, Harvard step test; ABI, pulses, postural BP, Buerger–Allen; stump and compression bandaging, stockings, burns splints, taping; sternal precautions, log-roll; suction and bagging (manikin); TENS; BLS.

Spots (5 × 2)

- **X-ray:** PA or AP; main finding (e.g. left lower lobe collapse: sail sign, raised left diaphragm); two features; physio meaning.
- **ABG:** pH → respiratory or metabolic → compensation → oxygenation → name it ("acute type 2 respiratory failure").
- **PFT:** FEV1, FVC, ratio → obstructive, restrictive or mixed; severity; reversibility; likely condition.
- **ECG:** rate, rhythm, P, QRS width, ST → name → safe to exercise?
- **Device:** name, parts, principle, one indication, one precaution.

Incentive spirometer	Feedback for SMI: post-op expansion; add-on only
PEP, Flutter, Acapella	10–20 cmH_2O (+ oscillation); not with undrained pneumothorax
MDI, spacer, DPI	Rinse after steroids; DPI needs a fast strong breath
Jet nebuliser	Narrow jet + baffle; air-driven if CO_2 risk
Cannula, simple, Venturi, reservoir	~24–40%, 35–60%, fixed 24–60%, 60–90%; simple mask ≥ 5 L
HME	Passive humidity; not with thick or bloody secretions or nebulisation
Suction catheter, Yankauer	Under 15 s, ≤ −200 mmHg
Ambu bag, MHI circuit	Slow breath, hold, quick release; ≤ ~40 cmH_2O; no undrained pneumothorax
ETT, tracheostomy (cuffed)	Cuff seals; secure on every turn
ICD with underwater seal	Water = one-way valve; bottle below chest; never clamp if bubbling
Peak flow meter, spirometer	PEF; FEV1, FVC; best of three

Pulse oximeter	Light absorption; errors: cold, movement, polish, CO
Doppler + cuff	ABI; above 1.40 unreliable
Threshold IMT	Spring valve, start ~30% MIP; not after pneumothorax, perforated eardrum
Stockings, bandages, garments, splints	ABPI before compression; check splint pressure

Journal, affective domain, last two weeks

Journal: each case under assessment, evaluation, ICF and functional diagnosis, prognosis, goals, intervention with doses and progression, outcome measures, SOAP note; examiners read ICF and doses first.

Affective domain: greeting, consent, respectful language, privacy; hand hygiene, infection control; safety checks (vitals, oxygen, lines, drains) and stopping when unwell; explain in Marathi or Hindi and check understanding; honest viva answers ("I don't know, I would find out by..."); leave the patient comfortable.

Last two weeks: present three timed practice long cases aloud (cardiac, respiratory, vascular or surgical); practise each skill on a friend with the 8 steps; spot drills (X-ray, ABG, PFT, ECG, device in 10 min); finish the journal.

Numbers to know

For the last day: the values examiners expect to see written, collected from all the sheets. Cover the right-hand column and recall. The sheet name in brackets tells you where the detail is.

Cardiac rehab (IHD, Cardiac rehab)

Phase I limits	**RPE below 13; HR ≤ rest + 20 and ≤ 120 (MI); ≤ rest + 30 (surgery)**; up to ~3 METs
Phase I dose	Bouts 3–5 min, rest 1–2 min, ~20 min total; 3–4 times a day (days 1–3), then twice
Phase II aerobic	40–80% HRR or RPE 11–16 (start 11–13); 3–5 days; 20–60 min; build to 150 min/week
Phase II resistance	2–3 days; 10–15 reps at RPE 11–13 (~40–60% 1RM); 8–10 exercises
Karvonen	Rest + % × (max – rest); max = 220 – age
Risk (AACVPR)	Low: EF ≥ 50%, ≥ 7 METs. High: EF below 40%, symptoms below 5 METs
Sternum	Precautions 6–8 weeks; no heavy lifting ~3 months; driving ~4 weeks after CABG; 1 week after PCI if EF ≥ 40%
Don't start	BP above 180/110; postural drop over 20 with symptoms; resting HR over 120
Stop	Diastolic ≥ 110; systolic falls over 10 as work rises. Exercise test: systolic over 250 or diastolic over 115
Angina	GTN; not better 5 min after first dose → emergency
CR benefit	MI ↓ 28%, admissions ↓ 35% (Cochrane 2026)

Heart, BP, metabolic

ECG paper	Small square 0.04 s, large 0.2 s; PR 0.12–0.20 s; QRS below 0.12 s; rate = 300 ÷ large squares
STEMI	≥ 1 mm in 2 neighbouring leads (V2–V3: 2 mm men ≥ 40, 2.5 mm men under 40, 1.5 mm women)
Primary PCI	Within 90 min (120 if transferred); thrombolysis within 30 min; O_2 only if SpO_2 below 90%
EF	Normal 55–70%; reduced ≤ 40%
Conduction	SA 60–100; AV backup 40–60; ventricles 20–40/min
MAP	≥ 65 mmHg (ICU target); 120/80 → ~93
Hypertension	≥ 140/90 (ISH, India); ACC/AHA ≥ 130/80; target below 130/80
Isometric handgrip	4 × 2 min at 30–40% max, 3 days a week
Salt	Below 5 g a day (sodium below 2,300 mg)
BMI (Asian Indian)	Normal 18.5–22.9; overweight 23–24.9; obese ≥ 25; waist ≥ 90 (M), ≥ 80 (F)
Metabolic syndrome	3 of 5: waist; TG ≥ 150; HDL below 40 (M) / 50 (F); BP ≥ 130/85; glucose ≥ 100
Diabetes	Fasting ≥ 126; 2-h ≥ 200; HbA1c ≥ 6.5%; hypo below 70 → 15 g, recheck in 15 min
Diabetes exercise	≥ 150 min/week, no more than 2 days off; resistance 2–3 days
Weight loss exercise	Over 250 min/week; lose 5–10%
Heart failure alarm	2 kg gain in 3 days

Vascular (PAD, Venous, Amputation)

ABI	1.00–1.40 normal; 0.91–0.99 borderline; **≤ 0.90 PAD**; below 0.40 severe; above 1.40 calcified (TBI ≤ 0.70)
Compression	ABPI ≥ 0.8 full (~40 mmHg at ankle); 0.5–0.8 reduced; below 0.5 none
Claudication walking	**30–45 min, ≥ 3 a week, ≥ 12 weeks**; to moderate pain (2–3 of 4)
Fontaine	IIa over ~200 m; IIb under ~200 m; III rest pain; IV ulcer, gangrene
Tests	Capillary refill over ~2 s; venous filling over ~20 s; Buerger's angle below 20° = severe
DVT	Ankle pumps 10 an hour; walk once anticoagulation is therapeutic
Varicose	Veins ≥ 3 mm; elevate 30 min 3–4 times a day; heel raises 2–3 × 10–15
Lymphoedema	≥ 2 cm circumference or ~10% volume; CDT intensive 2–4 weeks
Stump	Transtibial 12–15 cm below joint line; transfemoral ~25 cm from trochanter; bandage 10 / 15 cm wide, re-apply 4–6 h; prone 20–30 min 2–3 times a day

Lungs, PFT, ABG

SpO_2 targets	94–98%; **88–92%** if CO_2 risk
LTOT	PaO_2 ≤ 55 (SpO_2 ≤ 88%), or 56–59 with cor pulmonale or polycythaemia; ≥ 15 h a day
COPD	FEV1/FVC below 0.70; GOLD 1 ≥ 80, 2 50–79, 3 30–49, 4 below 30%
Reversibility	≥ 12% and 200 mL (2005); over 10% of predicted (2022)
Asthma	Life-threatening: SpO_2 below 92%, PEF below 33%; acute severe: PEF 33–50%, RR ≥ 25, HR ≥ 110. Zones 80 / 50%
ABG	pH 7.35–7.45; $PaCO_2$ 35–45; HCO_3^- 22–26; PaO_2 80–100; **failure below 60**
P/F	Normal 400–500; ARDS mild 201–300, moderate 101–200, severe ≤ 100; prone if below 150 (≥ 16 h)
Volumes	TV 0.5, IRV 3.0, ERV 1.1, RV 1.2, FRC 2.3, VC 4.6, TLC 5.8 L; dead space 150 mL
V/Q	Overall 0.8; apex ~3, base ~0.6
Children, fast breathing	≥ 60 (under 2 months), ≥ 50 (2–11 months), ≥ 40 (1–5 years)
Sweat chloride	≥ 60 mmol/L = CF

Techniques and devices

Deep breathing	5–10 breaths every hour, 3-s hold (SMI 3–5 s)
ACBT	3–4 expansion breaths, 1–2 huffs; stop after 2 dry cycles; 10–30 min once or twice a day
Postural drainage	~10 min per position; foot up ~15° (middle lobe, lingula), 20–30° (basal)
Sputum indication	Over ~25–30 mL a day
PEP	10–20 cmH_2O; 8–10 breaths then 2–3 huffs; 15–20 min
IMT	≥ 30% MIP; 30 breaths twice a day; ≥ 6–8 weeks
Suction	**≤ 15 s; ≤ –200 mmHg** (–120 children); catheter under ½ tube (8.0 → 12 Fr); ≤ 2–3 passes
Bagging	Hold 2–3 s, quick release; ≤ ~40 cmH_2O
Cough assist	~±40 cmH_2O
Humidity	Normal 44 mg/L at 37 °C; HME ≥ 30; heated 33–44 mg/L at 34–41 °C
Nebuliser	Droplets 1–5 µm; jet 6–8 L/min

Oxygen devices	Cannula 1–4 L (24–40%); simple mask 5–10 L (never below 5); reservoir 15 L (60–90%); Venturi fixed 24–60%

ICU and ventilation

Settings	VT 6–8 mL/kg (6 in ARDS); plateau ≤ 30; PEEP ~5
Mobilise (green)	**FiO_2 ≤ 0.6, SpO_2 ≥ 90%, RR ≤ 30, PEEP ≤ 10**; RASS –1 to +1
Weaning screen	SpO_2 ≥ 90% on FiO_2 ≤ 0.4–0.5 and PEEP ≤ 5–8; pH ≥ 7.25
Weaning indices	RSBI below 105; MIP stronger than –20 to –30; VC over 10 mL/kg; VE below 10 L/min
SBT	30–120 min; stop: RR over 35, SpO_2 below 90%, HR over 140, systolic over 180 or below 90
Neuromuscular	VC below 20 mL/kg (20/30/40 rule); peak cough flow below 270 (augment), below 160 ineffective
ICP	Normal 5–15; treat over ~22; CPP 60–70; head up 30°; GCS ≤ 8 intubate
ICU weakness	MRC sum score below 48/60
Positioning	Head of bed 30–45°; turns 2–4-hourly; Braden ≤ 18 at risk
Organophosphate	Intermediate syndrome 24–96 h

Exercise, testing, scales

MET	3.5 mL O_2/kg/min; light below 3, moderate 3–5.9, vigorous ≥ 6
Moderate intensity	40–59% HRR; 64–76% HRmax; RPE 12–13
WHO adults	150–300 min moderate (75–150 vigorous); strength ≥ 2 days
Exercise test, absolute no	MI within 2 days; unstable angina; symptomatic severe AS; acute PE, DVT, dissection
Bruce	3-min stages; stage 1 1.7 mph at 10%; modified adds 0% and 5%
6MWT	Course ≥ 30 m; MCID ~25–33 m; not within a month of MI
ISWT	10 m, 12 levels; MCID ~47.5 m
Harvard step	~50 cm, 30 steps/min, 5 min; index = s × 100 ÷ (2 × sum of 3 counts)
Pulmonary rehab	6–12 weeks, ≥ 2 supervised a week; walk 80% of 6MWT speed; Borg 4–6; 60–70% 1RM
Borg	RPE 6–20 (13 somewhat hard); CR10: 3 moderate, 5 severe, 7 very severe

Burns, surgery, BLS

Rule of nines	Head 9; each arm 9; trunk front 18, back 18; each leg 18; perineum 1; palm 1%
Burns	Cool water 20 min; Parkland 4 mL × kg × %TBSA, half in 8 h; urine 0.5 mL/kg/h; shock over ~20%
Grafts, scars	Immobilise graft 3–5 days; garments ~23 h a day for 12–18 months (20–25 mmHg)
Mastectomy	Shoulder ≤ 90° until drains out (~1 week); stretches 20–30 s × 3–5, 2–3 times a day
Abdominal surgery	No heavy lifting ~6 weeks; 30-min pre-op session halves PPCs
Pacemaker, ICD	Arm below shoulder 4–6 weeks; train ≥ 10 beats below the ICD rate
Flail chest	≥ 3 ribs, each broken in ≥ 2 places
Scoliosis	Cobb ≥ 10°
CPR	**100–120/min, 5–6 cm, 30:2** (15:2 two rescuers, children); pauses under 10 s; check ≤ 10 s; swap every 2 min; call 108 or 112
Choking	5 back blows + 5 abdominal thrusts (infant: chest thrusts)

Abbreviations

1RM: One-repetition maximum
6MWT: Six-minute walk test
AACVPR: American Association of Cardiovascular and Pulmonary Rehabilitation
AARC: American Association for Respiratory Care
ABG: Arterial blood gas
ABI: Ankle–brachial index
ABPI: Ankle–brachial pressure index
ACBT: Active cycle of breathing technique
ACC: American College of Cardiology
ACCP: American College of Chest Physicians
ACE: Angiotensin-converting enzyme
ACEi: ACE inhibitor
ACPRC: Association of Chartered Physiotherapists in Respiratory Care
ACS: Acute coronary syndrome
ACSM: American College of Sports Medicine
ACT: Asthma Control Test
AD: Autogenic drainage
ADA: American Diabetes Association
ADL: Activities of daily living
AED: Automated external defibrillator
AF: Atrial fibrillation
AHA: American Heart Association
AKA: Above-knee amputation
ALAT: Latin American Thoracic Association
AMPPRO: Amputee Mobility Predictor (with prosthesis)
AP: Anteroposterior (X-ray view; chest diameter)
APTA: American Physical Therapy Association
AQLQ: Asthma Quality of Life Questionnaire
ARB: Angiotensin receptor blocker
ARDS: Acute respiratory distress syndrome
ARNI: Angiotensin receptor–neprilysin inhibitor
AS: Aortic stenosis
ASD: Atrial septal defect
ATP: Adenosine triphosphate
ATS: American Thoracic Society
AV: Atrioventricular
AYUSH: Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (Indian ministry)
BACPAR: British Association of Chartered Physiotherapists in Amputee Rehabilitation
BC: Breathing control
BiPAP: Bilevel positive airway pressure
BK: Below-knee
BKA: Below-knee amputation
BLS: Basic life support
BMI: Body mass index
BNP: Brain natriuretic peptide
BP: Blood pressure
BRCA: Breast cancer genes 1 and 2
BTS: British Thoracic Society
CAAT: Chronic Airways Assessment Test (formerly the COPD Assessment Test, CAT)
CABG: Coronary artery bypass grafting
CAD: Coronary artery disease
CAT: COPD Assessment Test
CBNAAT: Cartridge-based nucleic acid amplification test (for TB)
CCS: Canadian Cardiovascular Society
CEAP: Clinical, aetiological, anatomical, pathophysiological classification (venous disease)
CF: Cystic fibrosis
CFTR: Cystic fibrosis transmembrane conductance regulator (gene and protein)
CHD: Coronary heart disease
CIVIQ: Chronic Venous Insufficiency Quality of Life Questionnaire
CK: Creatine kinase
CK-MB: Creatine kinase, heart muscle (MB) form
CMV: Controlled mandatory ventilation
CO: Carbon monoxide (also cardiac output)
COLS: Compression-only life support
COPD: Chronic obstructive pulmonary disease
CP: Cerebral palsy (also creatine phosphate)
CPAP: Continuous positive airway pressure
CPET: Cardiopulmonary exercise test
CPP: Cerebral perfusion pressure (MAP – ICP)
CPR: Cardiopulmonary resuscitation
CR: Cardiac rehabilitation
CR10: Borg category–ratio 0–10 scale
CRQ: Chronic Respiratory Questionnaire
CSF: Cerebrospinal fluid
CT: Computed tomography
CTR: Cardiothoracic ratio
CVC: Central venous catheter
CXR: Chest X-ray
D2: Diagonal 2 (PNF pattern)
DKA: Diabetic ketoacidosis
DLCO: Diffusing capacity of the lung for carbon monoxide
DM: Diabetes mellitus
DNA: Deoxyribonucleic acid
DNR: Do not resuscitate
DPI: Dry-powder inhaler
DPP-4: Dipeptidyl peptidase 4
DVT: Deep vein thrombosis
ECC: Emergency cardiovascular care
ECG: Electrocardiogram
ECMO: Extracorporeal membrane oxygenation
EDV: End-diastolic volume
EF: Ejection fraction
EIB: Exercise-induced bronchoconstriction
EMG: Electromyography
EORTC: European Organisation for Research and Treatment of Cancer
EPP: Equal pressure point
EQ-5D: EuroQol five-dimension questionnaire
ER: External rotation
ERAS: Enhanced recovery after surgery
ERS: European Respiratory Society
ERV: Expiratory reserve volume
ESC: European Society of Cardiology
ESVS: European Society for Vascular Surgery
ESWT: Endurance shuttle walk test
ETT: Endotracheal tube
EU: European Union
EVD: External ventricular drain
FeNO: Fractional exhaled nitric oxide
FET: Forced expiration technique
FEV1: Forced expiratory volume in 1 second
FIM: Functional Independence Measure
FiO_2: Fraction of inspired oxygen (0.21 = room air)
FITT: Frequency, intensity, time and type
FRC: Functional residual capacity
FVC: Forced vital capacity
GBS: Guillain–Barré syndrome
GCS: Glasgow Coma Scale
GINA: Global Initiative for Asthma
GLP-1: Glucagon-like peptide 1
GOLD: Global Initiative for Chronic Obstructive Lung Disease
GTN: Glyceryl trinitrate (nitroglycerin)
HADS: Hospital Anxiety and Depression Scale
HbA1c: Glycated haemoglobin (3-month blood sugar)
HDL: High-density lipoprotein (good cholesterol)
HF: Heart failure
HFCWO: High-frequency chest wall oscillation
HFNC: High-flow nasal cannula
HFrEF: Heart failure with reduced ejection fraction
HIV: Human immunodeficiency virus
HME: Heat and moisture exchanger
HR: Heart rate
HRCT: High-resolution CT scan
HRmax: Maximum heart rate
HRR: Heart-rate reserve
HTN: Hypertension
ICD: Implantable cardioverter-defibrillator; on the lung surgery and pleural sheets, intercostal (chest) drain
ICF: International Classification of Functioning, Disability and Health
ICMR: Indian Council of Medical Research
ICP: Intracranial pressure
ICS: Inhaled corticosteroid
ICU: Intensive care unit
ICU-AW: ICU-acquired weakness
IHD: Ischaemic heart disease
ILCOR: International Liaison Committee on Resuscitation
ILD: Interstitial lung disease
IMA: Internal mammary artery
IMAs: Internal mammary arteries
IMT: Inspiratory muscle training
INR: International normalised ratio (warfarin control)
IP: Interphalangeal
IPF: Idiopathic pulmonary fibrosis
IPPB: Intermittent positive pressure breathing
IPV: Intrapulmonary percussive ventilation
IRV: Inspiratory reserve volume
ISBI: International Society for Burn Injuries
ISH: International Society of Hypertension
ISWT: Incremental shuttle walk test
IWGDF: International Working Group on the Diabetic Foot
JRS: Japanese Respiratory Society
KCCQ: Kansas City Cardiomyopathy Questionnaire
LA: Left atrium
LABA: Long-acting β2-agonist
LAD: Left anterior descending artery
LAMA: Long-acting muscarinic antagonist
LAQ: Long-answer question
LDH: Lactate dehydrogenase
LDL: Low-density lipoprotein
LIMA: Left internal mammary artery
LOPS: Loss of protective sensation
LTG: Long-term goals
LTOT: Long-term oxygen therapy
LV: Left ventricle

LVH: Left ventricular hypertrophy
MAP: Mean arterial pressure
MCID: Minimal clinically important difference
MCP: Metacarpophalangeal
MDI: Metered-dose inhaler
MDR-TB: Multidrug-resistant tuberculosis
MEP: Maximal expiratory pressure
MET: Metabolic equivalent
METs: Metabolic equivalents (1 MET = resting energy use)
MHI: Manual hyperinflation (bagging)
MI: Myocardial infarction
MIP: Maximal inspiratory pressure
MLHFQ: Minnesota Living with Heart Failure Questionnaire
mMRC: Modified Medical Research Council dyspnoea scale
MODS: Multiple organ dysfunction syndrome
MONA: Morphine, oxygen, nitrates, aspirin
MR: Mitral regurgitation
MRC: Medical Research Council
MRI: Magnetic resonance imaging
MRM: Modified radical mastectomy
MS: Mitral stenosis (also multiple sclerosis)
NG: Nasogastric
NHS: National Health Service (UK)
NICE: National Institute for Health and Care Excellence (UK)
NIV: Non-invasive ventilation
NMES: Neuromuscular electrical stimulation
NPF: Neurophysiological facilitation of respiration
NRS: Numeric rating scale
NSAID: Non-steroidal anti-inflammatory drug
NSAIDs: Non-steroidal anti-inflammatory drugs
NSTEMI: Non-ST-elevation myocardial infarction
NTEP: National Tuberculosis Elimination Programme (India)
NYHA: New York Heart Association
PA: Posteroanterior (X-ray view); pulmonary artery
$PaCO_2$: Arterial carbon dioxide pressure
PAD: Peripheral arterial disease
PaO_2: Arterial oxygen pressure
PAQLQ: Paediatric Asthma Quality of Life Questionnaire
PCF: Peak cough flow
PCI: Percutaneous coronary intervention (angioplasty)
PDA: Patent ductus arteriosus
PE: Pulmonary embolism
PEEP: Positive end-expiratory pressure
PEF: Peak expiratory flow
PEP: Positive expiratory pressure
PFO: Patent foramen ovale
PFT: Pulmonary function test
PHQ-9: Patient Health Questionnaire (9 items)
PImax: Maximal inspiratory pressure
PIP: Proximal interphalangeal
PND: Paroxysmal nocturnal dyspnoea
PNF: Proprioceptive neuromuscular facilitation
PPCs: Post-operative pulmonary complications
PROM: Passive range of motion
PS: Pressure support
PSV: Pressure support ventilation
PT: Physiotherapy
PTB: Pulmonary tuberculosis
PTCA: Percutaneous transluminal coronary angioplasty
QoL: Quality of life
QRS: QRS complex (ventricular depolarisation on the ECG)
RA: Right atrium
RASS: Richmond Agitation–Sedation Scale
RCA: Right coronary artery
RDS: Respiratory distress syndrome
RH: Relative humidity
RHD: Rheumatic heart disease
RIMA: Right internal mammary artery
RLL: Right lower lobe
RMT: Respiratory muscle training
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RR: Respiratory rate
RSBI: Rapid shallow breathing index (breathing rate ÷ tidal volume in litres)
RV: Residual volume (also right ventricle)
RVH: Right ventricular hypertrophy
SA: Sinoatrial
SABA: Short-acting β2-agonist
SACH: Solid ankle cushion heel (prosthetic foot)
SBT: Spontaneous breathing trial
SCI: Spinal cord injury
SEAS: Scientific Exercise Approach to Scoliosis
SF-36: Short Form 36 health survey
SFJ: Saphenofemoral junction
SGLT2: Sodium–glucose co-transporter 2
SGRQ: St George's Respiratory Questionnaire
SIGAM: Special Interest Group in Amputee Medicine (mobility grades)
SIGN: Scottish Intercollegiate Guidelines Network
SIMV: Synchronised intermittent mandatory ventilation
SMART: Specific, measurable, achievable, relevant, time-bound
SMI: Sustained maximal inspiration
SOAP: Subjective, objective, assessment, plan
SOFA: Sequential Organ Failure Assessment score
SOS: When needed (si opus sit)
SOSORT: International Society on Scoliosis Orthopaedic and Rehabilitation Treatment
SpO_2: Oxygen saturation by pulse oximeter
STEMI: ST-elevation myocardial infarction
STG: Short-term goals
SV: Stroke volume
SVS: Society for Vascular Surgery
TAVI: Transcatheter aortic valve implantation
TB: Tuberculosis
TBSA: Total body surface area
TEE: Thoracic expansion exercises
TENS: Transcutaneous electrical nerve stimulation
TLC: Total lung capacity
TOF: Tetralogy of Fallot
TV: Tidal volume
UK: United Kingdom
UN: United Nations
UVC: Ultraviolet C
VAP: Ventilator-associated pneumonia
VAS: Visual analogue scale
VATS: Video-assisted thoracoscopic surgery
VC: Vital capacity
VF: Ventricular fibrillation
VHI: Ventilator hyperinflation
VO_2: Oxygen uptake
VO_2max: Maximal oxygen uptake
VSD: Ventricular septal defect
VT: Tidal volume on the ventilator sheets; ventricular tachycardia on the ECG sheets
WHO: World Health Organization
WHOQOL-BREF: WHO Quality of Life questionnaire, short version
WIQ: Walking Impairment Questionnaire